

T3 UV one-loop RGE term comparison

Each running coupling has its own table. Rows are model configurations and columns are additive structures in the beta function. Every populated cell contains the complete model-specific term, including its numerical coefficient. A dash means that the term is absent for that configuration. All equations use $16\pi^2 \mu \, dX/d\mu = \beta_X^{(1)}$. Gauge-coupling results use the same RGBeta $g^2 \rightarrow g$ conversion as in the previous report.

$\beta_{g_Y}^{(1)}$

$16\pi^2 \mu \frac{dg_Y}{d\mu} = \beta_{g_Y}^{(1)}$

Column Representative term structure	
T1	$\frac{55g_Y^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\frac{55g_Y^3}{6}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\frac{43g_Y^3}{6}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{59g_Y^3}{6}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\frac{103g_Y^3}{6}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$\frac{175g_Y^3}{6}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$\frac{17g_Y^3}{2}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{43g_Y^3}{6}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{17g_Y^3}{2}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$\frac{25g_Y^3}{2}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$\frac{115g_Y^3}{6}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$\frac{21g_Y^3}{2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{43g_Y^3}{6}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\frac{21g_Y^3}{2}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{41g_Y^3}{2}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$\frac{223g_Y^3}{6}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{59g_Y^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$\frac{43g_Y^3}{6}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{55g_Y^3}{6}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$\frac{95g_Y^3}{6}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$\frac{163g_Y^3}{6}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$\frac{59g_Y^3}{6}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\frac{22g_Y^3}{3}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{59g_Y^3}{6}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\frac{52g_Y^3}{3}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\frac{179g_Y^3}{6}$

$$\beta_{g_2}^{(1)}$$

$$16\pi^2\mu\frac{dg_2}{d\mu}=\beta_{g_2}^{(1)}$$

Column Representative term structure

T1	$-\frac{g_2^3}{2}$
T2	$-\frac{3g_2^3}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{g_2^3}{2}$	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-\frac{3g_2^3}{2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{g_2^3}{2}$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{g_2^3}{2}$	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-\frac{g_2^3}{2}$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$-\frac{17g_2^3}{6}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	$-\frac{17g_2^3}{6}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-\frac{17g_2^3}{6}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$-\frac{17g_2^3}{6}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$-\frac{17g_2^3}{6}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$\frac{31g_2^3}{6}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$\frac{7g_2^3}{6}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$\frac{31g_2^3}{6}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$\frac{31g_2^3}{6}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$\frac{31g_2^3}{6}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{g_2^3}{2}$	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$-\frac{3g_2^3}{2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{g_2^3}{2}$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-\frac{g_2^3}{2}$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-\frac{g_2^3}{2}$	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$\frac{g_2^3}{6}$	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-\frac{5g_2^3}{6}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{g_2^3}{6}$	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\frac{g_2^3}{6}$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\frac{g_2^3}{6}$	—

$$\beta_{g_3}^{(1)}$$

$$16\pi^2\mu\frac{dg_3}{d\mu}=\beta_{g_3}^{(1)}$$

Column Representative term structure

T1	$-7g_3^3$
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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-7g_3^3$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-7g_3^3$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-7g_3^3$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-7g_3^3$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-7g_3^3$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-7g_3^3$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-7g_3^3$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-7g_3^3$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-7g_3^3$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$-7g_3^3$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-7g_3^3$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-7g_3^3$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-7g_3^3$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-7g_3^3$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-7g_3^3$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-7g_3^3$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-7g_3^3$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-7g_3^3$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-7g_3^3$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-7g_3^3$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-7g_3^3$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-7g_3^3$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-7g_3^3$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-7g_3^3$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-7g_3^3$

$$\beta_{Y_u}^{(1)}$$

$$16\pi^2 \mu \frac{dY_u}{d\mu} = \beta_{Y_u}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$3 \operatorname{Tr}\left(Y_d \left(Y_d\right)^{\dagger}\right)\left(Y_u\right)_{i j}$
T2	$+ \operatorname{Tr}\left(Y_e \left(Y_e\right)^{\dagger}\right)\left(Y_u\right)_{i j}$
T3	$+ 3 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger}\right)\left(Y_u\right)_{i j}$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$3 \operatorname{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) \left(Y_u \right)_{ij}$	$+ \operatorname{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) \left(Y_u \right)_{ij}$	$+ 3 \operatorname{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) \left(Y_u \right)_{ij}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$3 \operatorname{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) \left(Y_u \right)_{ij}$	$+ \operatorname{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) \left(Y_u \right)_{ij}$	$+ 3 \operatorname{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) \left(Y_u \right)_{ij}$

Terms 2 of 3

Column	Representative term structure	
T1	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	
T2	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	
T3	$-8g_3^2(Y_u)_{ij}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{17}{12}g_Y^2(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{17}{12}g_Y{}^2(Y_u)_{ij}$

$\beta_{Y_d}^{(1)}$

$$16\pi^2 \mu \frac{dY_d}{d\mu} = \beta_{Y_d}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\text{Tr}\left(Y_e \left(Y_e\right)^\dagger\right)\left(Y_d\right)_{ij}$
T2	$+3 \text{Tr}\left(Y_u \left(Y_u\right)^\dagger\right)\left(Y_d\right)_{ij}$
T3	$+3 \text{Tr}\left(Y_d \left(Y_d\right)^\dagger\right)\left(Y_d\right)_{ij}$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)\left(Y_d\right)_{ij}$	$+3\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\left(Y_d\right)_{ij}$	$+3\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)\left(Y_d\right)_{ij}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)\left(Y_d\right)_{ij}$	$+3\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\left(Y_d\right)_{ij}$	$+3\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)\left(Y_d\right)_{ij}$

Terms 2 of 3

Column	Representative term structure	
T1	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	
T2	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	
T3	$-8g_3^2(Y_d)_{ij}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}\left(\left(Y_d\left(Y_d\right)^{\dagger} Y_d\right)_{ij}-\left(Y_u\left(Y_u\right)^{\dagger} Y_d\right)_{ij}\right)$	$-\frac{9}{4} g_2^2\left(Y_d\right)_{ij}$	$-8 g_3^2\left(Y_d\right)_{ij}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{3}{2}\left(\left(Y_d\left(Y_d\right)^{\dagger} Y_d\right)_{ij}-\left(Y_u\left(Y_u\right)^{\dagger} Y_d\right)_{ij}\right)$	$-\frac{9}{4} g_2^2\left(Y_d\right)_{ij}$	$-8 g_3^2\left(Y_d\right)_{ij}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\frac{3}{2}\left(\left(Y_d\left(Y_d\right)^{\dagger} Y_d\right)_{ij}-\left(Y_u\left(Y_u\right)^{\dagger} Y_d\right)_{ij}\right)$	$-\frac{9}{4} g_2^2\left(Y_d\right)_{ij}$	$-8 g_3^2\left(Y_d\right)_{ij}$

Terms 3 of 3

Column Representative term structure

T1	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{5}{12}g_Y^2(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{5}{12}g_Y{}^2(Y_d)_{ij}$

$$\beta_{Y_e}^{(1)}$$

$$16\pi^2 \mu \frac{dY_e}{d\mu} = \beta_{Y_e}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$3 \operatorname{Tr}\left(Y_d \left(Y_d\right)^{\dagger}\right)\left(Y_e\right)_{i j}$
T2	$+3 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger}\right)\left(Y_e\right)_{i j}$
T3	$+ \operatorname{Tr}\left(Y_e \left(Y_e\right)^{\dagger}\right)\left(Y_e\right)_{i j}$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$3 \operatorname{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) \left(Y_e \right)_{ij}$	$+3 \operatorname{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) \left(Y_e \right)_{ij}$	$+ \operatorname{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) \left(Y_e \right)_{ij}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$3 \operatorname{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) \left(Y_e \right)_{ij}$	$+3 \operatorname{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) \left(Y_e \right)_{ij}$	$+ \operatorname{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) \left(Y_e \right)_{ij}$

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{1}{4}(2\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+3\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$
T2	$-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$
T3	$-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T_1	T_2	T_3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+\frac{1}{4}(2\langle y_1(y_1)^\dagger Y_e \rangle_{ij} + 3\langle y_2(y_2)^\dagger Y_e \rangle_{ij} + 6\langle Y_e(Y_e)^\dagger Y_e \rangle_{ij})$	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{1}{4}(2\langle y_1(y_1)^\dagger Y_e \rangle_{ij} + 3\langle y_2(y_2)^\dagger Y_e \rangle_{ij} + 6\langle Y_e(Y_e)^\dagger Y_e \rangle_{ij})$	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{4}(2\langle y_1(y_1)^\dagger Y_e \rangle_{ij} + 3\langle y_2(y_2)^\dagger Y_e \rangle_{ij} + 6\langle Y_e(Y_e)^\dagger Y_e \rangle_{ij})$	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{1}{4}(2\langle y_1(y_1)^\dagger Y_e \rangle_{ij} + 3\langle y_2(y_2)^\dagger Y_e \rangle_{ij} + 6\langle Y_e(Y_e)^\dagger Y_e \rangle_{ij})$	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+\frac{1}{4}(2\langle y_1(y_1)^\dagger Y_e \rangle_{ij} + 3\langle y_2(y_2)^\dagger Y_e \rangle_{ij} + 6\langle Y_e(Y_e)^\dagger Y_e \rangle_{ij})$	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{1}{4}(3\langle y_1(y_1)^\dagger Y_e \rangle_{ij} + 2\langle y_2(y_2)^\dagger Y_e \rangle_{ij} + 6\langle Y_e(Y_e)^\dagger Y_e \rangle_{ij})$	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+\frac{1}{4}(3\langle y_1(y_1)^\dagger Y_e \rangle_{ij} + 2\langle y_2(y_2)^\dagger Y_e \rangle_{ij} + 6\langle Y_e(Y_e)^\dagger Y_e \rangle_{ij})$	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{1}{4}(3\langle y_1(y_1)^\dagger Y_e \rangle_{ij} + 2\langle y_2(y_2)^\dagger Y_e \rangle_{ij} + 6\langle Y_e(Y_e)^\dagger Y_e \rangle_{ij})$	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+\frac{1}{4}(3\langle y_1(y_1)^\dagger Y_e \rangle_{ij} + 2\langle y_2(y_2)^\dagger Y_e \rangle_{ij} + 6\langle Y_e(Y_e)^\dagger Y_e \rangle_{ij})$	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+\frac{1}{4}(3\langle y_1(y_1)^\dagger Y_e \rangle_{ij} + 2\langle y_2(y_2)^\dagger Y_e \rangle_{ij} + 6\langle Y_e(Y_e)^\dagger Y_e \rangle_{ij})$	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_Y^2(Y_e)_{ij}$

Terms 3 of 3

Column Representative term structure

$$T1 \quad +\frac{1}{2}((y_1 \ (y_1)^\dagger \ Y_e)_{ij} + (y_2 \ (y_2)^\dagger \ Y_e)_{ij} + 3 \ (Y_e \ (Y_e)^\dagger \ Y_e)_{ij})$$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+\frac{1}{2}((y_1 \ (y_1)^\dagger \ Y_e)_{ij} + (y_2 \ (y_2)^\dagger \ Y_e)_{ij} + 3 \ (Y_e \ (Y_e)^\dagger \ Y_e)_{ij})$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{1}{2}((y_1 \ (y_1)^\dagger \ Y_e)_{ij} + (y_2 \ (y_2)^\dagger \ Y_e)_{ij} + 3 \ (Y_e \ (Y_e)^\dagger \ Y_e)_{ij})$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{1}{2}((y_1 \ (y_1)^\dagger \ Y_e)_{ij} + (y_2 \ (y_2)^\dagger \ Y_e)_{ij} + 3 \ (Y_e \ (Y_e)^\dagger \ Y_e)_{ij})$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{1}{2}((y_1 \ (y_1)^\dagger \ Y_e)_{ij} + (y_2 \ (y_2)^\dagger \ Y_e)_{ij} + 3 \ (Y_e \ (Y_e)^\dagger \ Y_e)_{ij})$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+\frac{1}{2}((y_1 \ (y_1)^\dagger \ Y_e)_{ij} + (y_2 \ (y_2)^\dagger \ Y_e)_{ij} + 3 \ (Y_e \ (Y_e)^\dagger \ Y_e)_{ij})$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+\frac{3}{4}((y_1 \ (y_1)^\dagger \ Y_e)_{ij} + (y_2 \ (y_2)^\dagger \ Y_e)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ Y_e)_{ij})$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3}{4}((y_1 \ (y_1)^\dagger \ Y_e)_{ij} + (y_2 \ (y_2)^\dagger \ Y_e)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ Y_e)_{ij})$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{3}{4}((y_1 \ (y_1)^\dagger \ Y_e)_{ij} + (y_2 \ (y_2)^\dagger \ Y_e)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ Y_e)_{ij})$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+\frac{3}{4}((y_1 \ (y_1)^\dagger \ Y_e)_{ij} + (y_2 \ (y_2)^\dagger \ Y_e)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ Y_e)_{ij})$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+\frac{3}{4}((y_1 \ (y_1)^\dagger \ Y_e)_{ij} + (y_2 \ (y_2)^\dagger \ Y_e)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ Y_e)_{ij})$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\frac{3}{4}((y_1 \ (y_1)^\dagger \ Y_e)_{ij} + (y_2 \ (y_2)^\dagger \ Y_e)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ Y_e)_{ij})$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{3}{4}((y_1 \ (y_1)^\dagger \ Y_e)_{ij} + (y_2 \ (y_2)^\dagger \ Y_e)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ Y_e)_{ij})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{4}((y_1 \ (y_1)^\dagger \ Y_e)_{ij} + (y_2 \ (y_2)^\dagger \ Y_e)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ Y_e)_{ij})$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{3}{4}((y_1 \ (y_1)^\dagger \ Y_e)_{ij} + (y_2 \ (y_2)^\dagger \ Y_e)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ Y_e)_{ij})$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\frac{3}{4}((y_1 \ (y_1)^\dagger \ Y_e)_{ij} + (y_2 \ (y_2)^\dagger \ Y_e)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ Y_e)_{ij})$

$$\beta_{y_1}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dy_1}{d\mu}=\beta_{y_1}^{(1)}$$

Terms 1 of 5

Column	Representative term structure	
T1	$2\,\text{Tr}\big(y_1\,(y_1)^\dagger\big)\,(y_1)_{ij}$	
T2	$+\frac{3}{4}\,(y_2\,(y_2)^\dagger\,y_1)_{ij}$	
T3	$+\frac{1}{2}\,(Y_e\,(Y_e)^\dagger\,y_1)_{ij}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$2\,\text{Tr}\big(y_1\,(y_1)^\dagger\big)\,(y_1)_{ij}$	$+\frac{3}{4}\,(y_2\,(y_2)^\dagger\,y_1)_{ij}$	$+\frac{1}{2}\,(Y_e\,(Y_e)^\dagger\,y_1)_{ij}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$2\,\text{Tr}\big(y_1\,(y_1)^\dagger\big)\,(y_1)_{ij}$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$2\,\text{Tr}\big(y_1\,(y_1)^\dagger\big)\,(y_1)_{ij}$	$+\frac{3}{4}\,(y_2\,(y_2)^\dagger\,y_1)_{ij}$	$+\frac{1}{2}\,(Y_e\,(Y_e)^\dagger\,y_1)_{ij}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$2\,\text{Tr}\big(y_1\,(y_1)^\dagger\big)\,(y_1)_{ij}$	$+\frac{3}{4}\,(y_2\,(y_2)^\dagger\,y_1)_{ij}$	$+\frac{1}{2}\,(Y_e\,(Y_e)^\dagger\,y_1)_{ij}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$2\,\text{Tr}\big(y_1\,(y_1)^\dagger\big)\,(y_1)_{ij}$	$+\frac{3}{4}\,(y_2\,(y_2)^\dagger\,y_1)_{ij}$	$+\frac{1}{2}\,(Y_e\,(Y_e)^\dagger\,y_1)_{ij}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure	
T1	$+ \left(y_1 \left(y_1 \right)^\dagger y_1 \right)_{ij}$	
T2	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	
T3	$-\frac{3}{2} g_Y^2 \left(y_1 \right)_{ij}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+ \left(y_1 \left(y_1 \right)^\dagger y_1 \right)_{ij}$	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_Y^2 \left(y_1 \right)_{ij}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{4} g_Y^2 \left(y_1 \right)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+ \left(y_1 \left(y_1 \right)^\dagger y_1 \right)_{ij}$	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_Y^2 \left(y_1 \right)_{ij}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+ \left(y_1 \left(y_1 \right)^\dagger y_1 \right)_{ij}$	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{4} g_Y^2 \left(y_1 \right)_{ij}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+ \left(y_1 \left(y_1 \right)^\dagger y_1 \right)_{ij}$	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{2} g_Y^2 \left(y_1 \right)_{ij}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$-\frac{9}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_Y^2 \left(y_1 \right)_{ij}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	$-\frac{9}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{4} g_Y^2 \left(y_1 \right)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-\frac{9}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_Y^2 \left(y_1 \right)_{ij}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$-\frac{9}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{4} g_Y^2 \left(y_1 \right)_{ij}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$-\frac{9}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{2} g_Y^2 \left(y_1 \right)_{ij}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$-\frac{33}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_Y^2 \left(y_1 \right)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$-\frac{33}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{4} g_Y^2 \left(y_1 \right)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-\frac{33}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_Y^2 \left(y_1 \right)_{ij}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$-\frac{33}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{4} g_Y^2 \left(y_1 \right)_{ij}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$-\frac{33}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{2} g_Y^2 \left(y_1 \right)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_Y^2 \left(y_1 \right)_{ij}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{4} g_Y^2 \left(y_1 \right)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_Y^2 \left(y_1 \right)_{ij}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{4} g_Y^2 \left(y_1 \right)_{ij}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{2} g_Y^2 \left(y_1 \right)_{ij}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_Y^2 \left(y_1 \right)_{ij}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{4} g_Y^2 \left(y_1 \right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{9}{2}g_2^2(y_1)_{ij}$	$-\frac{3}{2}g_Y^2(y_1)_{ij}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$-\frac{9}{2}g_2^2(y_1)_{ij}$	$-\frac{15}{4}g_Y^2(y_1)_{ij}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$-\frac{9}{2}g_2^2(y_1)_{ij}$	$-\frac{15}{2}g_Y^2(y_1)_{ij}$

Terms 3 of 5

Column	Representative term structure	
T1	$+\frac{1}{4}(3(y_1(y_2)^\dagger y_2)_{ij} + 3(y_2(y_2)^\dagger y_1)_{ij} + 2(Y_e(Y_e)^\dagger y_1)_{ij} + 4(y_1(y_1)^\dagger y_1)_{ij})$	
T2	$\text{Tr}(y_1(y_1)^\dagger)(y_1)_{ij}$	
T3	$+\frac{1}{2}((y_2(y_2)^\dagger y_1)_{ij} + (Y_e(Y_e)^\dagger y_1)_{ij} + 3(y_1(y_1)^\dagger y_1)_{ij})$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{1}{4}(3(y_1(y_2)^\dagger y_2)_{ij} + 3(y_2(y_2)^\dagger y_1)_{ij} + 2(Y_e(Y_e)^\dagger y_1)_{ij} + 4(y_1(y_1)^\dagger y_1)_{ij})$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$\text{Tr}(y_1(y_1)^\dagger)(y_1)_{ij}$	$+\frac{1}{2}((y_2(y_2)^\dagger y_1)_{ij} + (Y_e(Y_e)^\dagger y_1)_{ij} + 3(y_1(y_1)^\dagger y_1)_{ij})$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	$+\text{Tr}(y_1(y_1)^\dagger)(y_1)_{ij}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$\text{Tr}(y_1(y_1)^\dagger)(y_1)_{ij}$	$+\frac{1}{2}((y_2(y_2)^\dagger y_1)_{ij} + (Y_e(Y_e)^\dagger y_1)_{ij} + 3(y_1(y_1)^\dagger y_1)_{ij})$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$\text{Tr}(y_1(y_1)^\dagger)(y_1)_{ij}$	$+\frac{1}{2}((y_2(y_2)^\dagger y_1)_{ij} + (Y_e(Y_e)^\dagger y_1)_{ij} + 3(y_1(y_1)^\dagger y_1)_{ij})$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$\text{Tr}(y_1(y_1)^\dagger)(y_1)_{ij}$	$+\frac{1}{2}((y_2(y_2)^\dagger y_1)_{ij} + (Y_e(Y_e)^\dagger y_1)_{ij} + 3(y_1(y_1)^\dagger y_1)_{ij})$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{1}{4}(2(y_1(y_2)^\dagger y_2)_{ij} + 3(y_2(y_2)^\dagger y_1)_{ij} + 2(Y_e(Y_e)^\dagger y_1)_{ij} + 5(y_1(y_1)^\dagger y_1)_{ij})$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$\text{Tr}(y_1(y_1)^\dagger)(y_1)_{ij}$	$+\frac{1}{2}((y_2(y_2)^\dagger y_1)_{ij} + (Y_e(Y_e)^\dagger y_1)_{ij} + 3(y_1(y_1)^\dagger y_1)_{ij})$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$\text{Tr}(y_1(y_1)^\dagger)(y_1)_{ij}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$\text{Tr}(y_1(y_1)^\dagger)(y_1)_{ij}$	$+\frac{1}{2}((y_2(y_2)^\dagger y_1)_{ij} + (Y_e(Y_e)^\dagger y_1)_{ij} + 3(y_1(y_1)^\dagger y_1)_{ij})$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$\text{Tr}(y_1(y_1)^\dagger)(y_1)_{ij}$	$+\frac{1}{2}((y_2(y_2)^\dagger y_1)_{ij} + (Y_e(Y_e)^\dagger y_1)_{ij} + 3(y_1(y_1)^\dagger y_1)_{ij})$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$\text{Tr}(y_1(y_1)^\dagger)(y_1)_{ij}$	$+\frac{1}{2}((y_2(y_2)^\dagger y_1)_{ij} + (Y_e(Y_e)^\dagger y_1)_{ij} + 3(y_1(y_1)^\dagger y_1)_{ij})$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$\text{Tr}(y_1(y_1)^\dagger)(y_1)_{ij}$	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{1}{4}(3(y_1(y_2)^\dagger y_2)_{ij} + 3(y_2(y_2)^\dagger y_1)_{ij} + 2(Y_e(Y_e)^\dagger y_1)_{ij} + 6(y_1(y_1)^\dagger y_1)_{ij})$	$+\text{Tr}(y_1(y_1)^\dagger)(y_1)_{ij}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$\text{Tr}(y_1(y_1)^\dagger)(y_1)_{ij}$	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$\text{Tr}(y_1(y_1)^\dagger)(y_1)_{ij}$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$\text{Tr}(y_1(y_1)^\dagger)(y_1)_{ij}$	—

Terms 4 of 5

Column	Representative term structure
T1	$\text{Tr}\left(y_1\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{ij}$
T2	$+\frac{1}{2}\left(2\left(y_1\left(y_2\right)^{\dagger} y_2\right)_{ij}+\left(y_2\left(y_2\right)^{\dagger} y_1\right)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger} y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger} y_1\right)_{ij}\right)$
T3	$-\frac{3}{2} \text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\text{Tr}\left(y_1\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{ij}$	$+\frac{1}{2}\left(2\left(y_1\left(y_2\right)^{\dagger} y_2\right)_{ij}+\left(y_2\left(y_2\right)^{\dagger} y_1\right)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger} y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger} y_1\right)_{ij}\right)$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$\frac{3}{2} \text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$+\frac{3}{2} \text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$\frac{3}{2} \text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$\frac{3}{2} \text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$\frac{3}{2} \text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\text{Tr}\left(y_1\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{ij}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure									
T1	$+ \frac{1}{4} (3 (y_2 (y_2)^\dagger y_1)_{ij} + 2 (Y_e (Y_e)^\dagger y_1)_{ij} + 5 (y_1 (y_1)^\dagger y_1)_{ij})$									
T2	$-\frac{3}{2} \text{Tr}(y_1 (y_2)^\dagger) (y_2)_{ij}$									
T3	$+ \frac{1}{2} ((y_1 (y_2)^\dagger y_2)_{ij} + (y_2 (y_2)^\dagger y_1)_{ij} + (Y_e (Y_e)^\dagger y_1)_{ij} + 3 (y_1 (y_1)^\dagger y_1)_{ij})$									

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2} + \frac{1}{4} (3 (y_2 (y_2)^\dagger y_1)_{ij} + 2 (Y_e (Y_e)^\dagger y_1)_{ij} + 5 (y_1 (y_1)^\dagger y_1)_{ij})$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$\frac{3}{2} \text{Tr}(y_1 (y_2)^\dagger) (y_2)_{ij}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2} + \frac{1}{4} (3 (y_2 (y_2)^\dagger y_1)_{ij} + 2 (Y_e (Y_e)^\dagger y_1)_{ij} + 5 (y_1 (y_1)^\dagger y_1)_{ij})$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	$1 + \frac{1}{4} (3 (y_2 (y_2)^\dagger y_1)_{ij} + 2 (Y_e (Y_e)^\dagger y_1)_{ij} + 5 (y_1 (y_1)^\dagger y_1)_{ij})$	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2} + \frac{1}{4} (3 (y_2 (y_2)^\dagger y_1)_{ij} + 2 (Y_e (Y_e)^\dagger y_1)_{ij} + 5 (y_1 (y_1)^\dagger y_1)_{ij})$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	$+\frac{1}{2} ((y_1 (y_2)^\dagger y_2)_{ij} + (y_2 (y_2)^\dagger y_1)_{ij} + (Y_e (Y_e)^\dagger y_1)_{ij} + 3 (y_1 (y_1)^\dagger y_1)_{ij})$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2} + \frac{1}{4} (3 (y_2 (y_2)^\dagger y_1)_{ij} + 2 (Y_e (Y_e)^\dagger y_1)_{ij} + 6 (y_1 (y_1)^\dagger y_1)_{ij})$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2} + \frac{1}{4} (3 (y_2 (y_2)^\dagger y_1)_{ij} + 2 (Y_e (Y_e)^\dagger y_1)_{ij} + 6 (y_1 (y_1)^\dagger y_1)_{ij})$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	$1 + \frac{1}{4} (3 (y_2 (y_2)^\dagger y_1)_{ij} + 2 (Y_e (Y_e)^\dagger y_1)_{ij} + 6 (y_1 (y_1)^\dagger y_1)_{ij})$	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2} + \frac{1}{4} (3 (y_2 (y_2)^\dagger y_1)_{ij} + 2 (Y_e (Y_e)^\dagger y_1)_{ij} + 6 (y_1 (y_1)^\dagger y_1)_{ij})$	—	—	—

$$\beta_{y_2}^{(1)}$$

$$16\pi^2\,\mu\,\frac{dy_2}{d\mu}=\beta_{y_2}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$
T2	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$
T3	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$	—	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$	$-\frac{9}{4}g_2^2\,(y_2)_{ij}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$	—	$-\frac{9}{4}g_2^2\,(y_2)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$	$-\frac{9}{4}g_2^2\,(y_2)_{ij}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$	$-\frac{9}{4}g_2^2\,(y_2)_{ij}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$	$-\frac{9}{4}g_2^2\,(y_2)_{ij}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$-\frac{33}{4}g_2^2\,(y_2)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$-\frac{33}{4}g_2^2\,(y_2)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$-\frac{33}{4}g_2^2\,(y_2)_{ij}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$-\frac{33}{4}g_2^2\,(y_2)_{ij}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$-\frac{33}{4}g_2^2\,(y_2)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$\text{Tr}\left(y_2 (y_2)^\dagger\right) (y_2)_{ij}$	—	$-\frac{9}{2} g_2^2 (y_2)_{ij}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\text{Tr}\left(y_2 (y_2)^\dagger\right) (y_2)_{ij}$	—	$-\frac{9}{2} g_2^2 (y_2)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\text{Tr}\left(y_2 (y_2)^\dagger\right) (y_2)_{ij}$	—	$-\frac{9}{2} g_2^2 (y_2)_{ij}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\text{Tr}\left(y_2 (y_2)^\dagger\right) (y_2)_{ij}$	—	$-\frac{9}{2} g_2^2 (y_2)_{ij}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\text{Tr}\left(y_2 (y_2)^\dagger\right) (y_2)_{ij}$	—	$-\frac{9}{2} g_2^2 (y_2)_{ij}$

Terms 2 of 5

Column	Representative term structure									
T1	$-\frac{3}{2}g_Y{}^2(y_2)_{ij}$									
T2	$+\frac{1}{2}((y_1(y_1)^\dagger y_2)_{ij}+(y_2(y_1)^\dagger y_1)_{ij}+(Y_e(Y_e)^\dagger y_2)_{ij}+3(y_2(y_2)^\dagger y_2)_{ij})$									
T3	$\text{Tr}(y_2(y_1)^\dagger)(y_1)_{ij}$									

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{3}{2}g_Y{}^2(y_2)_{ij}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{3}{4}g_Y{}^2(y_2)_{ij}$	$+\frac{1}{2}((y_1(y_1)^\dagger y_2)_{ij}+(y_2(y_1)^\dagger y_1)_{ij}+(Y_e(Y_e)^\dagger y_2)_{ij}+3(y_2(y_2)^\dagger y_2)_{ij})$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2}g_Y{}^2(y_2)_{ij}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{15}{4}g_Y{}^2(y_2)_{ij}$	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-\frac{15}{2}g_Y{}^2(y_2)_{ij}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-\frac{3}{2}g_Y{}^2(y_2)_{ij}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3}{4}g_Y{}^2(y_2)_{ij}$	—	$\text{Tr}(y_2(y_1)^\dagger)(y_1)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{3}{2}g_Y{}^2(y_2)_{ij}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-\frac{15}{4}g_Y{}^2(y_2)_{ij}$	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$-\frac{15}{2}g_Y{}^2(y_2)_{ij}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-\frac{3}{2}g_Y{}^2(y_2)_{ij}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{3}{4}g_Y{}^2(y_2)_{ij}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{3}{2}g_Y{}^2(y_2)_{ij}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{15}{4}g_Y{}^2(y_2)_{ij}$	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-\frac{15}{2}g_Y{}^2(y_2)_{ij}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{3}{2}g_Y{}^2(y_2)_{ij}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-\frac{3}{4}g_Y{}^2(y_2)_{ij}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{3}{2}g_Y{}^2(y_2)_{ij}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-\frac{15}{4}g_Y{}^2(y_2)_{ij}$	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-\frac{15}{2}g_Y{}^2(y_2)_{ij}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{3}{2}g_Y{}^2(y_2)_{ij}$	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{3}{4}g_Y{}^2(y_2)_{ij}$	—	$\text{Tr}(y_2(y_1)^\dagger)(y_1)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2}g_Y{}^2(y_2)_{ij}$	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{15}{4}g_Y{}^2(y_2)_{ij}$	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{15}{2}g_Y{}^2(y_2)_{ij}$	—	—

Terms 3 of 5

Column	Representative term structure									
T1	$+\frac{1}{2}((y_1(y_1)^\dagger y_2)_{ij}+2(y_2(y_1)^\dagger y_1)_{ij}+(Y_e(Y_e)^\dagger y_2)_{ij}+3(y_2(y_2)^\dagger y_2)_{ij})$									
T2	$-\frac{3}{2}\text{Tr}(y_2(y_2)^\dagger)(y_2)_{ij}$									
T3	$+\frac{1}{4}(3(y_1(y_1)^\dagger y_2)_{ij}+2(Y_e(Y_e)^\dagger y_2)_{ij}+5(y_2(y_2)^\dagger y_2)_{ij})$									

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{1}{2}((y_1(y_1)^\dagger y_2)_{ij}+2(y_2(y_1)^\dagger y_1)_{ij}+(Y_e(Y_e)^\dagger y_2)_{ij}+3(y_2(y_2)^\dagger y_2)_{ij})$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$\frac{3}{2}\text{Tr}(y_2(y_2)^\dagger)(y_2)_{ij}$	$+\frac{1}{4}(3(y_1(y_1)^\dagger y_2)_{ij}+2(Y_e(Y_e)^\dagger y_2)_{ij}+5(y_2(y_2)^\dagger y_2)_{ij})$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+\frac{3}{2}\text{Tr}(y_2(y_2)^\dagger)(y_2)_{ij}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$\frac{3}{2}\text{Tr}(y_2(y_2)^\dagger)(y_2)_{ij}$	$+\frac{1}{4}(3(y_1(y_1)^\dagger y_2)_{ij}+2(Y_e(Y_e)^\dagger y_2)_{ij}+5(y_2(y_2)^\dagger y_2)_{ij})$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$\frac{3}{2}\text{Tr}(y_2(y_2)^\dagger)(y_2)_{ij}$	$+\frac{1}{4}(3(y_1(y_1)^\dagger y_2)_{ij}+2(Y_e(Y_e)^\dagger y_2)_{ij}+5(y_2(y_2)^\dagger y_2)_{ij})$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$\frac{3}{2}\text{Tr}(y_2(y_2)^\dagger)(y_2)_{ij}$	$+\frac{1}{4}(3(y_1(y_1)^\dagger y_2)_{ij}+2(Y_e(Y_e)^\dagger y_2)_{ij}+5(y_2(y_2)^\dagger y_2)_{ij})$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	$+\frac{1}{4}(3(y_1(y_1)^\dagger y_2)_{ij}+2(Y_e(Y_e)^\dagger y_2)_{ij}+6(y_2(y_2)^\dagger y_2)_{ij})$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$+\frac{1}{4}(3(y_1(y_1)^\dagger y_2)_{ij}+2(Y_e(Y_e)^\dagger y_2)_{ij}+6(y_2(y_2)^\dagger y_2)_{ij})$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	$+\frac{1}{4}(3(y_1(y_1)^\dagger y_2)_{ij}+2(Y_e(Y_e)^\dagger y_2)_{ij}+6(y_2(y_2)^\dagger y_2)_{ij})$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	$+\frac{1}{4}(3(y_1(y_1)^\dagger y_2)_{ij}+2(Y_e(Y_e)^\dagger y_2)_{ij}+6(y_2(y_2)^\dagger y_2)_{ij})$

Terms 4 of 5

Column	Representative term structure
T1	$\frac{3}{2} \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right) (y_1)_{ij}$
T2	$+\frac{1}{4} (3 \left(y_1 \left(y_1 \right)^\dagger y_2 \right)_{ij} + 2 \left(y_2 \left(y_1 \right)^\dagger y_1 \right)_{ij} + 2 \left(Y_e \left(Y_e \right)^\dagger y_2 \right)_{ij} + 5 \left(y_2 \left(y_2 \right)^\dagger y_2 \right)_{ij})$
T3	$2 \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) (y_2)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{3}{2} \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right) (y_1)_{ij}$	$+\frac{1}{4} (3 \left(y_1 \left(y_1 \right)^\dagger y_2 \right)_{ij} + 2 \left(y_2 \left(y_1 \right)^\dagger y_1 \right)_{ij} + 2 \left(Y_e \left(Y_e \right)^\dagger y_2 \right)_{ij} + 5 \left(y_2 \left(y_2 \right)^\dagger y_2 \right)_{ij})$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$2 \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) (y_2)_{ij}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$+\frac{1}{4} (3 \left(y_1 \left(y_1 \right)^\dagger y_2 \right)_{ij} + 3 \left(y_2 \left(y_1 \right)^\dagger y_1 \right)_{ij} + 2 \left(Y_e \left(Y_e \right)^\dagger y_2 \right)_{ij} + 4 \left(y_2 \left(y_2 \right)^\dagger y_2 \right)_{ij})$	$2 \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) (y_2)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	$2 \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) (y_2)_{ij}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	$2 \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) (y_2)_{ij}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	$2 \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) (y_2)_{ij}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$+\frac{1}{4} (3 \left(y_1 \left(y_1 \right)^\dagger y_2 \right)_{ij} + 3 \left(y_2 \left(y_1 \right)^\dagger y_1 \right)_{ij} + 2 \left(Y_e \left(Y_e \right)^\dagger y_2 \right)_{ij} + 6 \left(y_2 \left(y_2 \right)^\dagger y_2 \right)_{ij})$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure						
T1	$+ \frac{3}{4} \left(y_1 \left(y_1 \right)^{\dagger} y_2 \right)_{ij}$						
T2	$+ \frac{1}{2} \left(Y_e \left(Y_e \right)^{\dagger} y_2 \right)_{ij}$						
T3	$+ \left(y_2 \left(y_2 \right)^{\dagger} y_2 \right)_{ij}$						

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+ \frac{3}{4} \left(y_1 \left(y_1 \right)^{\dagger} y_2 \right)_{ij}$	$+ \frac{1}{2} \left(Y_e \left(Y_e \right)^{\dagger} y_2 \right)_{ij}$	$+ \left(y_2 \left(y_2 \right)^{\dagger} y_2 \right)_{ij}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+ \frac{3}{4} \left(y_1 \left(y_1 \right)^{\dagger} y_2 \right)_{ij}$	$+ \frac{1}{2} \left(Y_e \left(Y_e \right)^{\dagger} y_2 \right)_{ij}$	$+ \left(y_2 \left(y_2 \right)^{\dagger} y_2 \right)_{ij}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+ \frac{3}{4} \left(y_1 \left(y_1 \right)^{\dagger} y_2 \right)_{ij}$	$+ \frac{1}{2} \left(Y_e \left(Y_e \right)^{\dagger} y_2 \right)_{ij}$	$+ \left(y_2 \left(y_2 \right)^{\dagger} y_2 \right)_{ij}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+ \frac{3}{4} \left(y_1 \left(y_1 \right)^{\dagger} y_2 \right)_{ij}$	$+ \frac{1}{2} \left(Y_e \left(Y_e \right)^{\dagger} y_2 \right)_{ij}$	$+ \left(y_2 \left(y_2 \right)^{\dagger} y_2 \right)_{ij}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

$$\beta_{M_F}^{(1)}$$

$$16\pi^2 \mu \frac{dM_F}{d\mu} = \beta_{M_F}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\frac{1}{4} (2 \left((y_1)^T (y_1)^* M_F \right)_{ij} + 3 \left(M_F (y_2)^\dagger y_2 \right)_{ij})$
T2	$-\frac{9}{2} g_2^2 (M_F)_{ij}$
T3	$-\frac{3}{2} g_Y^2 (M_F)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\frac{1}{4}(2\left((y_1)^T(y_1)^*M_F\right)_{ij}+3\left(M_F(y_2)^\dagger y_2\right)_{ij})-\frac{9}{2}g_2^2(M_F)_{ij}$		$-\frac{3}{2}g_Y^2(M_F)_{ij}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{1}{4}(2\left((y_1)^T(y_1)^*M_F\right)_{ij}+3\left(M_F(y_2)^\dagger y_2\right)_{ij})-\frac{9}{2}g_2^2(M_F)_{ij}$		$-\frac{3}{2}g_Y^2(M_F)_{ij}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\frac{1}{4}(2\left((y_1)^T(y_1)^*M_F\right)_{ij}+3\left(M_F(y_2)^\dagger y_2\right)_{ij})-\frac{9}{2}g_2^2(M_F)_{ij}$		—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$\frac{1}{4}(2\left((y_1)^T(y_1)^*M_F\right)_{ij}+3\left(M_F(y_2)^\dagger y_2\right)_{ij})-\frac{9}{2}g_2^2(M_F)_{ij}$		$-\frac{27}{2}g_Y^2(M_F)_{ij}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	$-\frac{3}{2}g_Y^2(M_F)_{ij}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$-\frac{3}{2}g_Y^2(M_F)_{ij}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	$-\frac{27}{2}g_Y^2(M_F)_{ij}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$-\frac{3}{2}g_Y^2(M_F)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$-\frac{3}{2}g_Y^2(M_F)_{ij}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$-\frac{27}{2}g_Y^2(M_F)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{1}{4}(3\left((y_1)^T(y_1)^*M_F\right)_{ij}+2\left(M_F(y_2)^\dagger y_2\right)_{ij})-\frac{9}{2}g_2^2(M_F)_{ij}$		$-\frac{3}{2}g_Y^2(M_F)_{ij}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{1}{4}(3\left((y_1)^T(y_1)^*M_F\right)_{ij}+2\left(M_F(y_2)^\dagger y_2\right)_{ij})-\frac{9}{2}g_2^2(M_F)_{ij}$		$-\frac{3}{2}g_Y^2(M_F)_{ij}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$\frac{1}{4}(3\left((y_1)^T(y_1)^*M_F\right)_{ij}+2\left(M_F(y_2)^\dagger y_2\right)_{ij})-\frac{9}{2}g_2^2(M_F)_{ij}$		—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$\frac{1}{4}(3\left((y_1)^T(y_1)^*M_F\right)_{ij}+2\left(M_F(y_2)^\dagger y_2\right)_{ij})-\frac{9}{2}g_2^2(M_F)_{ij}$		$-\frac{27}{2}g_Y^2(M_F)_{ij}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$-\frac{9}{2}g_2^2(M_F)_{ij}$	$-\frac{3}{2}g_Y^2(M_F)_{ij}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{9}{2}g_2^2(M_F)_{ij}$	$-\frac{3}{2}g_Y^2(M_F)_{ij}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$-\frac{9}{2}g_2^2(M_F)_{ij}$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$-\frac{9}{2}g_2^2(M_F)_{ij}$	$-\frac{27}{2}g_Y^2(M_F)_{ij}$

Terms 2 of 5

Column	Representative term structure
T1	$\frac{1}{4}(2((M_F)^T(y_1)^\dagger y_1)_{ij} + 2((y_1)^T(y_1)^* M_F)_{ij} + 3((M_F)^T(y_2)^\dagger y_2)_{ij} + 3((y_2)^T(y_2)^* M_F)_{ij})$
T2	$-\frac{9}{4}g_2^2((M_F)^T)_{ij} + (M_F)_{ij}$
T3	$-6g_Y^2(M_F)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\frac{1}{4}(2((M_F)^T(y_1)^\dagger y_1)_{ij} + 2((y_1)^T(y_1)^* M_F)_{ij} + 3((M_F)^T(y_2)^\dagger y_2)_{ij} + 3((y_2)^T(y_2)^* M_F)_{ij}) - \frac{9}{4}g_2^2((M_F)^T)_{ij} + (M_F)_{ij}$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	$-6g_Y^2(M_F)_{ij}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	$-6g_Y^2(M_F)_{ij}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$-6g_Y^2(M_F)_{ij}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$\frac{1}{4}(3((M_F)^T(y_1)^\dagger y_1)_{ij} + 3((y_1)^T(y_1)^* M_F)_{ij} + 2((M_F)^T(y_2)^\dagger y_2)_{ij} + 2((y_2)^T(y_2)^* M_F)_{ij}) - \frac{9}{4}g_2^2((M_F)^T)_{ij} + (M_F)_{ij}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	$-6g_Y^2(M_F)_{ij}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-\frac{9}{4}g_2^2((M_F)^T)_{ij} + (M_F)_{ij}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	$-6g_Y^2(M_F)_{ij}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$\left((y_1)^T (y_1)^* M_F \right)_{ij}$
T2	$+ \left(M_F (y_2)^\dagger y_2 \right)_{ij}$
T3	$\left((M_F)^T (y_1)^\dagger y_1 \right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$\left((y_1)^T (y_1)^* M_F \right)_{ij}$	$+ \left(M_F (y_2)^\dagger y_2 \right)_{ij}$	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \left((y_1)^T (y_1)^* M_F \right)_{ij}$	—	$\left((M_F)^T (y_1)^\dagger y_1 \right)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\left((y_1)^T (y_1)^* M_F \right)_{ij}$	$+ \left(M_F (y_2)^\dagger y_2 \right)_{ij}$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$\left((y_1)^T (y_1)^* M_F \right)_{ij}$	$+ \left(M_F (y_2)^\dagger y_2 \right)_{ij}$	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$\left((y_1)^T (y_1)^* M_F \right)_{ij}$	$+ \left(M_F (y_2)^\dagger y_2 \right)_{ij}$	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure	
T1	$+ \left((M_F)^T (y_2)^\dagger y_2 \right)_{ij}$	
T2	$+ \left((y_2)^T (y_2)^* M_F \right)_{ij}$	
T3	$\frac{1}{2} \left((y_1)^T (y_1)^* M_F \right)_{ij} + \left(M_F (y_2)^\dagger y_2 \right)_{ij}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \left((M_F)^T (y_2)^\dagger y_2 \right)_{ij}$	$+ \left((y_2)^T (y_2)^* M_F \right)_{ij}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$\frac{1}{2} \left((y_1)^T (y_1)^* M_F \right)_{ij} + \left(M_F (y_2)^\dagger y_2 \right)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$\frac{1}{2} \left((y_1)^T (y_1)^* M_F \right)_{ij} + \left(M_F (y_2)^\dagger y_2 \right)_{ij}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$\frac{1}{2} \left((y_1)^T (y_1)^* M_F \right)_{ij} + \left(M_F (y_2)^\dagger y_2 \right)_{ij}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$\frac{1}{2} \left((y_1)^T (y_1)^* M_F \right)_{ij} + \left(M_F (y_2)^\dagger y_2 \right)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	$\frac{3}{4} \left((y_1)^T (y_1)^* M_F \right)_{ij} + \left(M_F (y_2)^\dagger y_2 \right)_{ij}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$\frac{3}{4} \left((y_1)^T (y_1)^* M_F \right)_{ij} + \left(M_F (y_2)^\dagger y_2 \right)_{ij}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	$\frac{3}{4} \left((y_1)^T (y_1)^* M_F \right)_{ij} + \left(M_F (y_2)^\dagger y_2 \right)_{ij}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	$\frac{3}{4} \left((y_1)^T (y_1)^* M_F \right)_{ij} + \left(M_F (y_2)^\dagger y_2 \right)_{ij}$

Terms 5 of 5

Column	Representative term structure	
T1	$-12g_2^2 (M_F)_{ij}$	
T2	$\frac{1}{2}((M_F)^T (y_1)^\dagger y_1)_{ij} + ((y_1)^T (y_1)^* M_F)_{ij} + ((M_F)^T (y_2)^\dagger y_2)_{ij} + ((y_2)^T (y_2)^* M_F)_{ij}$	
T3	$-6g_2^2 ((M_F)^T)_{ij} + (M_F)_{ij}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-12g_2^2 (M_F)_{ij}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$\frac{1}{2}((M_F)^T (y_1)^\dagger y_1)_{ij} + ((y_1)^T (y_1)^* M_F)_{ij} + ((M_F)^T (y_2)^\dagger y_2)_{ij} + ((y_2)^T (y_2)^* M_F)_{ij}$	$-6g_2^2 ((M_F)^T)_{ij} + (M_F)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-12g_2^2 (M_F)_{ij}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-12g_2^2 (M_F)_{ij}$	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-12g_2^2 (M_F)_{ij}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$\frac{3}{4}((M_F)^T (y_1)^\dagger y_1)_{ij} + ((y_1)^T (y_1)^* M_F)_{ij} + ((M_F)^T (y_2)^\dagger y_2)_{ij} + ((y_2)^T (y_2)^* M_F)_{ij}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

$$\beta_{m_{S_1}^2}^{(1)}$$

$$16\pi^2\mu\frac{dm_{S_1}^2}{d\mu}=\beta_{m_{S_1}^2}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$m_{S_1}^2(4\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+4\lambda_{S_1})$
T2	$-8\text{Tr}\left(M_F\left(M_F\right)^\dagger\left(y_1\right)^T\left(y_1\right)^*\right)$
T3	$-6g_Y{}^2m_{S_1}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$m_{S_1}^2(4\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+4\lambda_{S_1})$	$-8\text{Tr}\left(M_F\left(M_F\right)^\dagger\left(y_1\right)^T\left(y_1\right)^*\right)$	$-6g_Y{}^2m_{S_1}^2$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$m_{S_1}^2(4\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+4\lambda_{S_1})$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$m_{S_1}^2(4\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+4\lambda_{S_1})$	$-8\text{Tr}\left(M_F\left(M_F\right)^\dagger\left(y_1\right)^T\left(y_1\right)^*\right)$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$m_{S_1}^2(4\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+4\lambda_{S_1})$	$-8\text{Tr}\left(M_F\left(M_F\right)^\dagger\left(y_1\right)^T\left(y_1\right)^*\right)$	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$m_{S_1}^2(4\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+4\lambda_{S_1})$	$-8\text{Tr}\left(M_F\left(M_F\right)^\dagger\left(y_1\right)^T\left(y_1\right)^*\right)$	$-6g_Y{}^2m_{S_1}^2$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$m_{S_1}^2(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+6\lambda_{S_1})$	$-4\text{Tr}\left(M_F\left(M_F\right)^\dagger\left(y_1\right)^T\left(y_1\right)^*\right)$	$-6g_Y{}^2m_{S_1}^2$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$m_{S_1}^2(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+6\lambda_{S_1})$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$m_{S_1}^2(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+6\lambda_{S_1})$	$-4\text{Tr}\left(M_F\left(M_F\right)^\dagger\left(y_1\right)^T\left(y_1\right)^*\right)$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$m_{S_1}^2(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+6\lambda_{S_1})$	$-4\text{Tr}\left(M_F\left(M_F\right)^\dagger\left(y_1\right)^T\left(y_1\right)^*\right)$	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$m_{S_1}^2(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+6\lambda_{S_1})$	$-4\text{Tr}\left(M_F\left(M_F\right)^\dagger\left(y_1\right)^T\left(y_1\right)^*\right)$	$-6g_Y{}^2m_{S_1}^2$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$m_{S_1}^2(3\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+6\lambda_{S_1})$	$-6\text{Tr}\left(M_F\left(M_F\right)^\dagger\left(y_1\right)^T\left(y_1\right)^*\right)$	$-6g_Y{}^2m_{S_1}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$m_{S_1}^2(3\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+6\lambda_{S_1})$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$m_{S_1}^2(3\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+6\lambda_{S_1})$	$-6\text{Tr}\left(M_F\left(M_F\right)^\dagger\left(y_1\right)^T\left(y_1\right)^*\right)$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$m_{S_1}^2(3\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+6\lambda_{S_1})$	$-6\text{Tr}\left(M_F\left(M_F\right)^\dagger\left(y_1\right)^T\left(y_1\right)^*\right)$	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$m_{S_1}^2(3\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+6\lambda_{S_1})$	$-6\text{Tr}\left(M_F\left(M_F\right)^\dagger\left(y_1\right)^T\left(y_1\right)^*\right)$	$-6g_Y{}^2m_{S_1}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-4\text{Tr}\left(M_F\left(M_F\right)^\dagger\left(y_1\right)^T\left(y_1\right)^*\right)$	$-6g_Y{}^2m_{S_1}^2$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-4\text{Tr}\left(M_F\left(M_F\right)^\dagger\left(y_1\right)^T\left(y_1\right)^*\right)$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$-4\text{Tr}\left(M_F\left(M_F\right)^\dagger\left(y_1\right)^T\left(y_1\right)^*\right)$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$-4\text{Tr}\left(M_F\left(M_F\right)^\dagger\left(y_1\right)^T\left(y_1\right)^*\right)$	$-6g_Y{}^2m_{S_1}^2$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$-4\text{Tr}\left(M_F\left(M_F\right)^\dagger\left(y_1\right)^T\left(y_1\right)^*\right)$	$-6g_Y{}^2m_{S_1}^2$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-4\text{Tr}\left(M_F\left(M_F\right)^\dagger\left(y_1\right)^T\left(y_1\right)^*\right)$	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$-4 \operatorname{Tr} \left(M_F (M_F)^\dagger (y_1)^T (y_1)^* \right)$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$-4 \operatorname{Tr} \left(M_F (M_F)^\dagger (y_1)^T (y_1)^* \right)$	$-6 g_Y^2 m_{S_1}^2$

Terms 2 of 4

Column	Representative term structure	
T1	$+6\lambda_{12}m_{S_2}^2$	
T2	$-16\text{Tr}\left(M_F(M_F)^*(y_1)^T(y_1)^*\right)$	
T3	$-16\text{Tr}\left(M_F(y_1)^\dagger y_1(M_F)^*\right)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+6\lambda_{12}m_{S_2}^2$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+6\lambda_{12}m_{S_2}^2$	$-16\text{Tr}\left(M_F(M_F)^*(y_1)^T(y_1)^*\right)$	$-16\text{Tr}\left(M_F(y_1)^\dagger y_1(M_F)^*\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{12}m_{S_2}^2$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+6\lambda_{12}m_{S_2}^2$	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+6\lambda_{12}m_{S_2}^2$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+4\lambda_{12}m_{S_2}^2$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{12}m_{S_2}^2$	$-8\text{Tr}\left(M_F(M_F)^*(y_1)^T(y_1)^*\right)$	$-8\text{Tr}\left(M_F(y_1)^\dagger y_1(M_F)^*\right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+4\lambda_{12}m_{S_2}^2$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+4\lambda_{12}m_{S_2}^2$	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+4\lambda_{12}m_{S_2}^2$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+4\lambda_{12}m_{S_2}^2$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{12}m_{S_2}^2$	$-12\text{Tr}\left(M_F(M_F)^*(y_1)^T(y_1)^*\right)$	$-12\text{Tr}\left(M_F(y_1)^\dagger y_1(M_F)^*\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+4\lambda_{12}m_{S_2}^2$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+4\lambda_{12}m_{S_2}^2$	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+4\lambda_{12}m_{S_2}^2$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+2\lambda_{12}m_{S_2}^2$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+2\lambda_{12}m_{S_2}^2$	$-8\text{Tr}\left(M_F(M_F)^*(y_1)^T(y_1)^*\right)$	$-8\text{Tr}\left(M_F(y_1)^\dagger y_1(M_F)^*\right)$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+2\lambda_{12}m_{S_2}^2$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+2\lambda_{12}m_{S_2}^2$	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+2\lambda_{12}m_{S_2}^2$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-8\text{Tr}\left(M_F(M_F)^*(y_1)^T(y_1)^*\right)$	$-8\text{Tr}\left(M_F(y_1)^\dagger y_1(M_F)^*\right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 3 of 4

Column	Representative term structure						
T1	$-\frac{3g_Y^2 m_{S_1}^2}{2}$						
T2	$-\frac{9g_2^2 m_{S_1}^2}{2}$						
T3	$m_{S_1}^2 (2 \text{Tr}(y_1 (y_1)^\dagger) + 8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$						

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{3g_Y^2 m_{S_1}^2}{2}$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{3g_Y^2 m_{S_1}^2}{2}$	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$-\frac{9g_2^2 m_{S_1}^2}{2}$	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_Y^2 m_{S_1}^2}{2}$	$-\frac{9g_2^2 m_{S_1}^2}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-\frac{9g_2^2 m_{S_1}^2}{2}$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-\frac{3g_Y^2 m_{S_1}^2}{2}$	$-\frac{9g_2^2 m_{S_1}^2}{2}$	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$-\frac{9g_2^2 m_{S_1}^2}{2}$	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$-\frac{9g_2^2 m_{S_1}^2}{2}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{3g_Y^2 m_{S_1}^2}{2}$	$-\frac{9g_2^2 m_{S_1}^2}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-\frac{9g_2^2 m_{S_1}^2}{2}$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{3g_Y^2 m_{S_1}^2}{2}$	$-\frac{9g_2^2 m_{S_1}^2}{2}$	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$-\frac{9g_2^2 m_{S_1}^2}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$m_{S_1}^2 (2 \text{Tr}(y_1 (y_1)^\dagger) + 8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-\frac{3g_Y^2 m_{S_1}^2}{2}$	—	$m_{S_1}^2 (2 \text{Tr}(y_1 (y_1)^\dagger) + 8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	$m_{S_1}^2 (2 \text{Tr}(y_1 (y_1)^\dagger) + 8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-\frac{3g_Y^2 m_{S_1}^2}{2}$	—	$m_{S_1}^2 (2 \text{Tr}(y_1 (y_1)^\dagger) + 8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	$m_{S_1}^2 (2 \text{Tr}(y_1 (y_1)^\dagger) + 8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	$m_{S_1}^2 (2 \text{Tr}(y_1 (y_1)^\dagger) + 8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{3g_Y^2 m_{S_1}^2}{2}$	—	$m_{S_1}^2 (2 \text{Tr} (y_1 (y_1)^\dagger) + 8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$m_{S_1}^2 (2 \text{Tr} (y_1 (y_1)^\dagger) + 8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{3g_Y^2 m_{S_1}^2}{2}$	—	$m_{S_1}^2 (2 \text{Tr} (y_1 (y_1)^\dagger) + 8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	$m_{S_1}^2 (2 \text{Tr} (y_1 (y_1)^\dagger) + 8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$

Terms 4 of 4

Column	Representative term structure
T1	$-12g_2^2 m_{S_1}^2$
T2	$+m_{S_2}^2 (6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-12g_2^2 m_{S_1}^2$	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-12g_2^2 m_{S_1}^2$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-12g_2^2 m_{S_1}^2$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-12g_2^2 m_{S_1}^2$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-12g_2^2 m_{S_1}^2$	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-12g_2^2 m_{S_1}^2$	$+m_{S_2}^2 (6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-12g_2^2 m_{S_1}^2$	$+m_{S_2}^2 (6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-12g_2^2 m_{S_1}^2$	$+m_{S_2}^2 (6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-12g_2^2 m_{S_1}^2$	$+m_{S_2}^2 (6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-12g_2^2 m_{S_1}^2$	$+m_{S_2}^2 (6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$

$$\beta_{m_{S_2}^2}^{(1)}$$

$$16\pi^2\mu\frac{dm_{S_2}^2}{d\mu}=\beta_{m_{S_2}^2}^{(1)}$$

Terms 1 of 4

Column	Representative term structure	
T1	$m_{S_2}^2(2\text{Tr}(y_2(y_2)^\dagger)+8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$	
T2	$-4\text{Tr}(M_F(y_2)^\dagger y_2(M_F)^\dagger)$	
T3	$-12g_2^2m_{S_2}^2$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$m_{S_2}^2(2\text{Tr}(y_2(y_2)^\dagger)+8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$	$-4\text{Tr}(M_F(y_2)^\dagger y_2(M_F)^\dagger)$	$-12g_2^2m_{S_2}^2$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$m_{S_2}^2(2\text{Tr}(y_2(y_2)^\dagger)+8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$	—	$-12g_2^2m_{S_2}^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$m_{S_2}^2(2\text{Tr}(y_2(y_2)^\dagger)+8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$	$-4\text{Tr}(M_F(y_2)^\dagger y_2(M_F)^\dagger)$	$-12g_2^2m_{S_2}^2$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$m_{S_2}^2(2\text{Tr}(y_2(y_2)^\dagger)+8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$	$-4\text{Tr}(M_F(y_2)^\dagger y_2(M_F)^\dagger)$	$-12g_2^2m_{S_2}^2$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$m_{S_2}^2(2\text{Tr}(y_2(y_2)^\dagger)+8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$	$-4\text{Tr}(M_F(y_2)^\dagger y_2(M_F)^\dagger)$	$-12g_2^2m_{S_2}^2$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$-4\text{Tr}(M_F(y_2)^\dagger y_2(M_F)^\dagger)$	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-4\text{Tr}(M_F(y_2)^\dagger y_2(M_F)^\dagger)$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$-4\text{Tr}(M_F(y_2)^\dagger y_2(M_F)^\dagger)$	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$-4\text{Tr}(M_F(y_2)^\dagger y_2(M_F)^\dagger)$	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$-6\text{Tr}(M_F(y_2)^\dagger y_2(M_F)^\dagger)$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-6\text{Tr}(M_F(y_2)^\dagger y_2(M_F)^\dagger)$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$-6\text{Tr}(M_F(y_2)^\dagger y_2(M_F)^\dagger)$	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$-6\text{Tr}(M_F(y_2)^\dagger y_2(M_F)^\dagger)$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-8\text{Tr}(M_F(y_2)^\dagger y_2(M_F)^\dagger)$	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-8\text{Tr}(M_F(y_2)^\dagger y_2(M_F)^\dagger)$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$-8\text{Tr}(M_F(y_2)^\dagger y_2(M_F)^\dagger)$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$-8\text{Tr}(M_F(y_2)^\dagger y_2(M_F)^\dagger)$	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$m_{S_2}^2(2\text{Tr}(y_2(y_2)^\dagger)+8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$	$-4\text{Tr}(M_F(y_2)^\dagger y_2(M_F)^\dagger)$	$-12g_2^2m_{S_2}^2$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$m_{S_2}^2(2\text{Tr}(y_2(y_2)^\dagger)+8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$	—	$-12g_2^2m_{S_2}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$m_{S_2}^2(2\text{Tr}(y_2(y_2)^\dagger)+8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$	$-4\text{Tr}(M_F(y_2)^\dagger y_2(M_F)^\dagger)$	$-12g_2^2m_{S_2}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$m_{S_2}^2 (2 \text{Tr} \left(y_2 (y_2)^\dagger \right) + 8\lambda_{S_2} + 4\lambda_{S_2}^{\text{Adj}})$	$-4 \text{Tr} \left(M_F (y_2)^\dagger y_2 (M_F)^\dagger \right)$	$-12g_2^2 m_{S_2}^2$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$m_{S_2}^2 (2 \text{Tr} \left(y_2 (y_2)^\dagger \right) + 8\lambda_{S_2} + 4\lambda_{S_2}^{\text{Adj}})$	$-4 \text{Tr} \left(M_F (y_2)^\dagger y_2 (M_F)^\dagger \right)$	$-12g_2^2 m_{S_2}^2$

Terms 2 of 4

Column	Representative term structure	
T1	$+2\lambda_{12}m_{S_1}^2$	
T2	$-8\text{Tr}\left(M_F(M_F)^*(y_2)^T(y_2)^*\right)$	
T3	$-8\text{Tr}\left(M_F(y_2)^\dagger y_2(M_F)^*\right)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+2\lambda_{12}m_{S_1}^2$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+2\lambda_{12}m_{S_1}^2$	$-8\text{Tr}\left(M_F(M_F)^*(y_2)^T(y_2)^*\right)$	$-8\text{Tr}\left(M_F(y_2)^\dagger y_2(M_F)^*\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{12}m_{S_1}^2$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+2\lambda_{12}m_{S_1}^2$	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+2\lambda_{12}m_{S_1}^2$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+4\lambda_{12}m_{S_1}^2$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{12}m_{S_1}^2$	$-8\text{Tr}\left(M_F(M_F)^*(y_2)^T(y_2)^*\right)$	$-8\text{Tr}\left(M_F(y_2)^\dagger y_2(M_F)^*\right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+4\lambda_{12}m_{S_1}^2$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+4\lambda_{12}m_{S_1}^2$	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+4\lambda_{12}m_{S_1}^2$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+4\lambda_{12}m_{S_1}^2$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{12}m_{S_1}^2$	$-12\text{Tr}\left(M_F(M_F)^*(y_2)^T(y_2)^*\right)$	$-12\text{Tr}\left(M_F(y_2)^\dagger y_2(M_F)^*\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+4\lambda_{12}m_{S_1}^2$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+4\lambda_{12}m_{S_1}^2$	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+4\lambda_{12}m_{S_1}^2$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+6\lambda_{12}m_{S_1}^2$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+6\lambda_{12}m_{S_1}^2$	$-16\text{Tr}\left(M_F(M_F)^*(y_2)^T(y_2)^*\right)$	$-16\text{Tr}\left(M_F(y_2)^\dagger y_2(M_F)^*\right)$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+6\lambda_{12}m_{S_1}^2$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+6\lambda_{12}m_{S_1}^2$	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+6\lambda_{12}m_{S_1}^2$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-8\text{Tr}\left(M_F(M_F)^*(y_2)^T(y_2)^*\right)$	$-8\text{Tr}\left(M_F(y_2)^\dagger y_2(M_F)^*\right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 3 of 4

Column	Representative term structure									
T1	$-\frac{3g_Y^2 m_{S_2}^2}{2}$									
T2	$-6g_Y^2 m_{S_2}^2$									
T3	$m_{S_2}^2 (2 \text{Tr}(y_2 (y_2)^\dagger) + 6\lambda_{S_2})$									

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{3g_Y^2 m_{S_2}^2}{2}$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$-6g_Y^2 m_{S_2}^2$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{27g_Y^2 m_{S_2}^2}{2}$	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	$-24g_Y^2 m_{S_2}^2$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	$m_{S_2}^2 (2 \text{Tr}(y_2 (y_2)^\dagger) + 6\lambda_{S_2})$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_Y^2 m_{S_2}^2}{2}$	—	$m_{S_2}^2 (2 \text{Tr}(y_2 (y_2)^\dagger) + 6\lambda_{S_2})$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-6g_Y^2 m_{S_2}^2$	$m_{S_2}^2 (2 \text{Tr}(y_2 (y_2)^\dagger) + 6\lambda_{S_2})$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-\frac{27g_Y^2 m_{S_2}^2}{2}$	—	$m_{S_2}^2 (2 \text{Tr}(y_2 (y_2)^\dagger) + 6\lambda_{S_2})$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$-24g_Y^2 m_{S_2}^2$	$m_{S_2}^2 (2 \text{Tr}(y_2 (y_2)^\dagger) + 6\lambda_{S_2})$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$m_{S_2}^2 (3 \text{Tr}(y_2 (y_2)^\dagger) + 6\lambda_{S_2})$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{3g_Y^2 m_{S_2}^2}{2}$	—	$m_{S_2}^2 (3 \text{Tr}(y_2 (y_2)^\dagger) + 6\lambda_{S_2})$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-6g_Y^2 m_{S_2}^2$	$m_{S_2}^2 (3 \text{Tr}(y_2 (y_2)^\dagger) + 6\lambda_{S_2})$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{27g_Y^2 m_{S_2}^2}{2}$	—	$m_{S_2}^2 (3 \text{Tr}(y_2 (y_2)^\dagger) + 6\lambda_{S_2})$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$-24g_Y^2 m_{S_2}^2$	$m_{S_2}^2 (3 \text{Tr}(y_2 (y_2)^\dagger) + 6\lambda_{S_2})$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$m_{S_2}^2 (4 \text{Tr}(y_2 (y_2)^\dagger) + 4\lambda_{S_2})$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-\frac{3g_Y^2 m_{S_2}^2}{2}$	—	$m_{S_2}^2 (4 \text{Tr}(y_2 (y_2)^\dagger) + 4\lambda_{S_2})$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-6g_Y^2 m_{S_2}^2$	$m_{S_2}^2 (4 \text{Tr}(y_2 (y_2)^\dagger) + 4\lambda_{S_2})$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-\frac{27g_Y^2 m_{S_2}^2}{2}$	—	$m_{S_2}^2 (4 \text{Tr}(y_2 (y_2)^\dagger) + 4\lambda_{S_2})$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$-24g_Y^2 m_{S_2}^2$	$m_{S_2}^2 (4 \text{Tr}(y_2 (y_2)^\dagger) + 4\lambda_{S_2})$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{3g_Y^2 m_{S_2}^2}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-6g_Y^2 m_{S_2}^2$	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{27g_Y^2 m_{S_2}^2}{2}$	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$-24g_Y^2 m_{S_2}^2$	—

Terms 4 of 4

Column	Representative term structure
T1	$-\frac{9g_2^2 m_{S_2}^2}{2}$
T2	$+m_{S_1}^2(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-\frac{9g_2^2 m_{S_2}^2}{2}$	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{9g_2^2 m_{S_2}^2}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{9g_2^2 m_{S_2}^2}{2}$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-\frac{9g_2^2 m_{S_2}^2}{2}$	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$-\frac{9g_2^2 m_{S_2}^2}{2}$	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-\frac{9g_2^2 m_{S_2}^2}{2}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{9g_2^2 m_{S_2}^2}{2}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{9g_2^2 m_{S_2}^2}{2}$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{9g_2^2 m_{S_2}^2}{2}$	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-\frac{9g_2^2 m_{S_2}^2}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$+m_{S_1}^2(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$+m_{S_1}^2(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+m_{S_1}^2(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$+m_{S_1}^2(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$+m_{S_1}^2(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$

$$\beta_{\lambda_H}^{(1)}$$

$$16\pi^2 \mu \frac{d\lambda_H}{d\mu} = \beta_{\lambda_H}^{(1)}$$

Terms 1 of 7

Column	Representative term structure
T1	$\lambda_H (12 \text{Tr} (Y_d (Y_d)^\dagger) + 4 \text{Tr} (Y_e (Y_e)^\dagger) + 12 \text{Tr} (Y_u (Y_u)^\dagger))$
T2	$-12 \text{Tr} (Y_d (Y_d)^\dagger Y_d (Y_d)^\dagger)$
T3	$-4 \text{Tr} (Y_e (Y_e)^\dagger Y_e (Y_e)^\dagger)$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_H(12\text{Tr}(Y_d(Y_d)^\dagger) + 4\text{Tr}(Y_e(Y_e)^\dagger) + 12\text{Tr}(Y_u(Y_u)^\dagger))$	$-12\text{Tr}(Y_d(Y_d)^\dagger Y_d(Y_d)^\dagger)$	$-4\text{Tr}(Y_e(Y_e)^\dagger Y_e(Y_e)^\dagger)$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\lambda_H(12\text{Tr}(Y_d(Y_d)^\dagger) + 4\text{Tr}(Y_e(Y_e)^\dagger) + 12\text{Tr}(Y_u(Y_u)^\dagger))$	$-12\text{Tr}(Y_d(Y_d)^\dagger Y_d(Y_d)^\dagger)$	$-4\text{Tr}(Y_e(Y_e)^\dagger Y_e(Y_e)^\dagger)$

Terms 2 of 7

Column	Representative term structure
T1	$-12 \operatorname{Tr} \left(Y_u \left(Y_u \right)^\dagger Y_u \left(Y_u \right)^\dagger \right)$
T2	$+4 \lambda_{T3} \left(\lambda_{T3} \right)^*$
T3	$+\frac{9 g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T_1	T_2	T_3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_{T3} (\lambda_{T3})^*$	$+\frac{9g_2^4}{4}$

Terms 3 of 7

Column	Representative term structure									
T1	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$									
T2	$-9g_2^2 \lambda_H$									
T3	$+\frac{3g_Y^4}{4}$									

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_H \right)$	$-9g_2^2 \lambda_H$	$+\frac{3g_Y^4}{4}$

Terms 4 of 7

Column	Representative term structure
T1	$+12\lambda_H^2$
T2	$+2\lambda_{H1}^2$
T3	$+6\lambda_{H2}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+12\lambda_H^2$	$+2\lambda_{H1}^2$	$+6\lambda_{H2}^2$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+12\lambda_H^2$	$+2\lambda_{H1}^2$	$+6\lambda_{H2}^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+12\lambda_H^2$	$+2\lambda_{H1}^2$	$+6\lambda_{H2}^2$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+12\lambda_H^2$	$+2\lambda_{H1}^2$	$+6\lambda_{H2}^2$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+12\lambda_H^2$	$+2\lambda_{H1}^2$	$+6\lambda_{H2}^2$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+12\lambda_H^2$	$+4\lambda_{H1}^2$	$+4\lambda_{H2}^2$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+12\lambda_H^2$	$+4\lambda_{H1}^2$	$+4\lambda_{H2}^2$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+12\lambda_H^2$	$+4\lambda_{H1}^2$	$+4\lambda_{H2}^2$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+12\lambda_H^2$	$+4\lambda_{H1}^2$	$+4\lambda_{H2}^2$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+12\lambda_H^2$	$+4\lambda_{H1}^2$	$+4\lambda_{H2}^2$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+12\lambda_H^2$	$+4\lambda_{H1}^2$	$+4\lambda_{H2}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+12\lambda_H^2$	$+4\lambda_{H1}^2$	$+4\lambda_{H2}^2$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+12\lambda_H^2$	$+4\lambda_{H1}^2$	$+4\lambda_{H2}^2$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+12\lambda_H^2$	$+4\lambda_{H1}^2$	$+4\lambda_{H2}^2$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+12\lambda_H^2$	$+4\lambda_{H1}^2$	$+4\lambda_{H2}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+12\lambda_H^2$	$+6\lambda_{H1}^2$	$+2\lambda_{H2}^2$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+12\lambda_H^2$	$+6\lambda_{H1}^2$	$+2\lambda_{H2}^2$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+12\lambda_H^2$	$+6\lambda_{H1}^2$	$+2\lambda_{H2}^2$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+12\lambda_H^2$	$+6\lambda_{H1}^2$	$+2\lambda_{H2}^2$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+12\lambda_H^2$	$+6\lambda_{H1}^2$	$+2\lambda_{H2}^2$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+12\lambda_H^2$	$+6\lambda_{H1}^2$	$+6\lambda_{H2}^2$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+12\lambda_H^2$	$+6\lambda_{H1}^2$	$+6\lambda_{H2}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+12\lambda_H^2$	$+6\lambda_{H1}^2$	$+6\lambda_{H2}^2$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+12\lambda_H^2$	$+6\lambda_{H1}^2$	$+6\lambda_{H2}^2$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+12\lambda_H^2$	$+6\lambda_{H1}^2$	$+6\lambda_{H2}^2$

Terms 5 of 7

Column	Representative term structure									
T1	$+3\lambda_{H2}^{\text{Adj}^2}$									
T2	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$									
T3	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$									

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+3\lambda_{H2}^{\text{Adj}^2}$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+3\lambda_{H2}^{\text{Adj}^2}$	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+3\lambda_{H2}^{\text{Adj}^2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+3\lambda_{H2}^{\text{Adj}^2}$	—	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	—	—

Terms 6 of 7

Column	Representative term structure
T1	$+8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$
T2	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$
T3	$+8\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	$+8\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	$+8\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Column	Representative term structure
T1	$+3\lambda_{H1}^{\text{Adj}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+3\lambda_{H1}^{\text{Adj}^2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+3\lambda_{H1}^{\text{Adj}^2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+3\lambda_{H1}^{\text{Adj}^2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+3\lambda_{H1}^{\text{Adj}^2}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+3\lambda_{H1}^{\text{Adj}^2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+3\lambda_{H1}^{\text{Adj}^2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+3\lambda_{H1}^{\text{Adj}^2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H1}^{\text{Adj}^2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+3\lambda_{H1}^{\text{Adj}^2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+3\lambda_{H1}^{\text{Adj}^2}$

$$\beta_{\lambda_{S_1}}^{(1)}$$

$$_{16\pi^2}\mu\frac{d\lambda_{S_1}}{d\mu}=\beta_{\lambda_{S_1}}^{(1)}$$

Terms 1 of 10

Column	Representative term structure	
T1	$8\lambda_{S_1}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	
T2	$-8\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	
T3	$+12g_Y^4$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$8\lambda_{S_1}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-8\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	$+12g_Y^4$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$8\lambda_{S_1}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-8\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$8\lambda_{S_1}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-8\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$8\lambda_{S_1}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-8\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$8\lambda_{S_1}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-8\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	$+12g_Y^4$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$4\lambda_{S_1}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	$+12g_Y^4$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$4\lambda_{S_1}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$4\lambda_{S_1}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$4\lambda_{S_1}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$4\lambda_{S_1}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-4\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	$+12g_Y^4$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$6\lambda_{S_1}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-5\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	$+12g_Y^4$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$6\lambda_{S_1}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-5\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$6\lambda_{S_1}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-5\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$6\lambda_{S_1}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-5\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$6\lambda_{S_1}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-5\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	$+12g_Y^4$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	$+12g_Y^4$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$-3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$-3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$-3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	$+12g_Y^4$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$-3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	$+12g_Y^4$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$-3 \operatorname{Tr} \left(y_1 \left(y_1 \right)^\dagger y_1 \left(y_1 \right)^\dagger \right)$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$-3 \operatorname{Tr} \left(y_1 \left(y_1 \right)^\dagger y_1 \left(y_1 \right)^\dagger \right)$	$+12 g_Y^4$

Terms 2 of 10

Column	Representative term structure
T1	$-12g_Y^2\lambda_{S_1}$
T2	$+6\lambda_{12}^2$
T3	$+4\lambda_{H1}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-12g_Y^2\lambda_{S_1}$	$+6\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-3g_Y^2\lambda_{S_1}$	$+6\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$+6\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-3g_Y^2\lambda_{S_1}$	$+6\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-12g_Y^2\lambda_{S_1}$	$+6\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$+4\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	$+4\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$+4\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$+4\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$+4\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$+4\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+4\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$+4\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$+4\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$+4\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$+2\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$+2\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$+2\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$+2\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$+2\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$+6\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$+6\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+6\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$+6\lambda_{12}^2$	$+4\lambda_{H1}^2$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$+6\lambda_{12}^2$	$+4\lambda_{H1}^2$

Terms 3 of 10

Column	Representative term structure
T1	$+10\lambda_{S_1}^2$
T2	$+\frac{3g_Y^4}{4}$
T3	$+\frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+10\lambda_{S_1}^2$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+10\lambda_{S_1}^2$	$+\frac{3g_Y^4}{4}$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+10\lambda_{S_1}^2$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+10\lambda_{S_1}^2$	$+\frac{3g_Y^4}{4}$	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+10\lambda_{S_1}^2$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+12\lambda_{S_1}^2$	—	$+\frac{9g_2^4}{4}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+12\lambda_{S_1}^2$	$+\frac{3g_Y^4}{4}$	$+\frac{9g_2^4}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+12\lambda_{S_1}^2$	—	$+\frac{9g_2^4}{4}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+12\lambda_{S_1}^2$	$+\frac{3g_Y^4}{4}$	$+\frac{9g_2^4}{4}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+12\lambda_{S_1}^2$	—	$+\frac{9g_2^4}{4}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+12\lambda_{S_1}^2$	—	$+\frac{9g_2^4}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+12\lambda_{S_1}^2$	$+\frac{3g_Y^4}{4}$	$+\frac{9g_2^4}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+12\lambda_{S_1}^2$	—	$+\frac{9g_2^4}{4}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+12\lambda_{S_1}^2$	$+\frac{3g_Y^4}{4}$	$+\frac{9g_2^4}{4}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+12\lambda_{S_1}^2$	—	$+\frac{9g_2^4}{4}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+14\lambda_{S_1}^2$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+14\lambda_{S_1}^2$	$+\frac{3g_Y^4}{4}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+14\lambda_{S_1}^2$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+14\lambda_{S_1}^2$	$+\frac{3g_Y^4}{4}$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+14\lambda_{S_1}^2$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+14\lambda_{S_1}^2$	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+14\lambda_{S_1}^2$	$+\frac{3g_Y^4}{4}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+14\lambda_{S_1}^2$	—	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+14\lambda_{S_1}^2$	$+\frac{3g_Y^4}{4}$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+14\lambda_{S_1}^2$	—	—

Terms 4 of 10

Column	Representative term structure
T1	$+g_Y^2 \left(6g_2^2 - 12\lambda_{S_1}\right)$
T2	$-9g_2^2 \lambda_{S_1}$
T3	$+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+g_Y^2 \left(6g_2^2 - 12\lambda_{S_1}\right)$	$-9g_2^2 \lambda_{S_1}$	$+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	$-9g_2^2 \lambda_{S_1}$	$+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-9g_2^2 \lambda_{S_1}$	$+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$-9g_2^2 \lambda_{S_1}$	$+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+g_Y^2 \left(6g_2^2 - 12\lambda_{S_1}\right)$	$-9g_2^2 \lambda_{S_1}$	$+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+g_Y^2 \left(6g_2^2 - 12\lambda_{S_1}\right)$	$-9g_2^2 \lambda_{S_1}$	$+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$-9g_2^2 \lambda_{S_1}$	$+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-9g_2^2 \lambda_{S_1}$	$+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$-9g_2^2 \lambda_{S_1}$	$+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+g_Y^2 \left(6g_2^2 - 12\lambda_{S_1}\right)$	$-9g_2^2 \lambda_{S_1}$	$+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 5 of 10

Column	Representative term structure	
T1	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	
T2	$+\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} (2 \text{Tr}(y_1 (y_2)^\dagger) + 24 \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*)$	
T3	$+2 \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \text{Tr}(y_2 (y_1)^\dagger)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	$+\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} (2 \text{Tr}(y_1 (y_2)^\dagger) + 24 \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*) + 2 \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \text{Tr}(y_2 (y_1)^\dagger)$	
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	$+\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} (3 \text{Tr}(y_1 (y_2)^\dagger) + 24 \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*) + 3 \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \text{Tr}(y_2 (y_1)^\dagger)$	
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 6 of 10

Column Representative term structure

T1	$+8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$
T2	$+8\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$
T3	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_1} \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$	$+8\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_1} \right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_1} \right)$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$	$+8\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_1} \right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_1} \right)$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 7 of 10

Column	Representative term structure	
T1	$\lambda_{S_1} (4 \text{Tr} (y_1 (y_1)^\dagger) + 14 \lambda_{S_1}^{\text{Adj}})$	
T2	$+2 \lambda_{S_1}^{\text{Adj}} \text{Tr} (y_1 (y_1)^\dagger)$	
T3	$+21 g_2^4$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_{S_1} (4 \text{Tr} (y_1 (y_1)^\dagger) + 14 \lambda_{S_1}^{\text{Adj}})$	$+2 \lambda_{S_1}^{\text{Adj}} \text{Tr} (y_1 (y_1)^\dagger)$	$+21 g_2^4$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$\lambda_{S_1} (4 \text{Tr} (y_1 (y_1)^\dagger) + 14 \lambda_{S_1}^{\text{Adj}})$	$+2 \lambda_{S_1}^{\text{Adj}} \text{Tr} (y_1 (y_1)^\dagger)$	$+21 g_2^4$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\lambda_{S_1} (4 \text{Tr} (y_1 (y_1)^\dagger) + 14 \lambda_{S_1}^{\text{Adj}})$	$+2 \lambda_{S_1}^{\text{Adj}} \text{Tr} (y_1 (y_1)^\dagger)$	$+21 g_2^4$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$\lambda_{S_1} (4 \text{Tr} (y_1 (y_1)^\dagger) + 14 \lambda_{S_1}^{\text{Adj}})$	$+2 \lambda_{S_1}^{\text{Adj}} \text{Tr} (y_1 (y_1)^\dagger)$	$+21 g_2^4$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$\lambda_{S_1} (4 \text{Tr} (y_1 (y_1)^\dagger) + 14 \lambda_{S_1}^{\text{Adj}})$	$+2 \lambda_{S_1}^{\text{Adj}} \text{Tr} (y_1 (y_1)^\dagger)$	$+21 g_2^4$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{S_1} (4 \text{Tr} (y_1 (y_1)^\dagger) + 14 \lambda_{S_1}^{\text{Adj}})$	$+2 \lambda_{S_1}^{\text{Adj}} \text{Tr} (y_1 (y_1)^\dagger)$	$+21 g_2^4$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{S_1} (4 \text{Tr} (y_1 (y_1)^\dagger) + 14 \lambda_{S_1}^{\text{Adj}})$	$+2 \lambda_{S_1}^{\text{Adj}} \text{Tr} (y_1 (y_1)^\dagger)$	$+21 g_2^4$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_1} (4 \text{Tr} (y_1 (y_1)^\dagger) + 14 \lambda_{S_1}^{\text{Adj}})$	$+2 \lambda_{S_1}^{\text{Adj}} \text{Tr} (y_1 (y_1)^\dagger)$	$+21 g_2^4$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{S_1} (4 \text{Tr} (y_1 (y_1)^\dagger) + 14 \lambda_{S_1}^{\text{Adj}})$	$+2 \lambda_{S_1}^{\text{Adj}} \text{Tr} (y_1 (y_1)^\dagger)$	$+21 g_2^4$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\lambda_{S_1} (4 \text{Tr} (y_1 (y_1)^\dagger) + 14 \lambda_{S_1}^{\text{Adj}})$	$+2 \lambda_{S_1}^{\text{Adj}} \text{Tr} (y_1 (y_1)^\dagger)$	$+21 g_2^4$

Terms 8 of 10

Column	Representative term structure						
T1	$+g_Y^2 \left(12g_2^2 - 12\lambda_{S_1} - 6\lambda_{S_1}^{\text{Adj}} \right)$						
T2	$+g_2^2 (-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$						
T3	$+\frac{3\lambda_{H1}^{\text{Adj}2}}{2}$						

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+g_Y^2 \left(12g_2^2 - 12\lambda_{S_1} - 6\lambda_{S_1}^{\text{Adj}} \right)$	$+g_2^2 (-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$	$+\frac{3\lambda_{H1}^{\text{Adj}2}}{2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$+g_2^2 (-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$	$+\frac{3\lambda_{H1}^{\text{Adj}2}}{2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$+g_2^2 (-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$	$+\frac{3\lambda_{H1}^{\text{Adj}2}}{2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$+g_2^2 (-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$	$+\frac{3\lambda_{H1}^{\text{Adj}2}}{2}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+g_Y^2 \left(12g_2^2 - 12\lambda_{S_1} - 6\lambda_{S_1}^{\text{Adj}} \right)$	$+g_2^2 (-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$	$+\frac{3\lambda_{H1}^{\text{Adj}2}}{2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+g_Y^2 \left(12g_2^2 - 12\lambda_{S_1} - 6\lambda_{S_1}^{\text{Adj}} \right)$	$+g_2^2 (-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$	$+\frac{3\lambda_{H1}^{\text{Adj}2}}{2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$+g_2^2 (-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$	$+\frac{3\lambda_{H1}^{\text{Adj}2}}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+g_2^2 (-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$	$+\frac{3\lambda_{H1}^{\text{Adj}2}}{2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$+g_2^2 (-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$	$+\frac{3\lambda_{H1}^{\text{Adj}2}}{2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+g_Y^2 \left(12g_2^2 - 12\lambda_{S_1} - 6\lambda_{S_1}^{\text{Adj}} \right)$	$+g_2^2 (-24\lambda_{S_1} - 12\lambda_{S_1}^{\text{Adj}})$	$+\frac{3\lambda_{H1}^{\text{Adj}2}}{2}$

Terms 9 of 10

Column	Representative term structure
T1	$+26\lambda_{S_1}^{\text{Adj}^2}$
T2	$+g_Y^2 \left(3g_2^2 - 3\lambda_{S_1} - \frac{3\lambda_{S_1}^{\text{Adj}}}{2} \right)$
T3	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+26\lambda_{S_1}^{\text{Adj}^2}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+26\lambda_{S_1}^{\text{Adj}^2}$	$+g_Y^2 \left(3g_2^2 - 3\lambda_{S_1} - \frac{3\lambda_{S_1}^{\text{Adj}}}{2} \right)$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+26\lambda_{S_1}^{\text{Adj}^2}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+26\lambda_{S_1}^{\text{Adj}^2}$	$+g_Y^2 \left(3g_2^2 - 3\lambda_{S_1} - \frac{3\lambda_{S_1}^{\text{Adj}}}{2} \right)$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+26\lambda_{S_1}^{\text{Adj}^2}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+26\lambda_{S_1}^{\text{Adj}^2}$	—	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+26\lambda_{S_1}^{\text{Adj}^2}$	$+g_Y^2 \left(3g_2^2 - 3\lambda_{S_1} - \frac{3\lambda_{S_1}^{\text{Adj}}}{2} \right)$	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+26\lambda_{S_1}^{\text{Adj}^2}$	—	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+26\lambda_{S_1}^{\text{Adj}^2}$	$+g_Y^2 \left(3g_2^2 - 3\lambda_{S_1} - \frac{3\lambda_{S_1}^{\text{Adj}}}{2} \right)$	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+26\lambda_{S_1}^{\text{Adj}^2}$	—	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$

Terms 10 of 10

Column	Representative term structure
T1	$+6\lambda_{12}^{\text{Adj}^2}$
T2	$+6\lambda_{12}^{\text{Cross}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+6\lambda_{12}^{\text{Adj}^2}$	$+6\lambda_{12}^{\text{Cross}^2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+6\lambda_{12}^{\text{Adj}^2}$	$+6\lambda_{12}^{\text{Cross}^2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{12}^{\text{Adj}^2}$	$+6\lambda_{12}^{\text{Cross}^2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+6\lambda_{12}^{\text{Adj}^2}$	$+6\lambda_{12}^{\text{Cross}^2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+6\lambda_{12}^{\text{Adj}^2}$	$+6\lambda_{12}^{\text{Cross}^2}$

$$\beta_{\lambda_{S_2}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{S_2}}{d\mu}=\beta_{\lambda_{S_2}}^{(1)}$$

Terms 1 of 10

Column	Representative term structure
T1	$\lambda_{S_2}(4\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+14\lambda_{S_2}^{\text{Adj}})$
T2	$+2\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$
T3	$-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{S_2}(4\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+14\lambda_{S_2}^{\text{Adj}})$	$+2\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$	$-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{S_2}(4\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+14\lambda_{S_2}^{\text{Adj}})$	$+2\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$	$-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_2}(4\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+14\lambda_{S_2}^{\text{Adj}})$	$+2\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$	$-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{S_2}(4\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+14\lambda_{S_2}^{\text{Adj}})$	$+2\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$	$-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$\lambda_{S_2}(4\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+14\lambda_{S_2}^{\text{Adj}})$	$+2\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$	$-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	$-4\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	$-4\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$-4\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	$-4\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	$-4\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$-5\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$-5\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$-5\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$-5\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$-5\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$-8\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	$-8\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	$-8\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	$-8\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	$-8\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{S_2}(4\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+14\lambda_{S_2}^{\text{Adj}})$	$+2\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$	$-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{S_2}(4\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+14\lambda_{S_2}^{\text{Adj}})$	$+2\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$	$-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_2}(4\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+14\lambda_{S_2}^{\text{Adj}})$	$+2\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$	$-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{S_2}(4 \text{Tr}(y_2 (y_2)^\dagger) + 14\lambda_{S_2}^{\text{Adj}})$	$+2\lambda_{S_2}^{\text{Adj}} \text{Tr}(y_2 (y_2)^\dagger)$	$-3 \text{Tr}(y_2 (y_2)^\dagger y_2 (y_2)^\dagger)$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\lambda_{S_2}(4 \text{Tr}(y_2 (y_2)^\dagger) + 14\lambda_{S_2}^{\text{Adj}})$	$+2\lambda_{S_2}^{\text{Adj}} \text{Tr}(y_2 (y_2)^\dagger)$	$-3 \text{Tr}(y_2 (y_2)^\dagger y_2 (y_2)^\dagger)$

Terms 2 of 10

Column	Representative term structure
T1	$+21g_2^4$
T2	$+g_2^2(-24\lambda_{S_2} - 12\lambda_{S_2}^{\text{Adj}})$
T3	$+2\lambda_{12}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+21g_2^4$	$+g_2^2(-24\lambda_{S_2} - 12\lambda_{S_2}^{\text{Adj}})$	$+2\lambda_{12}^2$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+21g_2^4$	$+g_2^2(-24\lambda_{S_2} - 12\lambda_{S_2}^{\text{Adj}})$	$+2\lambda_{12}^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+21g_2^4$	$+g_2^2(-24\lambda_{S_2} - 12\lambda_{S_2}^{\text{Adj}})$	$+2\lambda_{12}^2$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+21g_2^4$	$+g_2^2(-24\lambda_{S_2} - 12\lambda_{S_2}^{\text{Adj}})$	$+2\lambda_{12}^2$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+21g_2^4$	$+g_2^2(-24\lambda_{S_2} - 12\lambda_{S_2}^{\text{Adj}})$	$+2\lambda_{12}^2$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	$+4\lambda_{12}^2$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	$+4\lambda_{12}^2$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$+4\lambda_{12}^2$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	$+4\lambda_{12}^2$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	$+4\lambda_{12}^2$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$+4\lambda_{12}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$+4\lambda_{12}^2$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$+4\lambda_{12}^2$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$+4\lambda_{12}^2$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$+4\lambda_{12}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$+6\lambda_{12}^2$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	$+6\lambda_{12}^2$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	$+6\lambda_{12}^2$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	$+6\lambda_{12}^2$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	$+6\lambda_{12}^2$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+21g_2^4$	$+g_2^2(-24\lambda_{S_2} - 12\lambda_{S_2}^{\text{Adj}})$	$+6\lambda_{12}^2$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+21g_2^4$	$+g_2^2(-24\lambda_{S_2} - 12\lambda_{S_2}^{\text{Adj}})$	$+6\lambda_{12}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+21g_2^4$	$+g_2^2(-24\lambda_{S_2} - 12\lambda_{S_2}^{\text{Adj}})$	$+6\lambda_{12}^2$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+21g_2^4$	$+g_2^2(-24\lambda_{S_2} - 12\lambda_{S_2}^{\text{Adj}})$	$+6\lambda_{12}^2$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+21g_2^4$	$+g_2^2(-24\lambda_{S_2} - 12\lambda_{S_2}^{\text{Adj}})$	$+6\lambda_{12}^2$

Terms 3 of 10

Column	Representative term structure
T1	$+4\lambda_{H2}^2$
T2	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$
T3	$+14\lambda_{S2}^2$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+4\lambda_{H2}^2$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$	$+14\lambda_{S2}^2$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+4\lambda_{H2}^2$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$	$+14\lambda_{S2}^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H2}^2$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$	$+14\lambda_{S2}^2$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+4\lambda_{H2}^2$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$	$+14\lambda_{S2}^2$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+4\lambda_{H2}^2$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$	$+14\lambda_{S2}^2$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+4\lambda_{H2}^2$	—	$+12\lambda_{S2}^2$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{H2}^2$	—	$+12\lambda_{S2}^2$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+4\lambda_{H2}^2$	—	$+12\lambda_{S2}^2$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+4\lambda_{H2}^2$	—	$+12\lambda_{S2}^2$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+4\lambda_{H2}^2$	—	$+12\lambda_{S2}^2$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+4\lambda_{H2}^2$	—	$+12\lambda_{S2}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{H2}^2$	—	$+12\lambda_{S2}^2$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+4\lambda_{H2}^2$	—	$+12\lambda_{S2}^2$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+4\lambda_{H2}^2$	—	$+12\lambda_{S2}^2$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+4\lambda_{H2}^2$	—	$+12\lambda_{S2}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+4\lambda_{H2}^2$	—	$+10\lambda_{S2}^2$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+4\lambda_{H2}^2$	—	$+10\lambda_{S2}^2$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+4\lambda_{H2}^2$	—	$+10\lambda_{S2}^2$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+4\lambda_{H2}^2$	—	$+10\lambda_{S2}^2$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+4\lambda_{H2}^2$	—	$+10\lambda_{S2}^2$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+4\lambda_{H2}^2$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$	$+14\lambda_{S2}^2$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+4\lambda_{H2}^2$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$	$+14\lambda_{S2}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H2}^2$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$	$+14\lambda_{S2}^2$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+4\lambda_{H2}^2$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$	$+14\lambda_{S2}^2$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+4\lambda_{H2}^2$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$	$+14\lambda_{S2}^2$

Terms 4 of 10

Column	Representative term structure
T1	$+26\lambda_{S_2}^{\text{Adj}^2}$
T2	$+g_Y^2 \left(3g_2^2 - 3\lambda_{S_2} - \frac{3\lambda_{S_2}^{\text{Adj}}}{2} \right)$
T3	$+\frac{3g_Y^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+26\lambda_{S_2}^{\text{Adj}^2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+26\lambda_{S_2}^{\text{Adj}^2}$	$+g_Y^2 \left(3g_2^2 - 3\lambda_{S_2} - \frac{3\lambda_{S_2}^{\text{Adj}}}{2} \right)$	$+\frac{3g_Y^4}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+26\lambda_{S_2}^{\text{Adj}^2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+26\lambda_{S_2}^{\text{Adj}^2}$	$+g_Y^2 \left(27g_2^2 - 27\lambda_{S_2} - \frac{27\lambda_{S_2}^{\text{Adj}}}{2} \right)$	$+\frac{243g_Y^4}{4}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+26\lambda_{S_2}^{\text{Adj}^2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	$+\frac{3g_Y^4}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	$+\frac{243g_Y^4}{4}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$+\frac{3g_Y^4}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$+\frac{243g_Y^4}{4}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	$+\frac{3g_Y^4}{4}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	$+\frac{243g_Y^4}{4}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+26\lambda_{S_2}^{\text{Adj}^2}$	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+26\lambda_{S_2}^{\text{Adj}^2}$	$+g_Y^2 \left(3g_2^2 - 3\lambda_{S_2} - \frac{3\lambda_{S_2}^{\text{Adj}}}{2} \right)$	$+\frac{3g_Y^4}{4}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+26\lambda_{S_2}^{\text{Adj}^2}$	—	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+26\lambda_{S_2}^{\text{Adj}^2}$	$+g_Y^2 \left(27g_2^2 - 27\lambda_{S_2} - \frac{27\lambda_{S_2}^{\text{Adj}}}{2} \right)$	$+\frac{243g_Y^4}{4}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+26\lambda_{S_2}^{\text{Adj}^2}$	—	—

Terms 5 of 10

Column	Representative term structure						
T1	$+g_Y{}^2\left(12g_2{}^2-12\lambda_{S_2}-6\lambda_{S_2}^{\text{Adj}}\right)$						
T2	$+12g_Y{}^4$						
T3	$4\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$						

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+g_Y{}^2\left(12g_2{}^2-12\lambda_{S_2}-6\lambda_{S_2}^{\text{Adj}}\right)$	$+12g_Y{}^4$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+g_Y{}^2\left(48g_2{}^2-48\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}}\right)$	$+192g_Y{}^4$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	$4\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	$+4\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$+12g_Y{}^4$	$4\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	$4\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$+192g_Y{}^4$	$4\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$6\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$+6\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$+12g_Y{}^4$	$6\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$6\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$+192g_Y{}^4$	$6\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$8\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	$8\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$+12g_Y{}^4$	$8\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	$8\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$+192g_Y{}^4$	$8\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_Y{}^2\left(12g_2{}^2-12\lambda_{S_2}-6\lambda_{S_2}^{\text{Adj}}\right)$	$+12g_Y{}^4$	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+g_Y{}^2\left(48g_2{}^2-48\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}}\right)$	$+192g_Y{}^4$	—

Terms 6 of 10

Column	Representative term structure
T1	$+\frac{9g_2^4}{4}$
T2	$-9g_2^2\lambda_{S_2}$
T3	$+\frac{\lambda_{12}^{\text{Adj}2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{S_2}$	$+\frac{\lambda_{12}^{\text{Adj}2}}{4}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{S_2}$	$+\frac{\lambda_{12}^{\text{Adj}2}}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{S_2}$	$+\frac{\lambda_{12}^{\text{Adj}2}}{4}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{S_2}$	$+\frac{\lambda_{12}^{\text{Adj}2}}{4}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{S_2}$	$+\frac{\lambda_{12}^{\text{Adj}2}}{4}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{S_2}$	$+\frac{\lambda_{12}^{\text{Adj}2}}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{S_2}$	$+\frac{\lambda_{12}^{\text{Adj}2}}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{S_2}$	$+\frac{\lambda_{12}^{\text{Adj}2}}{4}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{S_2}$	$+\frac{\lambda_{12}^{\text{Adj}2}}{4}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{S_2}$	$+\frac{\lambda_{12}^{\text{Adj}2}}{4}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 7 of 10

Column	Representative term structure
T1	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$
T2	$\lambda_{S_1^\dagger S_2 S_2^\dagger 2} (2 \text{Tr}(y_1 (y_2)^\dagger) + 24 \left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^*)$
T3	$+2 \left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* \text{Tr}(y_2 (y_1)^\dagger)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$	$\lambda_{S_1^\dagger S_2 S_2^\dagger 2} (2 \text{Tr}(y_1 (y_2)^\dagger) + 24 \left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^*) + 2 \left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* \text{Tr}(y_2 (y_1)^\dagger)$	
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$	$\lambda_{S_1^\dagger S_2 S_2^\dagger 2} (3 \text{Tr}(y_1 (y_2)^\dagger) + 24 \left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^*) + 3 \left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* \text{Tr}(y_2 (y_1)^\dagger)$	
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 8 of 10

Column	Representative term structure
T1	$+8\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$
T2	$+8\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$
T3	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_2} \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$	$+8\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_2} \right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	$+g_Y^2 \left(\frac{27g_2^2}{2} - 27\lambda_{S_2} \right)$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$	$+8\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$	$+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_2} \right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$+g_Y^2 \left(\frac{27g_2^2}{2} - 27\lambda_{S_2} \right)$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 9 of 10

Column	Representative term structure	
T1	$+g_Y^2 (6g_2^2 - 12\lambda_{S_2})$	
T2	$-3g_Y^2 \lambda_{S_2}$	
T3	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+g_Y^2 (6g_2^2 - 12\lambda_{S_2})$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+g_Y^2 (24g_2^2 - 48\lambda_{S_2})$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+g_Y^2 (6g_2^2 - 12\lambda_{S_2})$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+g_Y^2 (24g_2^2 - 48\lambda_{S_2})$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$-3g_Y^2 \lambda_{S_2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-12g_Y^2 \lambda_{S_2}$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$-27g_Y^2 \lambda_{S_2}$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$-48g_Y^2 \lambda_{S_2}$	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	$+8\lambda_{12}\lambda_{12}^{\text{Cross}}$

Terms 10 of 10

Column	Representative term structure
T1	$+6\lambda_{12}^{\text{Adj}^2}$
T2	$+6\lambda_{12}^{\text{Cross}^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+6\lambda_{12}^{\text{Adj}^2}$	$+6\lambda_{12}^{\text{Cross}^2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+6\lambda_{12}^{\text{Adj}^2}$	$+6\lambda_{12}^{\text{Cross}^2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{12}^{\text{Adj}^2}$	$+6\lambda_{12}^{\text{Cross}^2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+6\lambda_{12}^{\text{Adj}^2}$	$+6\lambda_{12}^{\text{Cross}^2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+6\lambda_{12}^{\text{Adj}^2}$	$+6\lambda_{12}^{\text{Cross}^2}$

$$\beta_{\lambda_{H1}}^{(1)}$$

$$16\pi^2 \mu \frac{d\lambda_{H1}}{d\mu} = \beta_{\lambda_{H1}}^{(1)}$$

Terms 1 of 8

Column	Representative term structure
T1	$\lambda_{H1}(4\text{Tr}(y_1(y_1)^\dagger) + 6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$
T2	$-4\text{Tr}(y_1(y_1)^\dagger Y_e(Y_e)^\dagger)$
T3	$+12\lambda_{T3}(\lambda_{T3})^*$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$-2 \operatorname{Tr} \left(y_1 \left(y_1 \right)^\dagger Y_e \left(Y_e \right)^\dagger \right)$	$+4 \lambda_{T3} \left(\lambda_{T3} \right)^*$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$-2 \operatorname{Tr} \left(y_1 \left(y_1 \right)^\dagger Y_e \left(Y_e \right)^\dagger \right)$	$+4 \lambda_{T3} \left(\lambda_{T3} \right)^*$

Terms 2 of 8

Column	Representative term structure
T1	$-\frac{9g_2^2\lambda_{H1}}{2}$
T2	$+3g_Y^4$
T3	$-\frac{15g_Y^2\lambda_{H1}}{2}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{9g_2^2\lambda_{H1}}{2}$	$+3g_Y^4$	$-\frac{15g_Y^2\lambda_{H1}}{2}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{9g_2^2\lambda_{H1}}{2}$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{9g_2^2\lambda_{H1}}{2}$	—	$-\frac{3g_Y^2\lambda_{H1}}{2}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{9g_2^2\lambda_{H1}}{2}$	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-\frac{9g_2^2\lambda_{H1}}{2}$	$+3g_Y^4$	$-\frac{15g_Y^2\lambda_{H1}}{2}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$+3g_Y^4$	$-\frac{15g_Y^2\lambda_{H1}}{2}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$-\frac{3g_Y^2\lambda_{H1}}{2}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$+3g_Y^4$	$-\frac{15g_Y^2\lambda_{H1}}{2}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$+3g_Y^4$	$-\frac{15g_Y^2\lambda_{H1}}{2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$-\frac{3g_Y^2\lambda_{H1}}{2}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$+3g_Y^4$	$-\frac{15g_Y^2\lambda_{H1}}{2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{33g_2^2\lambda_{H1}}{2}$	$+3g_Y^4$	$-\frac{15g_Y^2\lambda_{H1}}{2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-\frac{33g_2^2\lambda_{H1}}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{H1}}{2}$	—	$-\frac{3g_Y^2\lambda_{H1}}{2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-\frac{33g_2^2\lambda_{H1}}{2}$	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-\frac{33g_2^2\lambda_{H1}}{2}$	$+3g_Y^4$	$-\frac{15g_Y^2\lambda_{H1}}{2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{33g_2^2\lambda_{H1}}{2}$	$+3g_Y^4$	$-\frac{15g_Y^2\lambda_{H1}}{2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{33g_2^2\lambda_{H1}}{2}$	—	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{H1}}{2}$	—	$-\frac{3g_Y^2\lambda_{H1}}{2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{33g_2^2\lambda_{H1}}{2}$	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{33g_2^2\lambda_{H1}}{2}$	$+3g_Y^4$	$-\frac{15g_Y^2\lambda_{H1}}{2}$

Terms 3 of 8

Column	Representative term structure
T1	$+6\lambda_{12}\lambda_{H2}$
T2	$+6\lambda_H\lambda_{H1}$
T3	$+4\lambda_{H1}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+6\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+6\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+6\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+6\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+4\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+4\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+4\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+4\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+4\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+4\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+4\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+4\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+2\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+2\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+2\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+2\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+2\lambda_{12}\lambda_{H2}$	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$+6\lambda_H\lambda_{H1}$	$+4\lambda_{H1}^2$

Terms 4 of 8

Column	Representative term structure						
T1	$+4\lambda_{H1}\lambda_{S1}$						
T2	$+\frac{3g_Y^4}{4}$						
T3	$-3g_Y^2\lambda_{H1}$						

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+4\lambda_{H1}\lambda_{S1}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+4\lambda_{H1}\lambda_{S1}$	$+\frac{3g_Y^4}{4}$	$-3g_Y^2\lambda_{H1}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H1}\lambda_{S1}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+4\lambda_{H1}\lambda_{S1}$	$+\frac{3g_Y^4}{4}$	$-3g_Y^2\lambda_{H1}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+4\lambda_{H1}\lambda_{S1}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+6\lambda_{H1}\lambda_{S1}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{H1}\lambda_{S1}$	$+\frac{3g_Y^4}{4}$	$-3g_Y^2\lambda_{H1}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+6\lambda_{H1}\lambda_{S1}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+6\lambda_{H1}\lambda_{S1}$	$+\frac{3g_Y^4}{4}$	$-3g_Y^2\lambda_{H1}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+6\lambda_{H1}\lambda_{S1}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+6\lambda_{H1}\lambda_{S1}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_{H1}\lambda_{S1}$	$+\frac{3g_Y^4}{4}$	$-3g_Y^2\lambda_{H1}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+6\lambda_{H1}\lambda_{S1}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+6\lambda_{H1}\lambda_{S1}$	$+\frac{3g_Y^4}{4}$	$-3g_Y^2\lambda_{H1}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+6\lambda_{H1}\lambda_{S1}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+8\lambda_{H1}\lambda_{S1}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+8\lambda_{H1}\lambda_{S1}$	$+\frac{3g_Y^4}{4}$	$-3g_Y^2\lambda_{H1}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+8\lambda_{H1}\lambda_{S1}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+8\lambda_{H1}\lambda_{S1}$	$+\frac{3g_Y^4}{4}$	$-3g_Y^2\lambda_{H1}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+8\lambda_{H1}\lambda_{S1}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+8\lambda_{H1}\lambda_{S1}$	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+8\lambda_{H1}\lambda_{S1}$	$+\frac{3g_Y^4}{4}$	$-3g_Y^2\lambda_{H1}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{H1}\lambda_{S1}$	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+8\lambda_{H1}\lambda_{S1}$	$+\frac{3g_Y^4}{4}$	$-3g_Y^2\lambda_{H1}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+8\lambda_{H1}\lambda_{S1}$	—	—

Terms 5 of 8

Column	Representative term structure	
T1	$+\frac{9g_2^4}{4}$	
T2	$-9g_2^2\lambda_{H1}$	
T3	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H1}$	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H1}$	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H1}$	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H1}$	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H1}$	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H1}$	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H1}$	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H1}$	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H1}$	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H1}$	$+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 6 of 8

Column	Representative term structure	
T1	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\frac{1}{2} \text{Tr}(y_1 (y_2)^\dagger) + 4 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + 3 \left(\lambda_{S_1 S_1^\dagger S_2^\dagger} \right)^* \right)$	
T2	$+ \frac{1}{2} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \text{Tr}(y_2 (y_1)^\dagger)$	
T3	$+ 24 \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\frac{1}{2} \text{Tr}(y_1 (y_2)^\dagger) + 4 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + 3 \left(\lambda_{S_1 S_1^\dagger S_2^\dagger} \right)^* \right) + \frac{1}{2} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \text{Tr}(y_2 (y_1)^\dagger)$		$+ 24 \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\frac{3}{4} \text{Tr}(y_1 (y_2)^\dagger) + 4 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + 3 \left(\lambda_{S_1 S_1^\dagger S_2^\dagger} \right)^* \right) + \frac{3}{4} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \text{Tr}(y_2 (y_1)^\dagger)$		$+ 24 \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 7 of 8

Column	Representative term structure
T1	$+3\lambda_{S_1 S_1^\dagger S_2^\dagger} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$
T2	$\lambda_{H1} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{\text{Adj}})$
T3	$+6g_2^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+3\lambda_{S_1 S_1^\dagger S_2^\dagger} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+3\lambda_{S_1 S_1^\dagger S_2^\dagger} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$\lambda_{H1} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{\text{Adj}}) + 6g_2^4$	
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$\lambda_{H1} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{\text{Adj}}) + 6g_2^4$	
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$\lambda_{H1} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{\text{Adj}}) + 6g_2^4$	
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$\lambda_{H1} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{\text{Adj}}) + 6g_2^4$	
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$\lambda_{H1} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{\text{Adj}}) + 6g_2^4$	
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$\lambda_{H1} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{\text{Adj}}) + 6g_2^4$	
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$\lambda_{H1} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{\text{Adj}}) + 6g_2^4$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$\lambda_{H1} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{\text{Adj}}) + 6g_2^4$	
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$\lambda_{H1} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{\text{Adj}}) + 6g_2^4$	
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$\lambda_{H1} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{\text{Adj}}) + 6g_2^4$	

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Column	Representative term structure
T1	$+6\lambda_{H1}^{\text{Adj}2}$
T2	$+\lambda_{H2}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+6\lambda_{H1}^{\text{Adj}2}$	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+6\lambda_{H1}^{\text{Adj}2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+6\lambda_{H1}^{\text{Adj}2}$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+6\lambda_{H1}^{\text{Adj}2}$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+6\lambda_{H1}^{\text{Adj}2}$	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+6\lambda_{H1}^{\text{Adj}2}$	$+\lambda_{H2}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+6\lambda_{H1}^{\text{Adj}2}$	$+\lambda_{H2}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{H1}^{\text{Adj}2}$	$+\lambda_{H2}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+6\lambda_{H1}^{\text{Adj}2}$	$+\lambda_{H2}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+6\lambda_{H1}^{\text{Adj}2}$	$+\lambda_{H2}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$

$$\beta_{\lambda_{H2}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{H2}}{d\mu}=\beta_{\lambda_{H2}}^{(1)}$$

Terms 1 of 8

Column	Representative term structure	
T1	$\lambda_{H2}(2\text{Tr}(y_2(y_2)^\dagger)+6\text{Tr}(Y_d(Y_d)^\dagger)+2\text{Tr}(Y_e(Y_e)^\dagger)+6\text{Tr}(Y_u(Y_u)^\dagger)+4\lambda_{S2}^{\text{Adj}})$	
T2	$-2\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	
T3	$+4\lambda_{T3}(\lambda_{T3})^*$	

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{H2}(2\text{Tr}(y_2(y_2)^\dagger)+6\text{Tr}(Y_d(Y_d)^\dagger)+2\text{Tr}(Y_e(Y_e)^\dagger)+6\text{Tr}(Y_u(Y_u)^\dagger)+4\lambda_{S2}^{\text{Adj}})$	$-2\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+4\lambda_{T3}(\lambda_{T3})^*$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{H2}(2\text{Tr}(y_2(y_2)^\dagger)+6\text{Tr}(Y_d(Y_d)^\dagger)+2\text{Tr}(Y_e(Y_e)^\dagger)+6\text{Tr}(Y_u(Y_u)^\dagger)+4\lambda_{S2}^{\text{Adj}})$	$-2\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+4\lambda_{T3}(\lambda_{T3})^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{H2}(2\text{Tr}(y_2(y_2)^\dagger)+6\text{Tr}(Y_d(Y_d)^\dagger)+2\text{Tr}(Y_e(Y_e)^\dagger)+6\text{Tr}(Y_u(Y_u)^\dagger)+4\lambda_{S2}^{\text{Adj}})$	$-2\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+4\lambda_{T3}(\lambda_{T3})^*$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{H2}(2\text{Tr}(y_2(y_2)^\dagger)+6\text{Tr}(Y_d(Y_d)^\dagger)+2\text{Tr}(Y_e(Y_e)^\dagger)+6\text{Tr}(Y_u(Y_u)^\dagger)+4\lambda_{S2}^{\text{Adj}})$	$-2\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+4\lambda_{T3}(\lambda_{T3})^*$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$\lambda_{H2}(2\text{Tr}(y_2(y_2)^\dagger)+6\text{Tr}(Y_d(Y_d)^\dagger)+2\text{Tr}(Y_e(Y_e)^\dagger)+6\text{Tr}(Y_u(Y_u)^\dagger)+4\lambda_{S2}^{\text{Adj}})$	$-2\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+4\lambda_{T3}(\lambda_{T3})^*$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$-2\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+6\lambda_{T3}(\lambda_{T3})^*$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	$-2\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+6\lambda_{T3}(\lambda_{T3})^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-2\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+6\lambda_{T3}(\lambda_{T3})^*$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$-2\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+6\lambda_{T3}(\lambda_{T3})^*$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$-2\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+6\lambda_{T3}(\lambda_{T3})^*$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$-3\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+6\lambda_{T3}(\lambda_{T3})^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$-3\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+6\lambda_{T3}(\lambda_{T3})^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-3\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+6\lambda_{T3}(\lambda_{T3})^*$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$-3\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+6\lambda_{T3}(\lambda_{T3})^*$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$-3\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+6\lambda_{T3}(\lambda_{T3})^*$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-4\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+12\lambda_{T3}(\lambda_{T3})^*$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$-4\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+12\lambda_{T3}(\lambda_{T3})^*$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-4\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+12\lambda_{T3}(\lambda_{T3})^*$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$-4\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+12\lambda_{T3}(\lambda_{T3})^*$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$-4\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+12\lambda_{T3}(\lambda_{T3})^*$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{H2}(2\text{Tr}(y_2(y_2)^\dagger)+6\text{Tr}(Y_d(Y_d)^\dagger)+2\text{Tr}(Y_e(Y_e)^\dagger)+6\text{Tr}(Y_u(Y_u)^\dagger)+4\lambda_{S2}^{\text{Adj}})$	$-2\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+4\lambda_{T3}(\lambda_{T3})^*$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{H2}(2\text{Tr}(y_2(y_2)^\dagger)+6\text{Tr}(Y_d(Y_d)^\dagger)+2\text{Tr}(Y_e(Y_e)^\dagger)+6\text{Tr}(Y_u(Y_u)^\dagger)+4\lambda_{S2}^{\text{Adj}})$	$-2\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+4\lambda_{T3}(\lambda_{T3})^*$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{H2}(2\text{Tr}(y_2(y_2)^\dagger)+6\text{Tr}(Y_d(Y_d)^\dagger)+2\text{Tr}(Y_e(Y_e)^\dagger)+6\text{Tr}(Y_u(Y_u)^\dagger)+4\lambda_{S2}^{\text{Adj}})$	$-2\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+4\lambda_{T3}(\lambda_{T3})^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{H2}(2\text{Tr}(y_2(y_2)^\dagger) + 6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{S_2}^{Ad})$	$-2\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+4\lambda_{T3}(\lambda_{T3})^*$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\lambda_{H2}(2\text{Tr}(y_2(y_2)^\dagger) + 6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{S_2}^{Ad})$	$-2\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+4\lambda_{T3}(\lambda_{T3})^*$

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Column	Representative term structure
T1	$+6g_2^4$
T2	$-\frac{33g_2^2\lambda_{H2}}{2}$
T3	$-\frac{3g_Y^2\lambda_{H2}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+6g_2^4$	$-\frac{33g_2^2\lambda_{H2}}{2}$	$-\frac{3g_Y^2\lambda_{H2}}{2}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+6g_2^4$	$-\frac{33g_2^2\lambda_{H2}}{2}$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+6g_2^4$	$-\frac{33g_2^2\lambda_{H2}}{2}$	$-\frac{15g_Y^2\lambda_{H2}}{2}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+6g_2^4$	$-\frac{33g_2^2\lambda_{H2}}{2}$	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+6g_2^4$	$-\frac{33g_2^2\lambda_{H2}}{2}$	$-\frac{51g_Y^2\lambda_{H2}}{2}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	$-\frac{3g_Y^2\lambda_{H2}}{2}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$-\frac{15g_Y^2\lambda_{H2}}{2}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	$-\frac{51g_Y^2\lambda_{H2}}{2}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$-\frac{3g_Y^2\lambda_{H2}}{2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$-\frac{15g_Y^2\lambda_{H2}}{2}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$-\frac{51g_Y^2\lambda_{H2}}{2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-\frac{9g_2^2\lambda_{H2}}{2}$	$-\frac{3g_Y^2\lambda_{H2}}{2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$-\frac{9g_2^2\lambda_{H2}}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-\frac{9g_2^2\lambda_{H2}}{2}$	$-\frac{15g_Y^2\lambda_{H2}}{2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$-\frac{9g_2^2\lambda_{H2}}{2}$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$-\frac{9g_2^2\lambda_{H2}}{2}$	$-\frac{51g_Y^2\lambda_{H2}}{2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+6g_2^4$	$-\frac{33g_2^2\lambda_{H2}}{2}$	$-\frac{3g_Y^2\lambda_{H2}}{2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+6g_2^4$	$-\frac{33g_2^2\lambda_{H2}}{2}$	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6g_2^4$	$-\frac{33g_2^2\lambda_{H2}}{2}$	$-\frac{15g_Y^2\lambda_{H2}}{2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+6g_2^4$	$-\frac{33g_2^2\lambda_{H2}}{2}$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+6g_2^4$	$-\frac{33g_2^2\lambda_{H2}}{2}$	$-\frac{51g_Y^2\lambda_{H2}}{2}$

Terms 3 of 8

Column	Representative term structure
T1	$+2\lambda_{12}\lambda_{H1}$
T2	$+6\lambda_H\lambda_{H2}$
T3	$+4\lambda_{H2}^2$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+2\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+2\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+2\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+2\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+4\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+4\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+4\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+4\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+4\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+4\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+4\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+4\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+6\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+6\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+6\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+6\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+6\lambda_{12}\lambda_{H1}$	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$+6\lambda_H\lambda_{H2}$	$+4\lambda_{H2}^2$

Terms 4 of 8

Column	Representative term structure	
T1	$+8\lambda_{H2}\lambda_{S2}$	
T2	$+6\lambda_{H2}^{\text{Adj}^2}$	
T3	$+\frac{3g_Y^4}{4}$	

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+8\lambda_{H2}\lambda_{S2}$	$+6\lambda_{H2}^{\text{Adj}^2}$	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+8\lambda_{H2}\lambda_{S2}$	$+6\lambda_{H2}^{\text{Adj}^2}$	$+\frac{3g_Y^4}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{H2}\lambda_{S2}$	$+6\lambda_{H2}^{\text{Adj}^2}$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+8\lambda_{H2}\lambda_{S2}$	$+6\lambda_{H2}^{\text{Adj}^2}$	$+\frac{27g_Y^4}{4}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+8\lambda_{H2}\lambda_{S2}$	$+6\lambda_{H2}^{\text{Adj}^2}$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+6\lambda_{H2}\lambda_{S2}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{H2}\lambda_{S2}$	—	$+\frac{3g_Y^4}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+6\lambda_{H2}\lambda_{S2}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+6\lambda_{H2}\lambda_{S2}$	—	$+\frac{27g_Y^4}{4}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+6\lambda_{H2}\lambda_{S2}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+6\lambda_{H2}\lambda_{S2}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_{H2}\lambda_{S2}$	—	$+\frac{3g_Y^4}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+6\lambda_{H2}\lambda_{S2}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+6\lambda_{H2}\lambda_{S2}$	—	$+\frac{27g_Y^4}{4}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+6\lambda_{H2}\lambda_{S2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+4\lambda_{H2}\lambda_{S2}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+4\lambda_{H2}\lambda_{S2}$	—	$+\frac{3g_Y^4}{4}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+4\lambda_{H2}\lambda_{S2}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+4\lambda_{H2}\lambda_{S2}$	—	$+\frac{27g_Y^4}{4}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+4\lambda_{H2}\lambda_{S2}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+8\lambda_{H2}\lambda_{S2}$	$+6\lambda_{H2}^{\text{Adj}^2}$	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+8\lambda_{H2}\lambda_{S2}$	$+6\lambda_{H2}^{\text{Adj}^2}$	$+\frac{3g_Y^4}{4}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{H2}\lambda_{S2}$	$+6\lambda_{H2}^{\text{Adj}^2}$	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+8\lambda_{H2}\lambda_{S2}$	$+6\lambda_{H2}^{\text{Adj}^2}$	$+\frac{27g_Y^4}{4}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+8\lambda_{H2}\lambda_{S2}$	$+6\lambda_{H2}^{\text{Adj}^2}$	—

Terms 5 of 8

Column	Representative term structure									
T1	$-3g_Y^2\lambda_{H2}$									
T2	$+3g_Y^4$									
T3	$\lambda_{H2}(2\text{Tr}(y_2(y_2)^\dagger) + 6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$									

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-3g_Y^2\lambda_{H2}$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$+3g_Y^4$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-15g_Y^2\lambda_{H2}$	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	$+12g_Y^4$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	$\lambda_{H2}(2\text{Tr}(y_2(y_2)^\dagger) + 6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-3g_Y^2\lambda_{H2}$	—	$\lambda_{H2}(2\text{Tr}(y_2(y_2)^\dagger) + 6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$+3g_Y^4$	$\lambda_{H2}(2\text{Tr}(y_2(y_2)^\dagger) + 6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-15g_Y^2\lambda_{H2}$	—	$\lambda_{H2}(2\text{Tr}(y_2(y_2)^\dagger) + 6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$+12g_Y^4$	$\lambda_{H2}(2\text{Tr}(y_2(y_2)^\dagger) + 6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$\lambda_{H2}(3\text{Tr}(y_2(y_2)^\dagger) + 6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-3g_Y^2\lambda_{H2}$	—	$\lambda_{H2}(3\text{Tr}(y_2(y_2)^\dagger) + 6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$+3g_Y^4$	$\lambda_{H2}(3\text{Tr}(y_2(y_2)^\dagger) + 6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-15g_Y^2\lambda_{H2}$	—	$\lambda_{H2}(3\text{Tr}(y_2(y_2)^\dagger) + 6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$+12g_Y^4$	$\lambda_{H2}(3\text{Tr}(y_2(y_2)^\dagger) + 6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$\lambda_{H2}(4\text{Tr}(y_2(y_2)^\dagger) + 6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-3g_Y^2\lambda_{H2}$	—	$\lambda_{H2}(4\text{Tr}(y_2(y_2)^\dagger) + 6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$+3g_Y^4$	$\lambda_{H2}(4\text{Tr}(y_2(y_2)^\dagger) + 6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-15g_Y^2\lambda_{H2}$	—	$\lambda_{H2}(4\text{Tr}(y_2(y_2)^\dagger) + 6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$+12g_Y^4$	$\lambda_{H2}(4\text{Tr}(y_2(y_2)^\dagger) + 6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger))$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-3g_Y^2\lambda_{H2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+3g_Y^4$	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-15g_Y^2\lambda_{H2}$	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$+12g_Y^4$	—

Terms 6 of 8

Column	Representative term structure	
T1	$+\frac{9g_2^4}{4}$	
T2	$-9g_2^2\lambda_{H2}$	
T3	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H2}$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H2}$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H2}$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H2}$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H2}$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H2}$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H2}$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H2}$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H2}$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{H2}$	$+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 7 of 8

Column	Representative term structure
T1	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\frac{1}{2} \text{Tr} (y_1 (y_2)^\dagger) + 4 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + 3 \left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* \right)$
T2	$+\frac{1}{2} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \text{Tr} (y_2 (y_1)^\dagger)$
T3	$+3\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\frac{1}{2} \text{Tr} \left(y_1 \left(y_2 \right)^\dagger \right) + 4 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + 3 \left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* \right) + \frac{1}{2} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right)$	$+3\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\frac{3}{4} \text{Tr} \left(y_1 \left(y_2 \right)^\dagger \right) + 4 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + 3 \left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* \right) + \frac{3}{4} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right)$	$+3\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 8 of 8

Column	Representative term structure
T1	$+24\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$
T2	$+\lambda_{H1}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+24\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+24\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$+\lambda_{H1}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$+\lambda_{H1}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+\lambda_{H1}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$+\lambda_{H1}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$+\lambda_{H1}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$

$$\beta_{\lambda_{12}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{12}}{d\mu}=\beta_{\lambda_{12}}^{(1)}$$

Terms 1 of 12

Column	Representative term structure
T1	$\lambda_{12}(4\text{Tr}(y_1(y_1)^\dagger)+2\text{Tr}(y_2(y_2)^\dagger)+4\lambda_{S_2}^{\text{Adj}})$
T2	$-4\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$
T3	$+4\lambda_{T3}(\lambda_{T3})^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{12}(4\text{Tr}(y_1(y_1)^\dagger)+2\text{Tr}(y_2(y_2)^\dagger)+4\lambda_{S_2}^{\text{Adj}})$	$-4\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+4\lambda_{T3}(\lambda_{T3})^*$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{12}(4\text{Tr}(y_1(y_1)^\dagger)+2\text{Tr}(y_2(y_2)^\dagger)+4\lambda_{S_2}^{\text{Adj}})$	$-4\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+4\lambda_{T3}(\lambda_{T3})^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{12}(4\text{Tr}(y_1(y_1)^\dagger)+2\text{Tr}(y_2(y_2)^\dagger)+4\lambda_{S_2}^{\text{Adj}})$	$-4\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+4\lambda_{T3}(\lambda_{T3})^*$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{12}(4\text{Tr}(y_1(y_1)^\dagger)+2\text{Tr}(y_2(y_2)^\dagger)+4\lambda_{S_2}^{\text{Adj}})$	$-4\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+4\lambda_{T3}(\lambda_{T3})^*$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$\lambda_{12}(4\text{Tr}(y_1(y_1)^\dagger)+2\text{Tr}(y_2(y_2)^\dagger)+4\lambda_{S_2}^{\text{Adj}})$	$-4\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+4\lambda_{T3}(\lambda_{T3})^*$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$-2\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+3\lambda_{T3}(\lambda_{T3})^*$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	$-2\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+3\lambda_{T3}(\lambda_{T3})^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-2\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+3\lambda_{T3}(\lambda_{T3})^*$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$-2\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+3\lambda_{T3}(\lambda_{T3})^*$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$-2\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+3\lambda_{T3}(\lambda_{T3})^*$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$+3\lambda_{T3}(\lambda_{T3})^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$+3\lambda_{T3}(\lambda_{T3})^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$+3\lambda_{T3}(\lambda_{T3})^*$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$+3\lambda_{T3}(\lambda_{T3})^*$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$+3\lambda_{T3}(\lambda_{T3})^*$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-4\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+4\lambda_{T3}(\lambda_{T3})^*$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$-4\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+4\lambda_{T3}(\lambda_{T3})^*$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-4\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+4\lambda_{T3}(\lambda_{T3})^*$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$-4\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+4\lambda_{T3}(\lambda_{T3})^*$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$-4\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+4\lambda_{T3}(\lambda_{T3})^*$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$-2\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-2\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-2\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$-2\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$-2 \operatorname{Tr} \left(y_1 \left(y_1 \right)^\dagger y_2 \left(y_2 \right)^\dagger \right)$	—

Terms 2 of 12

Column	Representative term structure	
T1	$-12g_2^2\lambda_{12}$	
T2	$-6g_Y^2\lambda_{12}$	
T3	$+4\lambda_{12}^2$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-12g_2^2\lambda_{12}$	$-6g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-12g_2^2\lambda_{12}$	$-3g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-12g_2^2\lambda_{12}$	$-6g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-12g_2^2\lambda_{12}$	$-15g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-12g_2^2\lambda_{12}$	$-30g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-9g_2^2\lambda_{12}$	$-6g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{12}$	$-3g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-9g_2^2\lambda_{12}$	$-6g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-9g_2^2\lambda_{12}$	$-15g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$-9g_2^2\lambda_{12}$	$-30g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-9g_2^2\lambda_{12}$	$-6g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_{12}$	$-3g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-9g_2^2\lambda_{12}$	$-6g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-9g_2^2\lambda_{12}$	$-15g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-9g_2^2\lambda_{12}$	$-30g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-12g_2^2\lambda_{12}$	$-6g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-12g_2^2\lambda_{12}$	$-3g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-12g_2^2\lambda_{12}$	$-6g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-12g_2^2\lambda_{12}$	$-15g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-12g_2^2\lambda_{12}$	$-30g_Y^2\lambda_{12}$	$+4\lambda_{12}^2$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	$+4\lambda_{12}^2$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	$+4\lambda_{12}^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$+4\lambda_{12}^2$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	$+4\lambda_{12}^2$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	$+4\lambda_{12}^2$

Terms 3 of 12

Column	Representative term structure
T1	$+4\lambda_{12}\lambda_{S_1}$
T2	$+8\lambda_{12}\lambda_{S_2}$
T3	$+4\lambda_{H1}\lambda_{H2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+4\lambda_{12}\lambda_{S_1}$	$+8\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+4\lambda_{12}\lambda_{S_1}$	$+8\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{12}\lambda_{S_1}$	$+8\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+4\lambda_{12}\lambda_{S_1}$	$+8\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+4\lambda_{12}\lambda_{S_1}$	$+8\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+6\lambda_{12}\lambda_{S_1}$	$+6\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{12}\lambda_{S_1}$	$+6\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+6\lambda_{12}\lambda_{S_1}$	$+6\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+6\lambda_{12}\lambda_{S_1}$	$+6\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+6\lambda_{12}\lambda_{S_1}$	$+6\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+6\lambda_{12}\lambda_{S_1}$	$+6\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_{12}\lambda_{S_1}$	$+6\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+6\lambda_{12}\lambda_{S_1}$	$+6\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+6\lambda_{12}\lambda_{S_1}$	$+6\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+6\lambda_{12}\lambda_{S_1}$	$+6\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+8\lambda_{12}\lambda_{S_1}$	$+4\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+8\lambda_{12}\lambda_{S_1}$	$+4\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+8\lambda_{12}\lambda_{S_1}$	$+4\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+8\lambda_{12}\lambda_{S_1}$	$+4\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+8\lambda_{12}\lambda_{S_1}$	$+4\lambda_{12}\lambda_{S_2}$	$+4\lambda_{H1}\lambda_{H2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	$+4\lambda_{H1}\lambda_{H2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	$+4\lambda_{H1}\lambda_{H2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$+4\lambda_{H1}\lambda_{H2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	$+4\lambda_{H1}\lambda_{H2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	$+4\lambda_{H1}\lambda_{H2}$

Terms 4 of 12

Column	Representative term structure
T1	$-4 \text{Tr} \left(y_1 \left(y_2 \right)^\dagger y_2 \left(y_1 \right)^\dagger \right)$
T2	$+\frac{3g_Y^4}{4}$
T3	$+48g_Y^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-4 \text{Tr} \left(y_1 \left(y_2 \right)^\dagger y_2 \left(y_1 \right)^\dagger \right)$	$+\frac{3g_Y^4}{4}$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$+\frac{27g_Y^4}{4}$	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	$+48g_Y^4$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-4 \text{Tr} \left(y_1 \left(y_2 \right)^\dagger y_2 \left(y_1 \right)^\dagger \right)$	$+\frac{3g_Y^4}{4}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$+\frac{27g_Y^4}{4}$	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	$+48g_Y^4$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-3 \text{Tr} \left(y_1 \left(y_2 \right)^\dagger y_2 \left(y_1 \right)^\dagger \right)$	$+\frac{3g_Y^4}{4}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$+\frac{27g_Y^4}{4}$	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$+48g_Y^4$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-4 \text{Tr} \left(y_1 \left(y_2 \right)^\dagger y_2 \left(y_1 \right)^\dagger \right)$	$+\frac{3g_Y^4}{4}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$+\frac{27g_Y^4}{4}$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	$+48g_Y^4$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-2 \text{Tr} \left(y_1 \left(y_2 \right)^\dagger y_2 \left(y_1 \right)^\dagger \right)$	$+\frac{3g_Y^4}{4}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$+\frac{27g_Y^4}{4}$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	$+48g_Y^4$

Terms 5 of 12

Column	Representative term structure	
T1	$\lambda_{12}(2 \text{Tr}(y_1 (y_1)^\dagger) + 2 \text{Tr}(y_2 (y_2)^\dagger))$	
T2	$+\frac{9g_2^4}{4}$	
T3	$+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$\lambda_{12}(2 \text{Tr}(y_1 (y_1)^\dagger) + 2 \text{Tr}(y_2 (y_2)^\dagger))$	$+\frac{9g_2^4}{4}$	$+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{12}(2 \text{Tr}(y_1 (y_1)^\dagger) + 2 \text{Tr}(y_2 (y_2)^\dagger))$	$+\frac{9g_2^4}{4}$	$+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\lambda_{12}(2 \text{Tr}(y_1 (y_1)^\dagger) + 2 \text{Tr}(y_2 (y_2)^\dagger))$	$+\frac{9g_2^4}{4}$	$+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$\lambda_{12}(2 \text{Tr}(y_1 (y_1)^\dagger) + 2 \text{Tr}(y_2 (y_2)^\dagger))$	$+\frac{9g_2^4}{4}$	$+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$\lambda_{12}(2 \text{Tr}(y_1 (y_1)^\dagger) + 2 \text{Tr}(y_2 (y_2)^\dagger))$	$+\frac{9g_2^4}{4}$	$+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$\lambda_{12}(3 \text{Tr}(y_1 (y_1)^\dagger) + 3 \text{Tr}(y_2 (y_2)^\dagger))$	$+\frac{9g_2^4}{4}$	$+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{12}(3 \text{Tr}(y_1 (y_1)^\dagger) + 3 \text{Tr}(y_2 (y_2)^\dagger))$	$+\frac{9g_2^4}{4}$	$+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\lambda_{12}(3 \text{Tr}(y_1 (y_1)^\dagger) + 3 \text{Tr}(y_2 (y_2)^\dagger))$	$+\frac{9g_2^4}{4}$	$+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\lambda_{12}(3 \text{Tr}(y_1 (y_1)^\dagger) + 3 \text{Tr}(y_2 (y_2)^\dagger))$	$+\frac{9g_2^4}{4}$	$+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$\lambda_{12}(3 \text{Tr}(y_1 (y_1)^\dagger) + 3 \text{Tr}(y_2 (y_2)^\dagger))$	$+\frac{9g_2^4}{4}$	$+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 6 of 12

Column	Representative term structure
T1	$+\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left(\frac{3}{2} \text{Tr}(y_1 (y_2)^\dagger) + 9(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*) \right)$
T2	$+\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left(\frac{3}{2} \text{Tr}(y_1 (y_2)^\dagger) + 9(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*) \right)$
T3	$+\frac{3}{2}(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*) \text{Tr}(y_2 (y_1)^\dagger)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left(\frac{3}{2} \text{Tr}(y_1 (y_2)^\dagger) + 9(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*) \right) + \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left(\frac{3}{2} \text{Tr}(y_1 (y_2)^\dagger) + 9(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*) \right) + \frac{3}{2}(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*) \text{Tr}(y_2 (y_1)^\dagger)$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left(\frac{9}{4} \text{Tr}(y_1 (y_2)^\dagger) + 9(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*) \right) + \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left(\frac{9}{4} \text{Tr}(y_1 (y_2)^\dagger) + 9(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*) \right) + \frac{9}{4}(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*) \text{Tr}(y_2 (y_1)^\dagger)$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 7 of 12

Column	Representative term structure
T1	$+2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$
T2	$+24\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$
T3	$-\frac{9}{2} \text{Tr} \left(y_1 (y_1)^\dagger y_2 (y_2)^\dagger \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	$+24\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$-\frac{9}{2} \text{Tr} \left(y_1 (y_1)^\dagger y_2 (y_2)^\dagger \right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	$+24\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$	$-\frac{9}{2} \text{Tr} \left(y_1 (y_1)^\dagger y_2 (y_2)^\dagger \right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$-\frac{9}{2} \text{Tr} \left(y_1 (y_1)^\dagger y_2 (y_2)^\dagger \right)$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$-\frac{9}{2} \text{Tr} \left(y_1 (y_1)^\dagger y_2 (y_2)^\dagger \right)$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$-\frac{9}{2} \text{Tr} \left(y_1 (y_1)^\dagger y_2 (y_2)^\dagger \right)$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 8 of 12

Column	Representative term structure									
T1	$\lambda_{12}(2 \text{Tr}(y_1 (y_1)^\dagger) + 4 \text{Tr}(y_2 (y_2)^\dagger) + 4\lambda_{S_1}^{\text{Adj}})$									
T2	$\lambda_{12} \left(2 \text{Tr}(y_1 (y_1)^\dagger) + 2 \text{Tr}(y_2 (y_2)^\dagger) + \frac{16\lambda_{12}^{\text{Cross}}}{3} + 4\lambda_{S_1}^{\text{Adj}} + 4\lambda_{S_2}^{\text{Adj}} \right)$									
T3	$+\lambda_{12}^{\text{Cross}} \left(\frac{4}{3} \text{Tr}(y_1 (y_1)^\dagger) + \frac{4}{3} \text{Tr}(y_2 (y_2)^\dagger) \right)$									

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_{12}(2 \text{Tr}(y_1 (y_1)^\dagger) + 4 \text{Tr}(y_2 (y_2)^\dagger) + 4\lambda_{S_1}^{\text{Adj}})$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$\lambda_{12}(2 \text{Tr}(y_1 (y_1)^\dagger) + 4 \text{Tr}(y_2 (y_2)^\dagger) + 4\lambda_{S_1}^{\text{Adj}})$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\lambda_{12}(2 \text{Tr}(y_1 (y_1)^\dagger) + 4 \text{Tr}(y_2 (y_2)^\dagger) + 4\lambda_{S_1}^{\text{Adj}})$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$\lambda_{12}(2 \text{Tr}(y_1 (y_1)^\dagger) + 4 \text{Tr}(y_2 (y_2)^\dagger) + 4\lambda_{S_1}^{\text{Adj}})$	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$\lambda_{12}(2 \text{Tr}(y_1 (y_1)^\dagger) + 4 \text{Tr}(y_2 (y_2)^\dagger) + 4\lambda_{S_1}^{\text{Adj}})$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$\lambda_{12} \left(2 \text{Tr}(y_1 (y_1)^\dagger) + 2 \text{Tr}(y_2 (y_2)^\dagger) + \frac{16\lambda_{12}^{\text{Cross}}}{3} + 4\lambda_{S_1}^{\text{Adj}} + 4\lambda_{S_2}^{\text{Adj}} \right) + \lambda_{12}^{\text{Cross}} \left(\frac{4}{3} \text{Tr}(y_1 (y_1)^\dagger) + \frac{4}{3} \text{Tr}(y_2 (y_2)^\dagger) \right)$	
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$\lambda_{12} \left(2 \text{Tr}(y_1 (y_1)^\dagger) + 2 \text{Tr}(y_2 (y_2)^\dagger) + \frac{16\lambda_{12}^{\text{Cross}}}{3} + 4\lambda_{S_1}^{\text{Adj}} + 4\lambda_{S_2}^{\text{Adj}} \right) + \lambda_{12}^{\text{Cross}} \left(\frac{4}{3} \text{Tr}(y_1 (y_1)^\dagger) + \frac{4}{3} \text{Tr}(y_2 (y_2)^\dagger) \right)$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$\lambda_{12} \left(2 \text{Tr}(y_1 (y_1)^\dagger) + 2 \text{Tr}(y_2 (y_2)^\dagger) + \frac{16\lambda_{12}^{\text{Cross}}}{3} + 4\lambda_{S_1}^{\text{Adj}} + 4\lambda_{S_2}^{\text{Adj}} \right) + \lambda_{12}^{\text{Cross}} \left(\frac{4}{3} \text{Tr}(y_1 (y_1)^\dagger) + \frac{4}{3} \text{Tr}(y_2 (y_2)^\dagger) \right)$	
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$\lambda_{12} \left(2 \text{Tr}(y_1 (y_1)^\dagger) + 2 \text{Tr}(y_2 (y_2)^\dagger) + \frac{16\lambda_{12}^{\text{Cross}}}{3} + 4\lambda_{S_1}^{\text{Adj}} + 4\lambda_{S_2}^{\text{Adj}} \right) + \lambda_{12}^{\text{Cross}} \left(\frac{4}{3} \text{Tr}(y_1 (y_1)^\dagger) + \frac{4}{3} \text{Tr}(y_2 (y_2)^\dagger) \right)$	
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$\lambda_{12} \left(2 \text{Tr}(y_1 (y_1)^\dagger) + 2 \text{Tr}(y_2 (y_2)^\dagger) + \frac{16\lambda_{12}^{\text{Cross}}}{3} + 4\lambda_{S_1}^{\text{Adj}} + 4\lambda_{S_2}^{\text{Adj}} \right) + \lambda_{12}^{\text{Cross}} \left(\frac{4}{3} \text{Tr}(y_1 (y_1)^\dagger) + \frac{4}{3} \text{Tr}(y_2 (y_2)^\dagger) \right)$	

Terms 9 of 12

Column	Representative term structure	
T1	$+\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$	
T2	$+16g_2^4$	
T3	$+g_2^2(-24\lambda_{12}-16\lambda_{12}^{\text{Cross}})$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$	$+16g_2^4$	$+g_2^2(-24\lambda_{12}-16\lambda_{12}^{\text{Cross}})$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$	$+16g_2^4$	$+g_2^2(-24\lambda_{12}-16\lambda_{12}^{\text{Cross}})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$	$+16g_2^4$	$+g_2^2(-24\lambda_{12}-16\lambda_{12}^{\text{Cross}})$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$	$+16g_2^4$	$+g_2^2(-24\lambda_{12}-16\lambda_{12}^{\text{Cross}})$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$	$+16g_2^4$	$+g_2^2(-24\lambda_{12}-16\lambda_{12}^{\text{Cross}})$

Terms 10 of 12

Column	Representative term structure	
T1	$+g_Y{}^2(-6\lambda_{12} - 4\lambda_{12}^{\text{Cross}})$	
T2	$+\lambda_{S_1}\left(8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$	
T3	$+\lambda_{S_2}\left(8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+g_Y{}^2(-6\lambda_{12} - 4\lambda_{12}^{\text{Cross}})$	$+\lambda_{S_1}\left(8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$	$+\lambda_{S_2}\left(8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+g_Y{}^2(-3\lambda_{12} - 2\lambda_{12}^{\text{Cross}})$	$+\lambda_{S_1}\left(8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$	$+\lambda_{S_2}\left(8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_Y{}^2(-6\lambda_{12} - 4\lambda_{12}^{\text{Cross}})$	$+\lambda_{S_1}\left(8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$	$+\lambda_{S_2}\left(8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+g_Y{}^2(-15\lambda_{12} - 10\lambda_{12}^{\text{Cross}})$	$+\lambda_{S_1}\left(8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$	$+\lambda_{S_2}\left(8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+g_Y{}^2(-30\lambda_{12} - 20\lambda_{12}^{\text{Cross}})$	$+\lambda_{S_1}\left(8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$	$+\lambda_{S_2}\left(8\lambda_{12} + \frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$

Terms 11 of 12

Column	Representative term structure
T1	$+16\lambda_{12}^{\text{Adj}2}$
T2	$+\frac{32\lambda_{12}^{\text{Cross}2}}{3}$
T3	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+16\lambda_{12}^{\text{Adj}2}$	$+\frac{32\lambda_{12}^{\text{Cross}2}}{3}$	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+16\lambda_{12}^{\text{Adj}2}$	$+\frac{32\lambda_{12}^{\text{Cross}2}}{3}$	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+16\lambda_{12}^{\text{Adj}2}$	$+\frac{32\lambda_{12}^{\text{Cross}2}}{3}$	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+16\lambda_{12}^{\text{Adj}2}$	$+\frac{32\lambda_{12}^{\text{Cross}2}}{3}$	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+16\lambda_{12}^{\text{Adj}2}$	$+\frac{32\lambda_{12}^{\text{Cross}2}}{3}$	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$

Column	Representative term structure
T1	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$

$$\beta_{\lambda_{T3}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{T3}}{d\mu}=\beta_{\lambda_{T3}}^{(1)}$$

Terms 1 of 6

Column	Representative term structure									
T1	$\lambda_{T3}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+\text{Tr}\big(y_2\ (y_2)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda_{H2}^{\text{Adj}}\big)$									
T2	$-\frac{21g_2^2\lambda_{T3}}{2}$									
T3	$-\frac{9g_Y^2\lambda_{T3}}{2}$									

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{T3}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+\text{Tr}\big(y_2\ (y_2)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda_{H2}^{\text{Adj}}\big)$	$-\frac{21g_2^2\lambda_{T3}}{2}$	$-\frac{9g_Y^2\lambda_{T3}}{2}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{T3}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+\text{Tr}\big(y_2\ (y_2)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda_{H2}^{\text{Adj}}\big)$	$-\frac{21g_2^2\lambda_{T3}}{2}$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{T3}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+\text{Tr}\big(y_2\ (y_2)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda_{H2}^{\text{Adj}}\big)$	$-\frac{21g_2^2\lambda_{T3}}{2}$	$-\frac{9g_Y^2\lambda_{T3}}{2}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{T3}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+\text{Tr}\big(y_2\ (y_2)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda_{H2}^{\text{Adj}}\big)$	$-\frac{21g_2^2\lambda_{T3}}{2}$	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$\lambda_{T3}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+\text{Tr}\big(y_2\ (y_2)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda_{H2}^{\text{Adj}}\big)$	$-\frac{21g_2^2\lambda_{T3}}{2}$	$-\frac{33g_Y^2\lambda_{T3}}{2}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	$-\frac{9g_Y^2\lambda_{T3}}{2}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$-\frac{9g_Y^2\lambda_{T3}}{2}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	$-\frac{33g_Y^2\lambda_{T3}}{2}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$-\frac{9g_Y^2\lambda_{T3}}{2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$-\frac{9g_Y^2\lambda_{T3}}{2}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$-\frac{33g_Y^2\lambda_{T3}}{2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-\frac{21g_2^2\lambda_{T3}}{2}$	$-\frac{9g_Y^2\lambda_{T3}}{2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$-\frac{21g_2^2\lambda_{T3}}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-\frac{21g_2^2\lambda_{T3}}{2}$	$-\frac{9g_Y^2\lambda_{T3}}{2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$-\frac{21g_2^2\lambda_{T3}}{2}$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$-\frac{21g_2^2\lambda_{T3}}{2}$	$-\frac{33g_Y^2\lambda_{T3}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$-\frac{33g_2^2\lambda_{T3}}{2}$	$-\frac{9g_Y^2\lambda_{T3}}{2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-\frac{33g_2^2\lambda_{T3}}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{33g_2^2\lambda_{T3}}{2}$	$-\frac{9g_Y^2\lambda_{T3}}{2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$-\frac{33g_2^2\lambda_{T3}}{2}$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$-\frac{33g_2^2\lambda_{T3}}{2}$	$-\frac{33g_Y^2\lambda_{T3}}{2}$

Terms 2 of 6

Column	Representative term structure
T1	$+2\lambda_{12}\lambda_{T3}$
T2	$+2\lambda_H\lambda_{T3}$
T3	$+4\lambda_{H1}\lambda_{T3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+2\lambda_{12}\lambda_{T3}$	$+2\lambda_H\lambda_{T3}$	$+4\lambda_{H1}\lambda_{T3}$

Terms 3 of 6

Column	Representative term structure
T1	$+4\lambda_{H2}\lambda_{T3}$
T2	$-3g_Y{}^2\lambda_{T3}$
T3	$\lambda_{T3}\left(\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+4\lambda_{H2}\lambda_{T3}$	$-3g_Y{}^2\lambda_{T3}$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+4\lambda_{H2}\lambda_{T3}$	$-9g_Y{}^2\lambda_{T3}$	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	$\lambda_{T3}\left(\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{H2}\lambda_{T3}$	$-3g_Y{}^2\lambda_{T3}$	$\lambda_{T3}\left(\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	$\lambda_{T3}\left(\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+4\lambda_{H2}\lambda_{T3}$	$-9g_Y{}^2\lambda_{T3}$	$\lambda_{T3}\left(\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	$\lambda_{T3}\left(\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{H2}\lambda_{T3}$	$-3g_Y{}^2\lambda_{T3}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+4\lambda_{H2}\lambda_{T3}$	$-9g_Y{}^2\lambda_{T3}$	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+4\lambda_{H2}\lambda_{T3}$	$-3g_Y{}^2\lambda_{T3}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+4\lambda_{H2}\lambda_{T3}$	$-9g_Y{}^2\lambda_{T3}$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+4\lambda_{H2}\lambda_{T3}$	$-3g_Y{}^2\lambda_{T3}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+4\lambda_{H2}\lambda_{T3}$	$-9g_Y{}^2\lambda_{T3}$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+4\lambda_{H2}\lambda_{T3}$	—	—

Terms 4 of 6

Column	Representative term structure	
T1	$-9g_2^2\lambda_{T3}$	
T2	$+2\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$	
T3	$+2\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-9g_2^2\lambda_{T3}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{T3}$	$+2\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$	$+2\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-9g_2^2\lambda_{T3}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-9g_2^2\lambda_{T3}$	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$-9g_2^2\lambda_{T3}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-9g_2^2\lambda_{T3}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_{T3}$	$+3\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$	$+3\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-9g_2^2\lambda_{T3}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-9g_2^2\lambda_{T3}$	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-9g_2^2\lambda_{T3}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 5 of 6

Column	Representative term structure	
T1	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$	
T2	$+4\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$	
T3	$+4 \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* \left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}} \right)^* - \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$	$+4\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$	$+4 \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* \left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}} \right)^* - \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$	$+4\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$	$+4 \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* \left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}} \right)^* - \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 6 of 6

Column	Representative term structure
T1	$\lambda_{T3} \left(\frac{3}{2} \text{Tr}(y_1 (y_1)^\dagger) + \frac{3}{2} \text{Tr}(y_2 (y_2)^\dagger) + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + \frac{\lambda_{12}^{\text{Adj}}}{6} + \frac{2\lambda_{H1}^{\text{Adj}}}{3} + 2\lambda_{H2}^{\text{Adj}} \right)$
T2	$\lambda_{T3} (\text{Tr}(y_1 (y_1)^\dagger) + 2 \text{Tr}(y_2 (y_2)^\dagger) + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) - 4\lambda_{H1}^{\text{Adj}})$
T3	$\lambda_{T3} \left(\text{Tr}(y_1 (y_1)^\dagger) + \text{Tr}(y_2 (y_2)^\dagger) + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) - \frac{4\lambda_{12}^{\text{Adj}}}{3} - 2\lambda_{12}^{\text{Cross}} - 2\lambda_{H1}^{\text{Adj}} - \frac{4\lambda_{H2}^{\text{Adj}}}{3} \right)$

[illegible]

$$\beta_{\lambda_{12}^{\text{Adj}}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{12}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{12}^{\text{Adj}}}^{(1)}$$

Terms 1 of 8

Column	Representative term structure	
T1	$\lambda_{12}^{\text{Adj}}(2\text{Tr}(y_1(y_1)^\dagger)+2\text{Tr}(y_2(y_2)^\dagger))$	
T2	$-\frac{8}{3}\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	
T3	$+\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$\lambda_{12}^{\text{Adj}}(2\text{Tr}(y_1(y_1)^\dagger)+2\text{Tr}(y_2(y_2)^\dagger))$	$-\frac{8}{3}\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{12}^{\text{Adj}}(2\text{Tr}(y_1(y_1)^\dagger)+2\text{Tr}(y_2(y_2)^\dagger))$	$-\frac{8}{3}\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\lambda_{12}^{\text{Adj}}(2\text{Tr}(y_1(y_1)^\dagger)+2\text{Tr}(y_2(y_2)^\dagger))$	$-\frac{8}{3}\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$\lambda_{12}^{\text{Adj}}(2\text{Tr}(y_1(y_1)^\dagger)+2\text{Tr}(y_2(y_2)^\dagger))$	$-\frac{8}{3}\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$\lambda_{12}^{\text{Adj}}(2\text{Tr}(y_1(y_1)^\dagger)+2\text{Tr}(y_2(y_2)^\dagger))$	$-\frac{8}{3}\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$\lambda_{12}^{\text{Adj}}(3\text{Tr}(y_1(y_1)^\dagger)+3\text{Tr}(y_2(y_2)^\dagger))$	$-\frac{2}{3}\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{12}^{\text{Adj}}(3\text{Tr}(y_1(y_1)^\dagger)+3\text{Tr}(y_2(y_2)^\dagger))$	$-\frac{2}{3}\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\lambda_{12}^{\text{Adj}}(3\text{Tr}(y_1(y_1)^\dagger)+3\text{Tr}(y_2(y_2)^\dagger))$	$-\frac{2}{3}\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\lambda_{12}^{\text{Adj}}(3\text{Tr}(y_1(y_1)^\dagger)+3\text{Tr}(y_2(y_2)^\dagger))$	$-\frac{2}{3}\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$\lambda_{12}^{\text{Adj}}(3\text{Tr}(y_1(y_1)^\dagger)+3\text{Tr}(y_2(y_2)^\dagger))$	$-\frac{2}{3}\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$+\frac{4}{3}\lambda_{T3}(\lambda_{T3})^*$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$+\frac{2}{3}\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$-\frac{2}{9}\lambda_{T3}(\lambda_{T3})^*$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$+\frac{2}{3}\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$-\frac{2}{9}\lambda_{T3}(\lambda_{T3})^*$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+\frac{2}{3}\text{Tr}(y_1(y_1)^\dagger y_2(y_2)^\dagger)$	$-\frac{2}{9}\lambda_{T3}(\lambda_{T3})^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$+\frac{2}{3}\text{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$	$-\frac{2}{9}\lambda_{T3}\ (\lambda_{T3})^*$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$+\frac{2}{3}\text{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$	$-\frac{2}{9}\lambda_{T3}\ (\lambda_{T3})^*$

Terms 2 of 8

Column	Representative term structure	
T1	$-9g_2^2\lambda_{12}^{\text{Adj}}$	
T2	$-6g_Y^2\lambda_{12}^{\text{Adj}}$	
T3	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-9g_2^2\lambda_{12}^{\text{Adj}}$	$-6g_Y^2\lambda_{12}^{\text{Adj}}$	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{12}^{\text{Adj}}$	—	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-9g_2^2\lambda_{12}^{\text{Adj}}$	$-6g_Y^2\lambda_{12}^{\text{Adj}}$	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-9g_2^2\lambda_{12}^{\text{Adj}}$	—	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$-9g_2^2\lambda_{12}^{\text{Adj}}$	—	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-9g_2^2\lambda_{12}^{\text{Adj}}$	$-6g_Y^2\lambda_{12}^{\text{Adj}}$	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_{12}^{\text{Adj}}$	—	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-9g_2^2\lambda_{12}^{\text{Adj}}$	$-6g_Y^2\lambda_{12}^{\text{Adj}}$	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-9g_2^2\lambda_{12}^{\text{Adj}}$	—	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-9g_2^2\lambda_{12}^{\text{Adj}}$	—	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-24g_2^2\lambda_{12}^{\text{Adj}}$	$-6g_Y^2\lambda_{12}^{\text{Adj}}$	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-24g_2^2\lambda_{12}^{\text{Adj}}$	—	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-24g_2^2\lambda_{12}^{\text{Adj}}$	$-6g_Y^2\lambda_{12}^{\text{Adj}}$	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-24g_2^2\lambda_{12}^{\text{Adj}}$	—	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-24g_2^2\lambda_{12}^{\text{Adj}}$	—	$+8\lambda_{12}\lambda_{12}^{\text{Adj}}$

Terms 3 of 8

Column	Representative term structure	
T1	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	
T2	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$	
T3	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$	$+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$	$+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$	—

Terms 4 of 8

Column	Representative term structure	
T1	$+\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left(\frac{2}{3} \text{Tr}(y_1 (y_2)^\dagger) + \frac{4}{3} (5 \left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*) \right)$	
T2	$+\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left(\frac{2}{3} \text{Tr}(y_1 (y_2)^\dagger) + \frac{4}{3} \left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + 5 \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \right)$	
T3	$+\frac{2}{3} \left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \text{Tr}(y_2 (y_1)^\dagger)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left(\frac{2}{3} \text{Tr}(y_1 (y_2)^\dagger) + \frac{4}{3} (5 \left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*) \right) + \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left(\frac{2}{3} \text{Tr}(y_1 (y_2)^\dagger) + \frac{4}{3} \left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + 5 \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \right) + \frac{2}{3} \left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \text{Tr}(y_2 (y_1)^\dagger)$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 5 of 8

Column	Representative term structure	
T1	$+\frac{64}{3}\lambda_{S_1^\dagger 2 S_2^\dagger 2}\left(\lambda_{S_1^\dagger 2 S_2^\dagger 2}\right)^*$	
T2	$+g_Y{}^2\left(-6g_2{}^2-3\lambda_{12}^{\text{Adj}}\right)$	
T3	$+g_Y{}^2\left(18g_2{}^2-15\lambda_{12}^{\text{Adj}}\right)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{64}{3}\lambda_{S_1^\dagger 2 S_2^\dagger 2}\left(\lambda_{S_1^\dagger 2 S_2^\dagger 2}\right)^*$	$+g_Y{}^2\left(-6g_2{}^2-3\lambda_{12}^{\text{Adj}}\right)$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	$+g_Y{}^2\left(18g_2{}^2-15\lambda_{12}^{\text{Adj}}\right)$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	$+g_Y{}^2\left(48g_2{}^2-30\lambda_{12}^{\text{Adj}}\right)$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{64}{3}\lambda_{S_1^\dagger 2 S_2^\dagger 2}\left(\lambda_{S_1^\dagger 2 S_2^\dagger 2}\right)^*$	$+g_Y{}^2\left(-6g_2{}^2-3\lambda_{12}^{\text{Adj}}\right)$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$+g_Y{}^2\left(18g_2{}^2-15\lambda_{12}^{\text{Adj}}\right)$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$+g_Y{}^2\left(48g_2{}^2-30\lambda_{12}^{\text{Adj}}\right)$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$+g_Y{}^2\left(-2g_2{}^2-3\lambda_{12}^{\text{Adj}}\right)$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	$+g_Y{}^2\left(6g_2{}^2-15\lambda_{12}^{\text{Adj}}\right)$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	$+g_Y{}^2\left(16g_2{}^2-30\lambda_{12}^{\text{Adj}}\right)$

Terms 6 of 8

Column	Representative term structure	
T1	$+ \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left(\text{Tr}(y_1 (y_2)^\dagger) + \frac{4}{3} (5 \left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*) \right)$	
T2	$+ \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left(\text{Tr}(y_1 (y_2)^\dagger) + \frac{4}{3} \left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + 5 \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \right)$	
T3	$+ \left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \text{Tr}(y_2 (y_1)^\dagger)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left(\text{Tr}(y_1 (y_2)^\dagger) + \frac{4}{3} (5 \left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^*) \right) + \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left(\text{Tr}(y_1 (y_2)^\dagger) + \frac{4}{3} \left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + 5 \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \right) + \left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \text{Tr}(y_2 (y_1)^\dagger)$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 7 of 8

Column	Representative term structure
T1	$-\frac{8}{3}\text{Tr}\left(y_1\left(y_2\right)^\dagger y_2\left(y_1\right)^\dagger\right)$
T2	$\lambda_{12}^{\text{Adj}}\left(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$
T3	$+16\lambda_{12}^{\text{Adj}}\lambda_{S_1}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{8}{3}\text{Tr}\left(y_1\left(y_2\right)^\dagger y_2\left(y_1\right)^\dagger\right)$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$\lambda_{12}^{\text{Adj}}\left(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)+16\lambda_{12}^{\text{Adj}}\lambda_{S_1}^{\text{Adj}}$	
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{2}{3}\text{Tr}\left(y_1\left(y_2\right)^\dagger y_2\left(y_1\right)^\dagger\right)$	$\lambda_{12}^{\text{Adj}}\left(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)+16\lambda_{12}^{\text{Adj}}\lambda_{S_1}^{\text{Adj}}$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$\lambda_{12}^{\text{Adj}}\left(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)+16\lambda_{12}^{\text{Adj}}\lambda_{S_1}^{\text{Adj}}$	
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$\lambda_{12}^{\text{Adj}}\left(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)+16\lambda_{12}^{\text{Adj}}\lambda_{S_1}^{\text{Adj}}$	
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$\lambda_{12}^{\text{Adj}}\left(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)+16\lambda_{12}^{\text{Adj}}\lambda_{S_1}^{\text{Adj}}$	

Terms 8 of 8

Column	Representative term structure
T1	$+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$
T2	$+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$	$+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$	$+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$	$+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$	$+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$	$+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$

$$\beta_{\lambda_{12}^{\text{Cross}}}^{(1)}$$

$$16\pi^2 \mu \frac{d\lambda_{12}^{\text{Cross}}}{d\mu} = \beta_{\lambda_{12}^{\text{Cross}}}^{(1)}$$

Terms 1 of 6

Column	Representative term structure
T1	$\lambda_{12} \left(\frac{1}{2} \text{Tr}(y_1 (y_1)^\dagger) + \frac{1}{2} \text{Tr}(y_2 (y_2)^\dagger) + 8\lambda_{12}^{\text{Cross}} + \lambda_{S_1}^{\text{Adj}} + \lambda_{S_2}^{\text{Adj}} \right)$
T2	$+\lambda_{12}^{\text{Cross}} (2 \text{Tr}(y_1 (y_1)^\dagger) + 2 \text{Tr}(y_2 (y_2)^\dagger))$
T3	$-\frac{7}{6} \text{Tr}(y_1 (y_1)^\dagger y_2 (y_2)^\dagger)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{12} \left(\frac{1}{2} \text{Tr} \left(y_1 \left(y_1 \right)^\dagger \right) + \frac{1}{2} \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) + 8 \lambda_{12}^{\text{Cross}} + \lambda_{S_1}^{\text{Adj}} + \lambda_{S_2}^{\text{Adj}} \right) + \lambda_{12}^{\text{Cross}} \left(2 \text{Tr} \left(y_1 \left(y_1 \right)^\dagger \right) + 2 \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) \right)$	$-\frac{7}{6} \text{Tr} \left(y_1 \left(y_1 \right)^\dagger y_2 \left(y_2 \right)^\dagger \right)$	
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{12} \left(\frac{1}{2} \text{Tr} \left(y_1 \left(y_1 \right)^\dagger \right) + \frac{1}{2} \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) + 8 \lambda_{12}^{\text{Cross}} + \lambda_{S_1}^{\text{Adj}} + \lambda_{S_2}^{\text{Adj}} \right) + \lambda_{12}^{\text{Cross}} \left(2 \text{Tr} \left(y_1 \left(y_1 \right)^\dagger \right) + 2 \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) \right)$	$-\frac{7}{6} \text{Tr} \left(y_1 \left(y_1 \right)^\dagger y_2 \left(y_2 \right)^\dagger \right)$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{12} \left(\frac{1}{2} \text{Tr} \left(y_1 \left(y_1 \right)^\dagger \right) + \frac{1}{2} \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) + 8 \lambda_{12}^{\text{Cross}} + \lambda_{S_1}^{\text{Adj}} + \lambda_{S_2}^{\text{Adj}} \right) + \lambda_{12}^{\text{Cross}} \left(2 \text{Tr} \left(y_1 \left(y_1 \right)^\dagger \right) + 2 \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) \right)$	$-\frac{7}{6} \text{Tr} \left(y_1 \left(y_1 \right)^\dagger y_2 \left(y_2 \right)^\dagger \right)$	
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{12} \left(\frac{1}{2} \text{Tr} \left(y_1 \left(y_1 \right)^\dagger \right) + \frac{1}{2} \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) + 8 \lambda_{12}^{\text{Cross}} + \lambda_{S_1}^{\text{Adj}} + \lambda_{S_2}^{\text{Adj}} \right) + \lambda_{12}^{\text{Cross}} \left(2 \text{Tr} \left(y_1 \left(y_1 \right)^\dagger \right) + 2 \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) \right)$	$-\frac{7}{6} \text{Tr} \left(y_1 \left(y_1 \right)^\dagger y_2 \left(y_2 \right)^\dagger \right)$	
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\lambda_{12} \left(\frac{1}{2} \text{Tr} \left(y_1 \left(y_1 \right)^\dagger \right) + \frac{1}{2} \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) + 8 \lambda_{12}^{\text{Cross}} + \lambda_{S_1}^{\text{Adj}} + \lambda_{S_2}^{\text{Adj}} \right) + \lambda_{12}^{\text{Cross}} \left(2 \text{Tr} \left(y_1 \left(y_1 \right)^\dagger \right) + 2 \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) \right)$	$-\frac{7}{6} \text{Tr} \left(y_1 \left(y_1 \right)^\dagger y_2 \left(y_2 \right)^\dagger \right)$	

Terms 2 of 6

Column	Representative term structure	
T1	$-\frac{1}{2}\lambda_{T3}(\lambda_{T3})^*$	
T2	$+\frac{33g_2^4}{4}$	
T3	$+g_2^2(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}})$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{1}{2}\lambda_{T3}(\lambda_{T3})^*$	$+\frac{33g_2^4}{4}$	$+g_2^2(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}})$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{1}{2}\lambda_{T3}(\lambda_{T3})^*$	$+\frac{33g_2^4}{4}$	$+g_2^2(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2}\lambda_{T3}(\lambda_{T3})^*$	$+\frac{33g_2^4}{4}$	$+g_2^2(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}})$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{1}{2}\lambda_{T3}(\lambda_{T3})^*$	$+\frac{33g_2^4}{4}$	$+g_2^2(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}})$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{1}{2}\lambda_{T3}(\lambda_{T3})^*$	$+\frac{33g_2^4}{4}$	$+g_2^2(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}})$

Terms 3 of 6

Column Representative term structure

T1	$+g_Y^2 \left(-\frac{3\lambda_{12}}{2} - 6\lambda_{12}^{\text{Cross}} \right)$
T2	$+\lambda_{12}^2$
T3	$+\lambda_{S_1}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+g_Y^2 \left(-\frac{3\lambda_{12}}{2} - 6\lambda_{12}^{\text{Cross}} \right)$	$+\lambda_{12}^2$	$+\lambda_{S_1}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+g_Y^2 \left(-\frac{3\lambda_{12}}{4} - 3\lambda_{12}^{\text{Cross}} \right)$	$+\lambda_{12}^2$	$+\lambda_{S_1}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_Y^2 \left(-\frac{3\lambda_{12}}{2} - 6\lambda_{12}^{\text{Cross}} \right)$	$+\lambda_{12}^2$	$+\lambda_{S_1}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+g_Y^2 \left(-\frac{15\lambda_{12}}{4} - 15\lambda_{12}^{\text{Cross}} \right)$	$+\lambda_{12}^2$	$+\lambda_{S_1}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+g_Y^2 \left(-\frac{15\lambda_{12}}{2} - 30\lambda_{12}^{\text{Cross}} \right)$	$+\lambda_{12}^2$	$+\lambda_{S_1}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$

Terms 4 of 6

Column	Representative term structure	
T1	$+\lambda_{S_2}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$	
T2	$+4\lambda_{12}^{\text{Adj}^2}$	
T3	$+11\lambda_{12}^{\text{Cross}^2}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\lambda_{S_2}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$	$+4\lambda_{12}^{\text{Adj}^2}$	$+11\lambda_{12}^{\text{Cross}^2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\lambda_{S_2}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$	$+4\lambda_{12}^{\text{Adj}^2}$	$+11\lambda_{12}^{\text{Cross}^2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\lambda_{S_2}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$	$+4\lambda_{12}^{\text{Adj}^2}$	$+11\lambda_{12}^{\text{Cross}^2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\lambda_{S_2}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$	$+4\lambda_{12}^{\text{Adj}^2}$	$+11\lambda_{12}^{\text{Cross}^2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\lambda_{S_2}(2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$	$+4\lambda_{12}^{\text{Adj}^2}$	$+11\lambda_{12}^{\text{Cross}^2}$

Terms 5 of 6

Column	Representative term structure
T1	$-\lambda_{12}^{\text{Cross}} \lambda_{S_1}^{\text{Adj}}$
T2	$-\lambda_{12}^{\text{Cross}} \lambda_{S_2}^{\text{Adj}}$
T3	$+\lambda_{H1} \lambda_{H2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\lambda_{12}^{\text{Cross}} \lambda_{S_1}^{\text{Adj}}$	$-\lambda_{12}^{\text{Cross}} \lambda_{S_2}^{\text{Adj}}$	$+\lambda_{H1} \lambda_{H2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\lambda_{12}^{\text{Cross}} \lambda_{S_1}^{\text{Adj}}$	$-\lambda_{12}^{\text{Cross}} \lambda_{S_2}^{\text{Adj}}$	$+\lambda_{H1} \lambda_{H2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\lambda_{12}^{\text{Cross}} \lambda_{S_1}^{\text{Adj}}$	$-\lambda_{12}^{\text{Cross}} \lambda_{S_2}^{\text{Adj}}$	$+\lambda_{H1} \lambda_{H2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\lambda_{12}^{\text{Cross}} \lambda_{S_1}^{\text{Adj}}$	$-\lambda_{12}^{\text{Cross}} \lambda_{S_2}^{\text{Adj}}$	$+\lambda_{H1} \lambda_{H2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\lambda_{12}^{\text{Cross}} \lambda_{S_1}^{\text{Adj}}$	$-\lambda_{12}^{\text{Cross}} \lambda_{S_2}^{\text{Adj}}$	$+\lambda_{H1} \lambda_{H2}$

Terms 6 of 6

Column	Representative term structure	
T1	$-\frac{7}{6}\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$	
T2	$+\frac{3g_Y^4}{16}$	
T3	$+12g_Y^4$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{7}{6}\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$	$+\frac{3g_Y^4}{16}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$+\frac{27g_Y^4}{16}$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	$+12g_Y^4$

$$\beta_{\lambda_{H1}^{\text{Adj}}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{H1}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{H1}^{\text{Adj}}}^{(1)}$$

Terms 1 of 8

Column	Representative term structure
T1	$\lambda_{H1}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$
T2	$+\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$
T3	$+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$\lambda_{H1}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H1}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\lambda_{H1}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$\lambda_{H1}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$\lambda_{H1}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$\lambda_{H1}^{\text{Adj}}(3\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{H1}^{\text{Adj}}(3\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\lambda_{H1}^{\text{Adj}}(3\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\lambda_{H1}^{\text{Adj}}(3\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$\lambda_{H1}^{\text{Adj}}(3\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$-\frac{8}{3}\lambda_{T3}\ (\lambda_{T3})^*$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$-\frac{8}{3}\lambda_{T3}\ (\lambda_{T3})^*$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$-\frac{8}{3}\lambda_{T3}\ (\lambda_{T3})^*$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$-\frac{8}{3}\lambda_{T3}\ (\lambda_{T3})^*$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$-\frac{8}{3}\lambda_{T3}\ (\lambda_{T3})^*$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$-\frac{4}{3}\lambda_{T3}\ (\lambda_{T3})^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-\frac{4}{3} \text{Tr} \left(y_1 (y_1)^\dagger Y_e (Y_e)^\dagger \right)$	$-\frac{4}{3} \lambda_{T3} (\lambda_{T3})^*$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{4}{3} \text{Tr} \left(y_1 (y_1)^\dagger Y_e (Y_e)^\dagger \right)$	$-\frac{4}{3} \lambda_{T3} (\lambda_{T3})^*$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$-\frac{4}{3} \text{Tr} \left(y_1 (y_1)^\dagger Y_e (Y_e)^\dagger \right)$	$-\frac{4}{3} \lambda_{T3} (\lambda_{T3})^*$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$-\frac{4}{3} \text{Tr} \left(y_1 (y_1)^\dagger Y_e (Y_e)^\dagger \right)$	$-\frac{4}{3} \lambda_{T3} (\lambda_{T3})^*$

Terms 2 of 8

Column	Representative term structure	
T1	$+g_Y^2 \left(-12g_2^2 - \frac{15\lambda_{H1}^{\text{Adj}}}{2} \right)$	
T2	$-9g_2^2 \lambda_{H1}^{\text{Adj}}$	
T3	$+\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+g_Y^2 \left(-12g_2^2 - \frac{15\lambda_{H1}^{\text{Adj}}}{2} \right)$	$-9g_2^2 \lambda_{H1}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	$-9g_2^2 \lambda_{H1}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-9g_2^2 \lambda_{H1}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$-9g_2^2 \lambda_{H1}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$-9g_2^2 \lambda_{H1}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+g_Y^2 \left(-12g_2^2 - \frac{15\lambda_{H1}^{\text{Adj}}}{2} \right)$	$-9g_2^2 \lambda_{H1}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$-9g_2^2 \lambda_{H1}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-9g_2^2 \lambda_{H1}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$-9g_2^2 \lambda_{H1}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$-9g_2^2 \lambda_{H1}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}} \lambda_{H2}^{\text{Adj}}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+g_Y^2 \left(-4g_2^2 - \frac{15\lambda_{H1}^{\text{Adj}}}{2} \right)$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+g_Y^2 \left(-4g_2^2 - \frac{15\lambda_{H1}^{\text{Adj}}}{2} \right)$	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 3 of 8

Column	Representative term structure	
T1	$+2\lambda_H \lambda_{H1}^{\text{Adj}}$	
T2	$+8\lambda_{H1} \lambda_{H1}^{\text{Adj}}$	
T3	$+2\lambda_{H1}^{\text{Adj}} \lambda_{S1}$	

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+2\lambda_H \lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1} \lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}} \lambda_{S1}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_H \lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1} \lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}} \lambda_{S1}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+2\lambda_H \lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1} \lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}} \lambda_{S1}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+2\lambda_H \lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1} \lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}} \lambda_{S1}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+2\lambda_H \lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1} \lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}} \lambda_{S1}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+2\lambda_H \lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1} \lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}} \lambda_{S1}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_H \lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1} \lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}} \lambda_{S1}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+2\lambda_H \lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1} \lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}} \lambda_{S1}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+2\lambda_H \lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1} \lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}} \lambda_{S1}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+2\lambda_H \lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1} \lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}} \lambda_{S1}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+2\lambda_H \lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1} \lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}} \lambda_{S1}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+2\lambda_H \lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1} \lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}} \lambda_{S1}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+2\lambda_H \lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1} \lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}} \lambda_{S1}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+2\lambda_H \lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1} \lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}} \lambda_{S1}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+2\lambda_H \lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1} \lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}} \lambda_{S1}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+2\lambda_H \lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1} \lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}} \lambda_{S1}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+2\lambda_H \lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1} \lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}} \lambda_{S1}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+2\lambda_H \lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1} \lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}} \lambda_{S1}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+2\lambda_H \lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1} \lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}} \lambda_{S1}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+2\lambda_H \lambda_{H1}^{\text{Adj}}$	$+8\lambda_{H1} \lambda_{H1}^{\text{Adj}}$	$+2\lambda_{H1}^{\text{Adj}} \lambda_{S1}$

Terms 4 of 8

Column	Representative term structure	
T1	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(-\frac{2}{3} \text{Tr}(y_1 (y_2)^\dagger) - \frac{4}{3} (2 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + \left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \right)^*) \right)$	
T2	$-\frac{2}{3} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \text{Tr}(y_2 (y_1)^\dagger)$	
T3	$+\frac{64}{3} \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(-\frac{2}{3} \text{Tr}(y_1 (y_2)^\dagger) - \frac{4}{3} (2 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + \left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \right)^*) \right)$	$-\frac{2}{3} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \text{Tr}(y_2 (y_1)^\dagger)$	$+\frac{64}{3} \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$+\frac{64}{3} \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 5 of 8

Column	Representative term structure	
T1	$-\frac{4}{3}\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	
T2	$+g_Y^2 \left(-6g_2^2 - 3\lambda_{H1}^{\text{Adj}} \right)$	
T3	$-\frac{3g_Y^2 \lambda_{H1}^{\text{Adj}}}{2}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{4}{3}\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	$+g_Y^2 \left(-6g_2^2 - 3\lambda_{H1}^{\text{Adj}} \right)$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$-\frac{3g_Y^2 \lambda_{H1}^{\text{Adj}}}{2}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{4}{3}\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	$+g_Y^2 \left(-6g_2^2 - 3\lambda_{H1}^{\text{Adj}} \right)$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$-\frac{3g_Y^2 \lambda_{H1}^{\text{Adj}}}{2}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$+g_Y^2 \left(-2g_2^2 - 3\lambda_{H1}^{\text{Adj}} \right)$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	$-\frac{3g_Y^2 \lambda_{H1}^{\text{Adj}}}{2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$+g_Y^2 \left(-2g_2^2 - 3\lambda_{H1}^{\text{Adj}} \right)$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$-\frac{3g_Y^2 \lambda_{H1}^{\text{Adj}}}{2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 6 of 8

Column	Representative term structure	
T1	$+g_Y^2 \left(6g_2^2 - 3\lambda_{H1}^{\text{Adj}} \right)$	
T2	$+g_Y^2 \left(12g_2^2 - \frac{15\lambda_{H1}^{\text{Adj}}}{2} \right)$	
T3	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(-\text{Tr} \left(y_1 (y_2)^\dagger \right) - \frac{4}{3} \left(2 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \right)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+g_Y^2 \left(6g_2^2 - 3\lambda_{H1}^{\text{Adj}} \right)$	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+g_Y^2 \left(12g_2^2 - \frac{15\lambda_{H1}^{\text{Adj}}}{2} \right)$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(-\text{Tr} \left(y_1 (y_2)^\dagger \right) - \frac{4}{3} \left(2 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+g_Y^2 \left(6g_2^2 - 3\lambda_{H1}^{\text{Adj}} \right)$	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+g_Y^2 \left(12g_2^2 - \frac{15\lambda_{H1}^{\text{Adj}}}{2} \right)$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+g_Y^2 \left(2g_2^2 - 3\lambda_{H1}^{\text{Adj}} \right)$	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+g_Y^2 \left(4g_2^2 - \frac{15\lambda_{H1}^{\text{Adj}}}{2} \right)$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+g_Y^2 \left(2g_2^2 - 3\lambda_{H1}^{\text{Adj}} \right)$	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+g_Y^2 \left(4g_2^2 - \frac{15\lambda_{H1}^{\text{Adj}}}{2} \right)$	—	—

Terms 7 of 8

Column	Representative term structure
T1	$-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* \text{Tr}\left(y_2 \left(y_1\right)^\dagger\right)$
T2	$\lambda_{H1}^{\text{Adj}}(2 \text{Tr}\left(y_1 \left(y_1\right)^\dagger\right) + 6 \text{Tr}\left(Y_d \left(Y_d\right)^\dagger\right) + 2 \text{Tr}\left(Y_e \left(Y_e\right)^\dagger\right) + 6 \text{Tr}\left(Y_u \left(Y_u\right)^\dagger\right) + 16\lambda_{S1}^{\text{Adj}})$
T3	$-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* \text{Tr}\left(y_2 \left(y_1\right)^\dagger\right)$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$\lambda_{H1}^{\text{Adj}}(2 \text{Tr}\left(y_1 \left(y_1\right)^\dagger\right) + 6 \text{Tr}\left(Y_d \left(Y_d\right)^\dagger\right) + 2 \text{Tr}\left(Y_e \left(Y_e\right)^\dagger\right) + 6 \text{Tr}\left(Y_u \left(Y_u\right)^\dagger\right) + 16\lambda_{S1}^{\text{Adj}})$	$-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$\lambda_{H1}^{\text{Adj}}(2 \text{Tr}\left(y_1 \left(y_1\right)^\dagger\right) + 6 \text{Tr}\left(Y_d \left(Y_d\right)^\dagger\right) + 2 \text{Tr}\left(Y_e \left(Y_e\right)^\dagger\right) + 6 \text{Tr}\left(Y_u \left(Y_u\right)^\dagger\right) + 16\lambda_{S1}^{\text{Adj}})$	$-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$\lambda_{H1}^{\text{Adj}}(2 \text{Tr}\left(y_1 \left(y_1\right)^\dagger\right) + 6 \text{Tr}\left(Y_d \left(Y_d\right)^\dagger\right) + 2 \text{Tr}\left(Y_e \left(Y_e\right)^\dagger\right) + 6 \text{Tr}\left(Y_u \left(Y_u\right)^\dagger\right) + 16\lambda_{S1}^{\text{Adj}})$	$-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$\lambda_{H1}^{\text{Adj}}(2 \text{Tr}\left(y_1 \left(y_1\right)^\dagger\right) + 6 \text{Tr}\left(Y_d \left(Y_d\right)^\dagger\right) + 2 \text{Tr}\left(Y_e \left(Y_e\right)^\dagger\right) + 6 \text{Tr}\left(Y_u \left(Y_u\right)^\dagger\right) + 16\lambda_{S1}^{\text{Adj}})$	$-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$\lambda_{H1}^{\text{Adj}}(2 \text{Tr}\left(y_1 \left(y_1\right)^\dagger\right) + 6 \text{Tr}\left(Y_d \left(Y_d\right)^\dagger\right) + 2 \text{Tr}\left(Y_e \left(Y_e\right)^\dagger\right) + 6 \text{Tr}\left(Y_u \left(Y_u\right)^\dagger\right) + 16\lambda_{S1}^{\text{Adj}})$	$-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$\lambda_{H1}^{\text{Adj}}(2 \text{Tr}\left(y_1 \left(y_1\right)^\dagger\right) + 6 \text{Tr}\left(Y_d \left(Y_d\right)^\dagger\right) + 2 \text{Tr}\left(Y_e \left(Y_e\right)^\dagger\right) + 6 \text{Tr}\left(Y_u \left(Y_u\right)^\dagger\right) + 16\lambda_{S1}^{\text{Adj}})$	$-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$\lambda_{H1}^{\text{Adj}}(2 \text{Tr}\left(y_1 \left(y_1\right)^\dagger\right) + 6 \text{Tr}\left(Y_d \left(Y_d\right)^\dagger\right) + 2 \text{Tr}\left(Y_e \left(Y_e\right)^\dagger\right) + 6 \text{Tr}\left(Y_u \left(Y_u\right)^\dagger\right) + 16\lambda_{S1}^{\text{Adj}})$	$-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$\lambda_{H1}^{\text{Adj}}(2 \text{Tr}\left(y_1 \left(y_1\right)^\dagger\right) + 6 \text{Tr}\left(Y_d \left(Y_d\right)^\dagger\right) + 2 \text{Tr}\left(Y_e \left(Y_e\right)^\dagger\right) + 6 \text{Tr}\left(Y_u \left(Y_u\right)^\dagger\right) + 16\lambda_{S1}^{\text{Adj}})$	$-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$\lambda_{H1}^{\text{Adj}}(2 \text{Tr}\left(y_1 \left(y_1\right)^\dagger\right) + 6 \text{Tr}\left(Y_d \left(Y_d\right)^\dagger\right) + 2 \text{Tr}\left(Y_e \left(Y_e\right)^\dagger\right) + 6 \text{Tr}\left(Y_u \left(Y_u\right)^\dagger\right) + 16\lambda_{S1}^{\text{Adj}})$	$-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$\lambda_{H1}^{\text{Adj}}(2 \text{Tr}\left(y_1 \left(y_1\right)^\dagger\right) + 6 \text{Tr}\left(Y_d \left(Y_d\right)^\dagger\right) + 2 \text{Tr}\left(Y_e \left(Y_e\right)^\dagger\right) + 6 \text{Tr}\left(Y_u \left(Y_u\right)^\dagger\right) + 16\lambda_{S1}^{\text{Adj}})$	$-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$

Terms 8 of 8

Column	Representative term structure
T1	$+12\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+12\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+12\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+12\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+12\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+12\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$

$$\beta_{\lambda_{H2}^{\text{Adj}}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{H2}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{H2}^{\text{Adj}}}^{(1)}$$

Terms 1 of 7

Column	Representative term structure
T1	$\lambda_{H2}^{\text{Adj}}(2\text{Tr}(y_2(y_2)^\dagger)+6\text{Tr}(Y_d(Y_d)^\dagger)+2\text{Tr}(Y_e(Y_e)^\dagger)+6\text{Tr}(Y_u(Y_u)^\dagger)+16\lambda_{S2}^{\text{Adj}})$
T2	$+\frac{4}{3}\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$
T3	$+\frac{8}{3}\lambda_{T3}(\lambda_{T3})^*$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{H2}^{\text{Adj}}(2\text{Tr}(y_2(y_2)^\dagger)+6\text{Tr}(Y_d(Y_d)^\dagger)+2\text{Tr}(Y_e(Y_e)^\dagger)+6\text{Tr}(Y_u(Y_u)^\dagger)+16\lambda_{S2}^{\text{Adj}})$	$+\frac{4}{3}\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+\frac{8}{3}\lambda_{T3}(\lambda_{T3})^*$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{H2}^{\text{Adj}}(2\text{Tr}(y_2(y_2)^\dagger)+6\text{Tr}(Y_d(Y_d)^\dagger)+2\text{Tr}(Y_e(Y_e)^\dagger)+6\text{Tr}(Y_u(Y_u)^\dagger)+16\lambda_{S2}^{\text{Adj}})$	$+\frac{4}{3}\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+\frac{8}{3}\lambda_{T3}(\lambda_{T3})^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{H2}^{\text{Adj}}(2\text{Tr}(y_2(y_2)^\dagger)+6\text{Tr}(Y_d(Y_d)^\dagger)+2\text{Tr}(Y_e(Y_e)^\dagger)+6\text{Tr}(Y_u(Y_u)^\dagger)+16\lambda_{S2}^{\text{Adj}})$	$+\frac{4}{3}\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+\frac{8}{3}\lambda_{T3}(\lambda_{T3})^*$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{H2}^{\text{Adj}}(2\text{Tr}(y_2(y_2)^\dagger)+6\text{Tr}(Y_d(Y_d)^\dagger)+2\text{Tr}(Y_e(Y_e)^\dagger)+6\text{Tr}(Y_u(Y_u)^\dagger)+16\lambda_{S2}^{\text{Adj}})$	$+\frac{4}{3}\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+\frac{8}{3}\lambda_{T3}(\lambda_{T3})^*$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$\lambda_{H2}^{\text{Adj}}(2\text{Tr}(y_2(y_2)^\dagger)+6\text{Tr}(Y_d(Y_d)^\dagger)+2\text{Tr}(Y_e(Y_e)^\dagger)+6\text{Tr}(Y_u(Y_u)^\dagger)+16\lambda_{S2}^{\text{Adj}})$	$+\frac{4}{3}\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$+\frac{8}{3}\lambda_{T3}(\lambda_{T3})^*$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{H2}^{\text{Adj}}(2\text{Tr}(y_2(y_2)^\dagger)+6\text{Tr}(Y_d(Y_d)^\dagger)+2\text{Tr}(Y_e(Y_e)^\dagger)+6\text{Tr}(Y_u(Y_u)^\dagger)+16\lambda_{S2}^{\text{Adj}})$	$+\frac{4}{3}\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$-\frac{8}{9}\lambda_{T3}(\lambda_{T3})^*$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{H2}^{\text{Adj}}(2\text{Tr}(y_2(y_2)^\dagger)+6\text{Tr}(Y_d(Y_d)^\dagger)+2\text{Tr}(Y_e(Y_e)^\dagger)+6\text{Tr}(Y_u(Y_u)^\dagger)+16\lambda_{S2}^{\text{Adj}})$	$+\frac{4}{3}\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$-\frac{8}{9}\lambda_{T3}(\lambda_{T3})^*$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{H2}^{\text{Adj}}(2\text{Tr}(y_2(y_2)^\dagger)+6\text{Tr}(Y_d(Y_d)^\dagger)+2\text{Tr}(Y_e(Y_e)^\dagger)+6\text{Tr}(Y_u(Y_u)^\dagger)+16\lambda_{S2}^{\text{Adj}})$	$+\frac{4}{3}\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$-\frac{8}{9}\lambda_{T3}(\lambda_{T3})^*$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{H2}^{\text{Adj}}(2\text{Tr}(y_2(y_2)^\dagger)+6\text{Tr}(Y_d(Y_d)^\dagger)+2\text{Tr}(Y_e(Y_e)^\dagger)+6\text{Tr}(Y_u(Y_u)^\dagger)+16\lambda_{S2}^{\text{Adj}})$	$+\frac{4}{3}\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$-\frac{8}{9}\lambda_{T3}(\lambda_{T3})^*$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\lambda_{H2}^{\text{Adj}}(2\text{Tr}(y_2(y_2)^\dagger)+6\text{Tr}(Y_d(Y_d)^\dagger)+2\text{Tr}(Y_e(Y_e)^\dagger)+6\text{Tr}(Y_u(Y_u)^\dagger)+16\lambda_{S2}^{\text{Adj}})$	$+\frac{4}{3}\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$	$-\frac{8}{9}\lambda_{T3}(\lambda_{T3})^*$

Terms 2 of 7

Column	Representative term structure	
T1	$-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$	
T2	$-\frac{3g_Y^2\lambda_{H2}^{\text{Adj}}}{2}$	
T3	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$	$-\frac{3g_Y^2\lambda_{H2}^{\text{Adj}}}{2}$	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$-\frac{3g_Y^2\lambda_{H2}^{\text{Adj}}}{2}$	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$-\frac{3g_Y^2\lambda_{H2}^{\text{Adj}}}{2}$	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$	$-\frac{3g_Y^2\lambda_{H2}^{\text{Adj}}}{2}$	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$	—	$+2\lambda_H\lambda_{H2}^{\text{Adj}}$

Terms 3 of 7

Column	Representative term structure
T1	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$
T2	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$
T3	$+g_Y^2\left(2g_2^2-3\lambda_{H2}^{\text{Adj}}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	$+g_Y^2\left(2g_2^2-3\lambda_{H2}^{\text{Adj}}\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	$+g_Y^2\left(6g_2^2-15\lambda_{H2}^{\text{Adj}}\right)$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	$+g_Y^2\left(6g_2^2-3\lambda_{H2}^{\text{Adj}}\right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	$+g_Y^2\left(18g_2^2-15\lambda_{H2}^{\text{Adj}}\right)$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	$+g_Y^2\left(6g_2^2-3\lambda_{H2}^{\text{Adj}}\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	$+g_Y^2\left(18g_2^2-15\lambda_{H2}^{\text{Adj}}\right)$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	$+g_Y^2\left(2g_2^2-3\lambda_{H2}^{\text{Adj}}\right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	$+g_Y^2\left(6g_2^2-15\lambda_{H2}^{\text{Adj}}\right)$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$	$+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$	—

Terms 4 of 7

Column	Representative term structure									
T1	$+g_Y^2 \left(4g_2^2 - \frac{15\lambda_{H2}^{\text{Adj}}}{2} \right)$									
T2	$\lambda_{H2}^{\text{Adj}} (2 \text{Tr}(y_2 (y_2)^\dagger) + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$									
T3	$+8 \text{Tr}(y_2 (y_2)^\dagger Y_e (Y_e)^\dagger)$									

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+g_Y^2 \left(4g_2^2 - \frac{15\lambda_{H2}^{\text{Adj}}}{2} \right)$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+g_Y^2 \left(8g_2^2 - \frac{51\lambda_{H2}^{\text{Adj}}}{2} \right)$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$\lambda_{H2}^{\text{Adj}} (2 \text{Tr}(y_2 (y_2)^\dagger) + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$	$+8 \text{Tr}(y_2 (y_2)^\dagger Y_e (Y_e)^\dagger)$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	$\lambda_{H2}^{\text{Adj}} (2 \text{Tr}(y_2 (y_2)^\dagger) + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$	$+8 \text{Tr}(y_2 (y_2)^\dagger Y_e (Y_e)^\dagger)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+g_Y^2 \left(12g_2^2 - \frac{15\lambda_{H2}^{\text{Adj}}}{2} \right)$	$\lambda_{H2}^{\text{Adj}} (2 \text{Tr}(y_2 (y_2)^\dagger) + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$	$+8 \text{Tr}(y_2 (y_2)^\dagger Y_e (Y_e)^\dagger)$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$\lambda_{H2}^{\text{Adj}} (2 \text{Tr}(y_2 (y_2)^\dagger) + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$	$+8 \text{Tr}(y_2 (y_2)^\dagger Y_e (Y_e)^\dagger)$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+g_Y^2 \left(24g_2^2 - \frac{51\lambda_{H2}^{\text{Adj}}}{2} \right)$	$\lambda_{H2}^{\text{Adj}} (2 \text{Tr}(y_2 (y_2)^\dagger) + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$	$+8 \text{Tr}(y_2 (y_2)^\dagger Y_e (Y_e)^\dagger)$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$\lambda_{H2}^{\text{Adj}} (3 \text{Tr}(y_2 (y_2)^\dagger) + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$	$-4 \text{Tr}(y_2 (y_2)^\dagger Y_e (Y_e)^\dagger)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$\lambda_{H2}^{\text{Adj}} (3 \text{Tr}(y_2 (y_2)^\dagger) + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$	$-4 \text{Tr}(y_2 (y_2)^\dagger Y_e (Y_e)^\dagger)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+g_Y^2 \left(12g_2^2 - \frac{15\lambda_{H2}^{\text{Adj}}}{2} \right)$	$\lambda_{H2}^{\text{Adj}} (3 \text{Tr}(y_2 (y_2)^\dagger) + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$	$-4 \text{Tr}(y_2 (y_2)^\dagger Y_e (Y_e)^\dagger)$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$\lambda_{H2}^{\text{Adj}} (3 \text{Tr}(y_2 (y_2)^\dagger) + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$	$-4 \text{Tr}(y_2 (y_2)^\dagger Y_e (Y_e)^\dagger)$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+g_Y^2 \left(24g_2^2 - \frac{51\lambda_{H2}^{\text{Adj}}}{2} \right)$	$\lambda_{H2}^{\text{Adj}} (3 \text{Tr}(y_2 (y_2)^\dagger) + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$	$-4 \text{Tr}(y_2 (y_2)^\dagger Y_e (Y_e)^\dagger)$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_Y^2 \left(4g_2^2 - \frac{15\lambda_{H2}^{\text{Adj}}}{2} \right)$	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+g_Y{}^2\left(8g_2{}^2-\frac{51\lambda_{H^2}^{\text{Adj}}}{2}\right)$	—	—

Terms 5 of 7

Column	Representative term structure						
T1	$+16\lambda_{T3}(\lambda_{T3})^*$						
T2	$-9g_2^2\lambda_{H2}^{\text{Adj}}$						
T3	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$						

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+16\lambda_{T3}(\lambda_{T3})^*$	$-9g_2^2\lambda_{H2}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+16\lambda_{T3}(\lambda_{T3})^*$	$-9g_2^2\lambda_{H2}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+16\lambda_{T3}(\lambda_{T3})^*$	$-9g_2^2\lambda_{H2}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+16\lambda_{T3}(\lambda_{T3})^*$	$-9g_2^2\lambda_{H2}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+16\lambda_{T3}(\lambda_{T3})^*$	$-9g_2^2\lambda_{H2}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+16\lambda_{T3}(\lambda_{T3})^*$	$-9g_2^2\lambda_{H2}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+16\lambda_{T3}(\lambda_{T3})^*$	$-9g_2^2\lambda_{H2}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+16\lambda_{T3}(\lambda_{T3})^*$	$-9g_2^2\lambda_{H2}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+16\lambda_{T3}(\lambda_{T3})^*$	$-9g_2^2\lambda_{H2}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+16\lambda_{T3}(\lambda_{T3})^*$	$-9g_2^2\lambda_{H2}^{\text{Adj}}$	$+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 6 of 7

Column	Representative term structure
T1	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(-2\text{Tr}(y_1(y_2)^\dagger) - 4(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* + \left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*))$
T2	$-2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}(y_2(y_1)^\dagger)$
T3	$-4\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(-2\text{Tr}(y_1(y_2)^\dagger) - 4(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* + \left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*))$	$-2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}(y_2(y_1)^\dagger)$	$-4\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(-3\text{Tr}(y_1(y_2)^\dagger) - 4(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* + \left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*))$	$-3\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}(y_2(y_1)^\dagger)$	$-4\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 7 of 7

Column	Representative term structure
T1	$+64\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$
T2	$+12\lambda_{12}^{\text{Adj}} \lambda_{H1}^{\text{Adj}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+64\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+64\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$+12\lambda_{12}^{\text{Adj}} \lambda_{H1}^{\text{Adj}}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$+12\lambda_{12}^{\text{Adj}} \lambda_{H1}^{\text{Adj}}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+12\lambda_{12}^{\text{Adj}} \lambda_{H1}^{\text{Adj}}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$+12\lambda_{12}^{\text{Adj}} \lambda_{H1}^{\text{Adj}}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$+12\lambda_{12}^{\text{Adj}} \lambda_{H1}^{\text{Adj}}$

$$\beta_{\lambda}^{(1)}{}_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$$

$$16\pi^2\mu\frac{d\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{d\mu}=\beta_{\lambda}^{(1)}{}_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} (2\text{Tr}\big(y_1\ (y_1)^\dagger\big) + 6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$
T2	$+ (\lambda_{T3})^* \text{Tr}\big(y_2\ (y_1)^\dagger\big)$
T3	$-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (\lambda_{T3})^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} (2\text{Tr}\big(y_1\ (y_1)^\dagger\big) + 6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)) + (\lambda_{T3})^* \text{Tr}\big(y_2\ (y_1)^\dagger\big)$		$-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (\lambda_{T3})^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} (3\text{Tr}\big(y_1\ (y_1)^\dagger\big) + 6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	—	$-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (\lambda_{T3})^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 2 of 4

Column	Representative term structure
T1	$+2\lambda_{S_1 S_1^\dagger{}^2 S_2^\dagger} (\lambda_{T3})^*$
T2	$-9g_2{}^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$
T3	$-3g_Y{}^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{S_1 S_1^\dagger{}^2 S_2^\dagger} (\lambda_{T3})^*$	$-9g_2{}^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$	$-3g_Y{}^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{S_1 S_1^\dagger{}^2 S_2^\dagger} (\lambda_{T3})^*$	$-9g_2{}^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$	$-3g_Y{}^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 3 of 4

Column	Representative term structure
T1	$+2\lambda_H \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$
T2	$+8\lambda_{H1} \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$
T3	$+\frac{4\lambda_{H1}^{\text{Adj}} \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_H \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$	$+8\lambda_{H1} \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$	$+\frac{4\lambda_{H1}^{\text{Adj}} \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{3}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_H \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$	$+8\lambda_{H1} \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$	$+\frac{4\lambda_{H1}^{\text{Adj}} \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{3}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 4 of 4

Column	Representative term structure	
T1	$+2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \lambda_{S_1}$	
T2	$+4\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$	
T3	$+\frac{3}{2}(\lambda_{T3})^* \text{Tr}(y_2 (y_1)^\dagger)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \lambda_{S_1}$	$+4\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \lambda_{S_1}$	$+4\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$	$+\frac{3}{2}(\lambda_{T3})^* \text{Tr}(y_2 (y_1)^\dagger)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

$$\beta_{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{d\mu}=\beta_{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}^{(1)}$$

Terms 1 of 5

Column	Representative term structure	
T1	$\lambda_{H1}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+3\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)$	
T2	$+\lambda_{H1}^{\text{Adj}}\left(-\frac{1}{8}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)-\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}-\frac{\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}}{4}\right)$	
T3	$+\lambda_{H2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+3\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H1}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+3\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)+\lambda_{H1}^{\text{Adj}}\left(-\frac{1}{8}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)-\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}-\frac{\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}}{4}\right)+\lambda_{H2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+3\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)$		
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{H1}\left(\frac{3}{4}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+3\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)+\lambda_{H1}^{\text{Adj}}\left(-\frac{3}{16}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)-\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}-\frac{\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}}{4}\right)+\lambda_{H2}\left(\frac{3}{4}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+3\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)$		
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 2 of 5

Column Representative term structure

T1	$+\lambda_{H2}^{\text{Adj}} \left(-\frac{3}{8} \text{Tr}(y_2 (y_1)^\dagger) - \frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{3\lambda_{S_1^\dagger S_2 S_2^\dagger^2}}{4} \right)$
T2	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (\text{Tr}(y_1 (y_1)^\dagger) + \text{Tr}(y_2 (y_2)^\dagger) + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$
T3	$-4 \text{Tr}(y_2 (y_1)^\dagger Y_e (Y_e)^\dagger)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{H2}^{\text{Adj}} \left(-\frac{3}{8} \text{Tr}(y_2 (y_1)^\dagger) - \frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{3\lambda_{S_1^\dagger S_2 S_2^\dagger^2}}{4} \right) + \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (\text{Tr}(y_1 (y_1)^\dagger) + \text{Tr}(y_2 (y_2)^\dagger) + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$	$-4 \text{Tr}(y_2 (y_1)^\dagger Y_e (Y_e)^\dagger)$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{H2}^{\text{Adj}} \left(-\frac{9}{16} \text{Tr}(y_2 (y_1)^\dagger) - \frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{3\lambda_{S_1^\dagger S_2 S_2^\dagger^2}}{4} \right)$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure	
T1	$+6\lambda_{S_1^\dagger 2 S_2^\dagger 2} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	
T2	$-6 \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* (\lambda_{T3})^*$	
T3	$-9g_2^2 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{S_1^\dagger 2 S_2^\dagger 2} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	$-6 \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* (\lambda_{T3})^*$	$-9g_2^2 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_{S_1^\dagger 2 S_2^\dagger 2} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	$-6 \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* (\lambda_{T3})^*$	$-9g_2^2 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure	
T1	$-3g_Y^2 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	
T2	$+2\lambda_{12} \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	
T3	$+3\lambda_H \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-3g_Y^2 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+2\lambda_{12} \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+3\lambda_H \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-3g_Y^2 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+2\lambda_{12} \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+3\lambda_H \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$-6\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}^{\lambda_{T3}}$
T2	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\frac{3}{2} \text{Tr} \left(y_1 (y_1)^\dagger \right) + \frac{3}{2} \text{Tr} \left(y_2 (y_2)^\dagger \right) + 6 \text{Tr} \left(Y_d (Y_d)^\dagger \right) + 2 \text{Tr} \left(Y_e (Y_e)^\dagger \right) + 6 \text{Tr} \left(Y_u (Y_u)^\dagger \right) \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-6\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}^{\lambda_{T3}}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-6\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}^{\lambda_{T3}}$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\frac{3}{2} \text{Tr} \left(y_1 (y_1)^\dagger \right) + \frac{3}{2} \text{Tr} \left(y_2 (y_2)^\dagger \right) + 6 \text{Tr} \left(Y_d (Y_d)^\dagger \right) + 2 \text{Tr} \left(Y_e (Y_e)^\dagger \right) + 6 \text{Tr} \left(Y_u (Y_u)^\dagger \right) \right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—

$$\beta_{\lambda_{H^\dagger H^\dagger S_2 S_2}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{H^\dagger H^\dagger S_2 S_2}}{d\mu}=\beta_{\lambda_{H^\dagger H^\dagger S_2 S_2}}^{(1)}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_{H^\dagger H^\dagger S_2 S_2} (2 \operatorname{Tr} (y_2 \ (y_2)^\dagger) + 6 \operatorname{Tr} (Y_d \ (Y_d)^\dagger) + 2 \operatorname{Tr} (Y_e \ (Y_e)^\dagger) + 6 \operatorname{Tr} (Y_u \ (Y_u)^\dagger))$
T2	$+ (\lambda_{T3})^* \operatorname{Tr} (y_1 \ (y_2)^\dagger)$
T3	$+ 4 \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{H^\dagger H^\dagger S_2 S_2} (2 \operatorname{Tr} (y_2 \ (y_2)^\dagger) + 6 \operatorname{Tr} (Y_d \ (Y_d)^\dagger) + 2 \operatorname{Tr} (Y_e \ (Y_e)^\dagger) + 6 \operatorname{Tr} (Y_u \ (Y_u)^\dagger)) + (\lambda_{T3})^* \operatorname{Tr} (y_1 \ (y_2)^\dagger)$		$+ 4 \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{H^\dagger H^\dagger S_2 S_2} (3 \operatorname{Tr} (y_2 \ (y_2)^\dagger) + 6 \operatorname{Tr} (Y_d \ (Y_d)^\dagger) + 2 \operatorname{Tr} (Y_e \ (Y_e)^\dagger) + 6 \operatorname{Tr} (Y_u \ (Y_u)^\dagger))$	—	$+ 4 \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 2 of 4

Column	Representative term structure	
T1	$+2(\lambda_{T3})^* \left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* - \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \right)$	
T2	$-9g_2^2 \lambda_{H^\dagger H^\dagger S_2 S_2}$	
T3	$-3g_Y^2 \lambda_{H^\dagger H^\dagger S_2 S_2}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2(\lambda_{T3})^* \left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* - \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \right)$	$-9g_2^2 \lambda_{H^\dagger H^\dagger S_2 S_2}$	$-3g_Y^2 \lambda_{H^\dagger H^\dagger S_2 S_2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2(\lambda_{T3})^* \left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* - \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \right)$	$-9g_2^2 \lambda_{H^\dagger H^\dagger S_2 S_2}$	$-3g_Y^2 \lambda_{H^\dagger H^\dagger S_2 S_2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 3 of 4

Column	Representative term structure
T1	$+2\lambda_H \lambda_{H^\dagger H^\dagger S_2 S_2}$
T2	$+8\lambda_{H2} \lambda_{H^\dagger H^\dagger S_2 S_2}$
T3	$+4\lambda_{H2}^{\text{Adj}} \lambda_{H^\dagger H^\dagger S_2 S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_H \lambda_{H^\dagger H^\dagger S_2 S_2}$	$+8\lambda_{H2} \lambda_{H^\dagger H^\dagger S_2 S_2}$	$+4\lambda_{H2}^{\text{Adj}} \lambda_{H^\dagger H^\dagger S_2 S_2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_H \lambda_{H^\dagger H^\dagger S_2 S_2}$	$+8\lambda_{H2} \lambda_{H^\dagger H^\dagger S_2 S_2}$	$+4\lambda_{H2}^{\text{Adj}} \lambda_{H^\dagger H^\dagger S_2 S_2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 4 of 4

Column	Representative term structure
T1	$+2\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_2}$
T2	$+\frac{3}{2}(\lambda_{T3})^* \text{Tr}\left(y_1 \left(y_2\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_2}$	$+\frac{3}{2}(\lambda_{T3})^* \text{Tr}\left(y_1 \left(y_2\right)^\dagger\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—

$$\beta_{\lambda_{S_1}^{\text{Adj}}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{S_1}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{S_1}^{\text{Adj}}}^{(1)}$$

Terms 1 of 6

Column	Representative term structure	
T1	$\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+13\lambda_{S_1}^{\text{Adj}}\right)$	
T2	$+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	
T3	$-\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+13\lambda_{S_1}^{\text{Adj}}\right)$	$+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+13\lambda_{S_1}^{\text{Adj}}\right)$	$+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+13\lambda_{S_1}^{\text{Adj}}\right)$	$+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+13\lambda_{S_1}^{\text{Adj}}\right)$	$+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+13\lambda_{S_1}^{\text{Adj}}\right)$	$+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+13\lambda_{S_1}^{\text{Adj}}\right)$	$+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+13\lambda_{S_1}^{\text{Adj}}\right)$	$+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+13\lambda_{S_1}^{\text{Adj}}\right)$	$+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+13\lambda_{S_1}^{\text{Adj}}\right)$	$+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)$	$-\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_1\left(y_1\right)^{\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\lambda_{S_1} \left(\frac{1}{2} \text{Tr} \left(y_1 (y_1)^\dagger \right) + 13 \lambda_{S_1}^{\text{Adj}} \right)$	$+4 \lambda_{S_1}^{\text{Adj}} \text{Tr} \left(y_1 (y_1)^\dagger \right)$	$-\text{Tr} \left(y_1 (y_1)^\dagger y_1 (y_1)^\dagger \right)$

Terms 2 of 6

Column	Representative term structure	
T1	$+ \frac{3g_2^4}{4}$	
T2	$+g_Y^2 \left(9g_2^2 - \frac{3\lambda_{S_1}}{2} - 12\lambda_{S_1}^{\text{Adj}} \right)$	
T3	$+g_2^2(-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+ \frac{3g_2^4}{4}$	$+g_Y^2 \left(9g_2^2 - \frac{3\lambda_{S_1}}{2} - 12\lambda_{S_1}^{\text{Adj}} \right)$	$+g_2^2(-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+ \frac{3g_2^4}{4}$	—	$+g_2^2(-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+ \frac{3g_2^4}{4}$	—	$+g_2^2(-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+ \frac{3g_2^4}{4}$	—	$+g_2^2(-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+ \frac{3g_2^4}{4}$	$+g_Y^2 \left(9g_2^2 - \frac{3\lambda_{S_1}}{2} - 12\lambda_{S_1}^{\text{Adj}} \right)$	$+g_2^2(-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+ \frac{3g_2^4}{4}$	$+g_Y^2 \left(9g_2^2 - \frac{3\lambda_{S_1}}{2} - 12\lambda_{S_1}^{\text{Adj}} \right)$	$+g_2^2(-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+ \frac{3g_2^4}{4}$	—	$+g_2^2(-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+ \frac{3g_2^4}{4}$	—	$+g_2^2(-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+ \frac{3g_2^4}{4}$	—	$+g_2^2(-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+ \frac{3g_2^4}{4}$	$+g_Y^2 \left(9g_2^2 - \frac{3\lambda_{S_1}}{2} - 12\lambda_{S_1}^{\text{Adj}} \right)$	$+g_2^2(-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$

Terms 3 of 6

Column	Representative term structure
T1	$+ \frac{3g_Y^4}{2}$
T2	$+ \frac{\lambda_{12}^2}{4}$
T3	$+ \frac{\lambda_{H1}^2}{2}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+ \frac{3g_Y^4}{2}$	$+ \frac{\lambda_{12}^2}{4}$	$+ \frac{\lambda_{H1}^2}{2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+ \frac{3g_Y^4}{32}$	$+ \frac{\lambda_{12}^2}{4}$	$+ \frac{\lambda_{H1}^2}{2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$+ \frac{\lambda_{12}^2}{4}$	$+ \frac{\lambda_{H1}^2}{2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+ \frac{3g_Y^4}{32}$	$+ \frac{\lambda_{12}^2}{4}$	$+ \frac{\lambda_{H1}^2}{2}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+ \frac{3g_Y^4}{2}$	$+ \frac{\lambda_{12}^2}{4}$	$+ \frac{\lambda_{H1}^2}{2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+ \frac{3g_Y^4}{2}$	—	$+ \frac{\lambda_{H1}^2}{2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+ \frac{3g_Y^4}{32}$	—	$+ \frac{\lambda_{H1}^2}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$+ \frac{\lambda_{H1}^2}{2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+ \frac{3g_Y^4}{32}$	—	$+ \frac{\lambda_{H1}^2}{2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+ \frac{3g_Y^4}{2}$	—	$+ \frac{\lambda_{H1}^2}{2}$

Terms 4 of 6

Column	Representative term structure	
T1	$+3\lambda_{H1}^{\text{Adj}2}$	
T2	$+\frac{7\lambda_{S1}^2}{4}$	
T3	$+22\lambda_{S1}^{\text{Adj}2}$	

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+3\lambda_{H1}^{\text{Adj}2}$	$+\frac{7\lambda_{S1}^2}{4}$	$+22\lambda_{S1}^{\text{Adj}2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+3\lambda_{H1}^{\text{Adj}2}$	$+\frac{7\lambda_{S1}^2}{4}$	$+22\lambda_{S1}^{\text{Adj}2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+3\lambda_{H1}^{\text{Adj}2}$	$+\frac{7\lambda_{S1}^2}{4}$	$+22\lambda_{S1}^{\text{Adj}2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+3\lambda_{H1}^{\text{Adj}2}$	$+\frac{7\lambda_{S1}^2}{4}$	$+22\lambda_{S1}^{\text{Adj}2}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+3\lambda_{H1}^{\text{Adj}2}$	$+\frac{7\lambda_{S1}^2}{4}$	$+22\lambda_{S1}^{\text{Adj}2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+3\lambda_{H1}^{\text{Adj}2}$	$+\frac{7\lambda_{S1}^2}{4}$	$+22\lambda_{S1}^{\text{Adj}2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+3\lambda_{H1}^{\text{Adj}2}$	$+\frac{7\lambda_{S1}^2}{4}$	$+22\lambda_{S1}^{\text{Adj}2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{H1}^{\text{Adj}2}$	$+\frac{7\lambda_{S1}^2}{4}$	$+22\lambda_{S1}^{\text{Adj}2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+3\lambda_{H1}^{\text{Adj}2}$	$+\frac{7\lambda_{S1}^2}{4}$	$+22\lambda_{S1}^{\text{Adj}2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+3\lambda_{H1}^{\text{Adj}2}$	$+\frac{7\lambda_{S1}^2}{4}$	$+22\lambda_{S1}^{\text{Adj}2}$

Terms 5 of 6

Column	Representative term structure	
T1	$+g_Y^2 \left(\frac{9g_2^2}{4} - \frac{3\lambda_{S_1}}{8} - 3\lambda_{S_1}^{\text{Adj}} \right)$	
T2	$+\frac{3\lambda_{12}^2}{4}$	
T3	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+g_Y^2 \left(\frac{9g_2^2}{4} - \frac{3\lambda_{S_1}}{8} - 3\lambda_{S_1}^{\text{Adj}} \right)$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+g_Y^2 \left(\frac{9g_2^2}{4} - \frac{3\lambda_{S_1}}{8} - 3\lambda_{S_1}^{\text{Adj}} \right)$	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$+\frac{3\lambda_{12}^2}{4}$	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+g_Y^2 \left(\frac{9g_2^2}{4} - \frac{3\lambda_{S_1}}{8} - 3\lambda_{S_1}^{\text{Adj}} \right)$	$+\frac{3\lambda_{12}^2}{4}$	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+\frac{3\lambda_{12}^2}{4}$	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+g_Y^2 \left(\frac{9g_2^2}{4} - \frac{3\lambda_{S_1}}{8} - 3\lambda_{S_1}^{\text{Adj}} \right)$	$+\frac{3\lambda_{12}^2}{4}$	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$+\frac{3\lambda_{12}^2}{4}$	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$

Terms 6 of 6

Column	Representative term structure
T1	$+12\lambda_{12}^{\text{Adj}2}$
T2	$-\frac{\lambda_{12}^{\text{Cross}2}}{2}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+12\lambda_{12}^{\text{Adj}2}$	$-\frac{\lambda_{12}^{\text{Cross}2}}{2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+12\lambda_{12}^{\text{Adj}2}$	$-\frac{\lambda_{12}^{\text{Cross}2}}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+12\lambda_{12}^{\text{Adj}2}$	$-\frac{\lambda_{12}^{\text{Cross}2}}{2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+12\lambda_{12}^{\text{Adj}2}$	$-\frac{\lambda_{12}^{\text{Cross}2}}{2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+12\lambda_{12}^{\text{Adj}2}$	$-\frac{\lambda_{12}^{\text{Cross}2}}{2}$

$\beta_{\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}}^{(1)}$

$16\pi^2\,\mu\frac{d\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}}{d\mu}=\beta_{\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{12}(\text{Tr}\big(y_2\ (y_1)^{\dagger}\big)+6\lambda_{S_1^{\dagger}S_2S_2^{\dagger 2}}+6\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}})$
T2	$+ \lambda_{12}^{\text{Adj}}\left(\frac{1}{12}\text{Tr}\big(y_2\ (y_1)^{\dagger}\big)+\frac{\lambda_{S_1^{\dagger}S_2S_2^{\dagger 2}}}{6}+\frac{5\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}}{6}\right)$
T3	$+ \lambda_{S_1}(\text{Tr}\big(y_2\ (y_1)^{\dagger}\big)+12\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{12}(\text{Tr}\big(y_2\ (y_1)^{\dagger}\big)+6\lambda_{S_1^{\dagger}S_2S_2^{\dagger 2}}+6\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}})+\lambda_{12}^{\text{Adj}}\left(\frac{1}{12}\text{Tr}\big(y_2\ (y_1)^{\dagger}\big)+\frac{\lambda_{S_1^{\dagger}S_2S_2^{\dagger 2}}}{6}+\frac{5\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}}{6}\right)+\lambda_{S_1}(\text{Tr}\big(y_2\ (y_1)^{\dagger}\big)+12\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}})$		
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+ \lambda_{12}^{\text{Adj}}\left(\frac{1}{8}\text{Tr}\big(y_2\ (y_1)^{\dagger}\big)+\frac{\lambda_{S_1^{\dagger}S_2S_2^{\dagger 2}}}{6}+\frac{5\lambda_{S_1S_1^{\dagger 2}S_2^{\dagger}}}{6}\right)-$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure	
T1	$+ \lambda_{S_1^\dagger 2 S_2^\dagger 2} (2 \text{Tr}(y_1 (y_2)^\dagger) + 4(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^* + 5\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*))$	
T2	$+ \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} (3 \text{Tr}(y_1 (y_1)^\dagger) + \text{Tr}(y_2 (y_2)^\dagger))$	
T3	$- 4 \text{Tr}(y_1 (y_1)^\dagger y_2 (y_1)^\dagger)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \lambda_{S_1^\dagger 2 S_2^\dagger 2} (2 \text{Tr}(y_1 (y_2)^\dagger) + 4(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^* + 5\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*)) + \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} (3 \text{Tr}(y_1 (y_1)^\dagger) + \text{Tr}(y_2 (y_2)^\dagger))$	$- 4 \text{Tr}(y_1 (y_1)^\dagger y_2 (y_1)^\dagger)$	
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \lambda_{S_1^\dagger 2 S_2^\dagger 2} (3 \text{Tr}(y_1 (y_2)^\dagger) + 4(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^* + 5\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*))$	$- 5 \text{Tr}(y_1 (y_1)^\dagger y_2 (y_1)^\dagger)$	
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure	
T1	$-9g_2^2\lambda_{S_1S_1^\dagger S_2^\dagger}$	
T2	$-3g_Y^2\lambda_{S_1S_1^\dagger S_2^\dagger}$	
T3	$+2\lambda_{H1}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{S_1S_1^\dagger S_2^\dagger}$	$-3g_Y^2\lambda_{S_1S_1^\dagger S_2^\dagger}$	$+2\lambda_{H1}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_{S_1S_1^\dagger S_2^\dagger}$	$-3g_Y^2\lambda_{S_1S_1^\dagger S_2^\dagger}$	$+2\lambda_{H1}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$-\frac{\lambda_{H1}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$
T2	$+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}^{\lambda T3}$
T3	$\lambda_{12}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+6\lambda_{S_1^\dagger S_2 S_2^\dagger 2}+6\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{\lambda_{H1}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$	$+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}^{\lambda T3}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\lambda_{H1}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$	$+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}^{\lambda T3}$	$\lambda_{12}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+6\lambda_{S_1^\dagger S_2 S_2^\dagger 2}+6\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$+ \lambda_{S_1} \left(\frac{3}{2} \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right) + 12 \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)$
T2	$+ \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left(\frac{9}{2} \text{Tr} \left(y_1 \left(y_1 \right)^\dagger \right) + \frac{3}{2} \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \lambda_{S_1} \left(\frac{3}{2} \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right) + 12 \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)$	$+ \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left(\frac{9}{2} \text{Tr} \left(y_1 \left(y_1 \right)^\dagger \right) + \frac{3}{2} \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) \right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—

$$\beta_{\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}{d\mu}=\beta_{\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}^{(1)}$$

Terms 1 of 6

Column	Representative term structure
T1	$\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} (2\text{Tr}\big(y_1\ (y_1)^{\dagger}\big) + 2\text{Tr}\big(y_2\ (y_2)^{\dagger}\big))$
T2	$+ \lambda_{S_1^{\dagger} S_2 S_2^{\dagger}} 2\text{Tr}\big(y_2\ (y_1)^{\dagger}\big) + 2\lambda_{S_1 S_1^{\dagger} S_2^{\dagger}} S_2^{\dagger}$
T3	$+ \lambda_{S_1 S_1^{\dagger} S_2^{\dagger}} 2\text{Tr}\big(y_2\ (y_1)^{\dagger}\big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} (2\text{Tr}\big(y_1\ (y_1)^{\dagger}\big) + 2\text{Tr}\big(y_2\ (y_2)^{\dagger}\big))$	$+ \lambda_{S_1^{\dagger} S_2 S_2^{\dagger}} 2\text{Tr}\big(y_2\ (y_1)^{\dagger}\big) + 2\lambda_{S_1 S_1^{\dagger} S_2^{\dagger}} S_2^{\dagger}$	$+ \lambda_{S_1 S_1^{\dagger} S_2^{\dagger}} 2\text{Tr}\big(y_2\ (y_1)^{\dagger}\big)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} (3\text{Tr}\big(y_1\ (y_1)^{\dagger}\big) + 3\text{Tr}\big(y_2\ (y_2)^{\dagger}\big))$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 2 of 6

Column	Representative term structure
T1	$-2 \text{Tr} \left(y_2 \left(y_1 \right)^\dagger y_2 \left(y_1 \right)^\dagger \right)$
T2	$+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$
T3	$-9g_2^2 \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-2 \text{Tr} \left(y_2 \left(y_1 \right)^\dagger y_2 \left(y_1 \right)^\dagger \right)$	$+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$	$-9g_2^2 \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$	$-9g_2^2 \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 3 of 6

Column	Representative term structure	
T1	$-3g_Y^2 \lambda_{S_1^\dagger 2 S_2^\dagger 2}$	
T2	$+8\lambda_{12} \lambda_{S_1^\dagger 2 S_2^\dagger 2}$	
T3	$+\frac{4\lambda_{12}^{\text{Adj}} \lambda_{S_1^\dagger 2 S_2^\dagger 2}}{3}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-3g_Y^2 \lambda_{S_1^\dagger 2 S_2^\dagger 2}$	$+8\lambda_{12} \lambda_{S_1^\dagger 2 S_2^\dagger 2}$	$+\frac{4\lambda_{12}^{\text{Adj}} \lambda_{S_1^\dagger 2 S_2^\dagger 2}}{3}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-3g_Y^2 \lambda_{S_1^\dagger 2 S_2^\dagger 2}$	$+8\lambda_{12} \lambda_{S_1^\dagger 2 S_2^\dagger 2}$	$+\frac{4\lambda_{12}^{\text{Adj}} \lambda_{S_1^\dagger 2 S_2^\dagger 2}}{3}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 4 of 6

Column	Representative term structure	
T1	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}{}^2$	
T2	$+2\lambda_{S_1} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$	
T3	$+2\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \lambda_{S_2}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}{}^2$	$+2\lambda_{S_1} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$	$+2\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \lambda_{S_2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}{}^2$	$+2\lambda_{S_1} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$	$+2\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \lambda_{S_2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 5 of 6

Column	Representative term structure
T1	$+5\lambda_{S_1^\dagger S_2 S_2^\dagger}{}^2$
T2	$+5\lambda_{S_1 S_1^\dagger S_2^\dagger}{}^2$
T3	$+\lambda_{S_1^\dagger S_2 S_2^\dagger}{}^2 \left(\frac{3}{2} \text{Tr} \left(y_2 (y_1)^\dagger \right) + 2\lambda_{S_1 S_1^\dagger S_2^\dagger}{}^2 \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+5\lambda_{S_1^\dagger S_2 S_2^\dagger}{}^2$	$+5\lambda_{S_1 S_1^\dagger S_2^\dagger}{}^2$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+5\lambda_{S_1^\dagger S_2 S_2^\dagger}{}^2$	$+5\lambda_{S_1 S_1^\dagger S_2^\dagger}{}^2$	$+\lambda_{S_1^\dagger S_2 S_2^\dagger}{}^2 \left(\frac{3}{2} \text{Tr} \left(y_2 (y_1)^\dagger \right) + 2\lambda_{S_1 S_1^\dagger S_2^\dagger}{}^2 \right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 6 of 6

Column	Representative term structure
T1	$+\frac{3}{2}\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \text{Tr}(y_2 (y_1)^\dagger)$
T2	$-\frac{5}{2} \text{Tr}(y_2 (y_1)^\dagger y_2 (y_1)^\dagger)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3}{2}\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \text{Tr}(y_2 (y_1)^\dagger)$	$-\frac{5}{2} \text{Tr}(y_2 (y_1)^\dagger y_2 (y_1)^\dagger)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—

$$\beta_{\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}^{(1)}$$

$$16\pi^2\,\mu\,\frac{d\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}{d\mu}=\beta_{\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{12}(\text{Tr}\big(y_2\ (y_1)^\dagger\big)+6\lambda_{S_1^\dagger S_2 S_2^\dagger 2}+6\lambda_{S_1 S_1^\dagger 2 S_2^\dagger})$
T2	$+ \lambda_{12}^{\text{Adj}}\left(\frac{1}{12}\text{Tr}\big(y_2\ (y_1)^\dagger\big)+\frac{5\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}{6}+\frac{\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}}{6}\right)$
T3	$+ \lambda_{S_1^\dagger 2 S_2^\dagger 2}(2\text{Tr}\big(y_1\ (y_2)^\dagger\big)+4(5\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*+\left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{12}(\text{Tr}(y_2 (y_1)^\dagger) + 6\lambda_{S_1^\dagger S_2 S_2^\dagger 2} + 6\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}) + \lambda_{12}^{\text{Adj}}\left(\frac{1}{12}\text{Tr}(y_2 (y_1)^\dagger) + \frac{5\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}{6} + \frac{\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}}{6}\right) + \lambda_{S_1^\dagger 2 S_2^\dagger 2}(2\text{Tr}(y_1 (y_2)^\dagger) + 4(5\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*))$		
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \lambda_{12}^{\text{Adj}}\left(\frac{1}{8}\text{Tr}(y_2 (y_1)^\dagger) + \frac{5\lambda_{S_1^\dagger S_2 S_2^\dagger 2}}{6} + \frac{\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}}{6}\right) + \lambda_{S_1^\dagger 2 S_2^\dagger 2}(3\text{Tr}(y_1 (y_2)^\dagger) + 4(5\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^* + \left(\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)^*))$		
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$+\lambda_{S_2}(\text{Tr}(y_2 (y_1)^\dagger) + 12\lambda_{S_1^\dagger S_2 S_2^\dagger 2})$
T2	$+\lambda_{S_1^\dagger S_2 S_2^\dagger 2}(\text{Tr}(y_1 (y_1)^\dagger) + 3 \text{Tr}(y_2 (y_2)^\dagger))$
T3	$-4 \text{Tr}(y_2 (y_1)^\dagger y_2 (y_2)^\dagger)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{S_2}(\text{Tr}(y_2 (y_1)^\dagger) + 12\lambda_{S_1^\dagger S_2 S_2^\dagger 2})$	$+\lambda_{S_1^\dagger S_2 S_2^\dagger 2}(\text{Tr}(y_1 (y_1)^\dagger) + 3 \text{Tr}(y_2 (y_2)^\dagger))$	$-4 \text{Tr}(y_2 (y_1)^\dagger y_2 (y_2)^\dagger)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$-5 \text{Tr}(y_2 (y_1)^\dagger y_2 (y_2)^\dagger)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$+4 \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* (\lambda_{T3})^*$
T2	$-9g_2^2 \lambda_{S_1^\dagger S_2 S_2^\dagger 2}$
T3	$-3g_Y^2 \lambda_{S_1^\dagger S_2 S_2^\dagger 2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4 \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* (\lambda_{T3})^*$	$-9g_2^2 \lambda_{S_1^\dagger S_2 S_2^\dagger 2}$	$-3g_Y^2 \lambda_{S_1^\dagger S_2 S_2^\dagger 2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4 \left(\lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* (\lambda_{T3})^*$	$-9g_2^2 \lambda_{S_1^\dagger S_2 S_2^\dagger 2}$	$-3g_Y^2 \lambda_{S_1^\dagger S_2 S_2^\dagger 2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$+2\lambda_{H2}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T2	$-\frac{\lambda_{H2}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$
T3	$\lambda_{12}\left(\frac{3}{2}\text{Tr}\left(y_2\ (y_1)^\dagger\right)+6\lambda_{S_1^\dagger S_2 S_2^\dagger 2}+6\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{H2}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$-\frac{\lambda_{H2}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{H2}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$-\frac{\lambda_{H2}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$	$\lambda_{12}\left(\frac{3}{2}\text{Tr}\left(y_2\ (y_1)^\dagger\right)+6\lambda_{S_1^\dagger S_2 S_2^\dagger 2}+6\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$+ \lambda_{S_2} \left(\frac{3}{2} \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right) + 12 \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)$
T2	$+ \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left(\frac{3}{2} \text{Tr} \left(y_1 \left(y_1 \right)^\dagger \right) + \frac{9}{2} \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \lambda_{S_2} \left(\frac{3}{2} \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right) + 12 \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)$	$+ \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left(\frac{3}{2} \text{Tr} \left(y_1 \left(y_1 \right)^\dagger \right) + \frac{9}{2} \text{Tr} \left(y_2 \left(y_2 \right)^\dagger \right) \right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—

$$\beta_{\lambda_{S_2}^{\text{Adj}}}^{(1)}$$

$$16\pi^2\mu\frac{d\lambda_{S_2}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{S_2}^{\text{Adj}}}^{(1)}$$

Terms 1 of 7

Column	Representative term structure
T1	$\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$
T2	$+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$
T3	$-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$	$+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$	$-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$	$+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$	$-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$	$+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$	$-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$	$+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$	$-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$	$+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$	$-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$	$+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$	$-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$	$+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$	$-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$	$+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$	$-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$	$+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$	$-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\lambda_{S_2} \left(\frac{1}{2} \text{Tr} \left(y_2 (y_2)^\dagger \right) + 13 \lambda_{S_2}^{\text{Adj}} \right)$	$+4 \lambda_{S_2}^{\text{Adj}} \text{Tr} \left(y_2 (y_2)^\dagger \right)$	$-\text{Tr} \left(y_2 (y_2)^\dagger y_2 (y_2)^\dagger \right)$

Terms 2 of 7

Column	Representative term structure
T1	$+\frac{3g_2^4}{4}$
T2	$+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$
T3	$+\frac{\lambda_{12}^2}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+\frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$	$+\frac{\lambda_{12}^2}{4}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$	$+\frac{\lambda_{12}^2}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$	$+\frac{\lambda_{12}^2}{4}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$	$+\frac{\lambda_{12}^2}{4}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+\frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$	$+\frac{\lambda_{12}^2}{4}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$	—

Terms 3 of 7

Column	Representative term structure	
T1	$+\frac{\lambda_{H2}^2}{2}$	
T2	$+3\lambda_{H2}^{\text{Adj}^2}$	
T3	$+\frac{7\lambda_{S2}^2}{4}$	

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+\frac{\lambda_{H2}^2}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	$+\frac{7\lambda_{S2}^2}{4}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{\lambda_{H2}^2}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	$+\frac{7\lambda_{S2}^2}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{\lambda_{H2}^2}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	$+\frac{7\lambda_{S2}^2}{4}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{\lambda_{H2}^2}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	$+\frac{7\lambda_{S2}^2}{4}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+\frac{\lambda_{H2}^2}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	$+\frac{7\lambda_{S2}^2}{4}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\frac{\lambda_{H2}^2}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	$+\frac{7\lambda_{S2}^2}{4}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{\lambda_{H2}^2}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	$+\frac{7\lambda_{S2}^2}{4}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{\lambda_{H2}^2}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	$+\frac{7\lambda_{S2}^2}{4}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{\lambda_{H2}^2}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	$+\frac{7\lambda_{S2}^2}{4}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\frac{\lambda_{H2}^2}{2}$	$+3\lambda_{H2}^{\text{Adj}^2}$	$+\frac{7\lambda_{S2}^2}{4}$

Terms 4 of 7

Column	Representative term structure
T1	$+22\lambda_{S_2}^{\text{Adj}^2}$
T2	$+g_Y^2 \left(\frac{9g_2^2}{4} - \frac{3\lambda_{S_2}}{8} - 3\lambda_{S_2}^{\text{Adj}} \right)$
T3	$+\frac{3g_Y^4}{32}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+22\lambda_{S_2}^{\text{Adj}^2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+22\lambda_{S_2}^{\text{Adj}^2}$	$+g_Y^2 \left(\frac{9g_2^2}{4} - \frac{3\lambda_{S_2}}{8} - 3\lambda_{S_2}^{\text{Adj}} \right)$	$+\frac{3g_Y^4}{32}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+22\lambda_{S_2}^{\text{Adj}^2}$	—	$+\frac{3g_Y^4}{2}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+22\lambda_{S_2}^{\text{Adj}^2}$	$+g_Y^2 \left(\frac{81g_2^2}{4} - \frac{27\lambda_{S_2}}{8} - 27\lambda_{S_2}^{\text{Adj}} \right)$	$+\frac{243g_Y^4}{32}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+22\lambda_{S_2}^{\text{Adj}^2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+22\lambda_{S_2}^{\text{Adj}^2}$	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+22\lambda_{S_2}^{\text{Adj}^2}$	$+g_Y^2 \left(\frac{9g_2^2}{4} - \frac{3\lambda_{S_2}}{8} - 3\lambda_{S_2}^{\text{Adj}} \right)$	$+\frac{3g_Y^4}{32}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+22\lambda_{S_2}^{\text{Adj}^2}$	—	$+\frac{3g_Y^4}{2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+22\lambda_{S_2}^{\text{Adj}^2}$	$+g_Y^2 \left(\frac{81g_2^2}{4} - \frac{27\lambda_{S_2}}{8} - 27\lambda_{S_2}^{\text{Adj}} \right)$	$+\frac{243g_Y^4}{32}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+22\lambda_{S_2}^{\text{Adj}^2}$	—	—

Terms 5 of 7

Column	Representative term structure	
T1	$+g_Y{}^2\left(9g_2{}^2-\frac{3\lambda_{S_2}}{2}-12\lambda_{S_2}^{\text{Adj}}\right)$	
T2	$+g_Y{}^2\left(36g_2{}^2-6\lambda_{S_2}-48\lambda_{S_2}^{\text{Adj}}\right)$	
T3	$+24g_Y{}^4$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+g_Y{}^2\left(9g_2{}^2-\frac{3\lambda_{S_2}}{2}-12\lambda_{S_2}^{\text{Adj}}\right)$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	$+g_Y{}^2\left(36g_2{}^2-6\lambda_{S_2}-48\lambda_{S_2}^{\text{Adj}}\right)$	$+24g_Y{}^4$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_Y{}^2\left(9g_2{}^2-\frac{3\lambda_{S_2}}{2}-12\lambda_{S_2}^{\text{Adj}}\right)$	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$+g_Y{}^2\left(36g_2{}^2-6\lambda_{S_2}-48\lambda_{S_2}^{\text{Adj}}\right)$	$+24g_Y{}^4$

Terms 6 of 7

Column	Representative term structure	
T1	$+\frac{3\lambda_{12}^2}{4}$	
T2	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$	
T3	$+12\lambda_{12}^{\text{Adj}^2}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\frac{3\lambda_{12}^2}{4}$	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$	$+12\lambda_{12}^{\text{Adj}^2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{3\lambda_{12}^2}{4}$	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$	$+12\lambda_{12}^{\text{Adj}^2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3\lambda_{12}^2}{4}$	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$	$+12\lambda_{12}^{\text{Adj}^2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{3\lambda_{12}^2}{4}$	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$	$+12\lambda_{12}^{\text{Adj}^2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\frac{3\lambda_{12}^2}{4}$	$+\lambda_{12}\lambda_{12}^{\text{Cross}}$	$+12\lambda_{12}^{\text{Adj}^2}$

Column		Representative term structure
T1	$-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$